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ON THE ARCHITECTURE OF DOMINANCE REVERSAL

By

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ABSTRACT OF THE DISSERTATION

On the Architecture of Dominance Reversal

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This dissertation investigates dominance reversal in ATR vowel harmony, a pattern in which a language otherwise characterized by [+ATR] dominance also exhibits instances where underlyingly [+ATR] vowels surface as [−ATR] in a restricted set of contexts. Existing analyses of such systems have differed substantially in their assumptions and formalisms, sometimes even disagreeing over whether the same empirical pattern instantiates dominance reversal at all. This dissertation argues that dominance reversal corresponds to a formal grammatical class best understood as a ranking architecture {Blocker, Harmony-Driver} >> IDENT-TRIGGER >> IDENT-TARGET rather than as an effect of any particular constraint, process or feature. This configuration is identified through a factorial typology over a minimal ATR system, and validated cross-linguistically across typologically diverse phonological systems.

This ranking architecture thus allows the unification of empirically disparate patterns, including non-ATR harmony systems and conspiracy effects, into a single grammatical class, establishing that dominance reversal is more general than its treatment in the ATR harmony literature has suggested. Additionally, the same architecture provides a diagnostic for the pattern grounded in the relational structure of the grammar rather than in surface description, abstracting over specific constraints, theoretical commitments and features, allowing comparison across different OT anal-

yses that are otherwise incomparable. Finally, variation in how dominance reversal surfaces and what resolutions it coexists with is shown to reduce to independently motivated differences within the same configuration, yielding a typology predicted by the architecture.

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"You must do the thing you think you cannot do."

— Eleanor Roosevelt, *You Learn by Living*

I recall starting this program five years ago and picturing what I would say in this section and to whom I would extend my most heartfelt thanks. Understandably, at the time, while I had an idea of who I would thank, the exact wording escaped me. And while the concept of an "acknowledgement section" remained diffusely present in my mind as this big step that would mark the completion of my studies, I realize now I should have perhaps thought about the content and wording a little bit more concretely and definitely earlier.

Whoever will someday write an acknowledgement of their own and is presently reading my own, I urge you to not make the mistake I am making now and start *earlier* than you think appropriate. Since it is too late for me to follow my own advice, I will approach the problem as I approach all problems: with a little panic and a lot of coffee. In the spirit, then, of trying to reassess the chronology of my life, let's start with who made it possible for me to embark on this journey.

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CHAPTER 1
INTRODUCTION

1.1 Introduction

Dominance reversal (Baković 2000) is a pattern attested in a small set of ATR vowel harmony languages where [+ATR] normally behaves as the dominant feature value. In these systems, although [−ATR] vowels typically surface as [+ATR] when they occur in the harmony domain of a [+ATR] vowel, an underlyingly [+ATR] vowel can also surface as [−ATR] in a restricted set of contexts. This pattern has been documented in typologically diverse languages, including, but not limited to, Kinande (Bantu; Schlindwein 1987; Mutaka 1994, 1995, 2007; Archangeli and Pulleyblank 1994, 2002; Hyman 2002; Downing and Krämer 2022), Turkana (Eastern Nilotic; *e.g.,* Dimmendaal 1983; Hulst and Smith 1986; Vago and Leder 1987; Noske 1990, 1996, 2000; Baković 2000; McCollum et al. 2020; Meinhardt et al. 2024), Komo (Koman; Otero 2015; 2018; Olejarczuk et al. 2019; Leduc and McCollum 2023), Liko (Bantu; Kutsch Lojenga 2008; Wit 2015a, 2015b, 2016; McCollum et al. 2020), and Ateso (Kitching 1915; Kiggen 1953a, 1953b; Hilders and Lawrance 1956, 1958; Otaala 1981; Noske and Otaala 2006; Barasa 2012, 2015, 2016, 2017, 2018). Consider, for example, Kinande (1) and Komo (2) below.

Kinande exhibits [+ATR] harmony triggered by a [+ATR] vowel anywhere in the prosodic word. As (1a) shows, in the absence of a [+ATR] vowel, all [−ATR] vowels surface faithfully. However, when a [+ATR] vowel is present, harmony spreads regressively onto preceding vowels (1b), changing their [−ATR] specification to [+ATR], though it generally does not spread rightward (1c). Kinande also features a pattern of progressive parasitic harmony, where a sequence of high vowels /i…ɪ/ causes the

[−ATR] vowel to surface as [+ATR] (1d). If, however, the high [+ATR] vowel is in a prefix position, and the [−ATR] vowel is in a root, then the prefix [+ATR] vowel surfaces as [−ATR] instead (1e).

(1) Kinande Harmony (Kenstowicz 2009:23,26; Downing and Krämer 2022:4)

	UR	SR	Gloss
No Harmony			
a.	/a-ba-√lʊmɛ/	[a-ba-√lʊmɛ]	'man.PL'
/+Atr/ Harmony			
b.	/a-ba-√kali/	[ə-bə-√kəli]	'woman/PL'
c.	/ɛ-ri-√hɛmbɛ/	[e-ri-√hɛmbɛ]	'horn'
d.	/ɛ-ri-√himb-ɪr-a/	[e-ri-√himb-ir-a]	'to build for'
Dominance Reversal			
e.	/ɛ-ri-√hɪrɪ/	[ɛ-rɪ-√hɪrɪ]	'crab'

Komo, on the other hand, exhibits regressive [+ATR] harmony triggered by high [+ATR] vowels. As (2a-c) shows, a high [+ATR] vowel causes the closest preceding vowel to surface as [+ATR], although harmony does not spread further, as shown in (2c). If, however, the preceding vowel is [+HI,−ATR], then the [+ATR] vowel itself surfaces as [−ATR] instead (2d).

(2) Komo Harmony (Otero 2015:215-217)

	UR	SR	Gloss
a.	/√ham-i/	[√həm-i]	'you yawn'
b.	/√kiʃ-a/	[√kiʃ-a]	'I harvest'
c.	/√maza-i/	[√mazə-i]	'be tired-2SG'
d.	/√ʃɪt-i/	[√ʃɪt-ɪ]	'you whistle'

Both languages, then, exhibit similar surface generalizations: a broad pattern of [+ATR] harmony coexisting with a restricted context in which [+ATR] vowels surface

as [−ATR]. It is the two mappings where [+ATR] → [−ATR] and [−ATR] → [+ATR] occurring in the same grammar that makes Kinande and Komo dominance reversal languages.

In Kinande, dominance reversal is restricted to a configuration where a high [+ATR] prefix precedes a high [−ATR] root. In Komo, all [−ATR] high vowels induce dominance reversal if they immediately precede a [+ATR] vowel. The question is whether these languages share a formal property that produces the surface pattern described above, or if their similarities are merely coincidental.

Dominance reversal was initially introduced by Baković (2000) as a ranking outcome in Optimality Theory (Prince and Smolensky 1993/2004) where a harmony-driving constraint and a higher-ranked constraint protecting the recessive feature value jointly dominate the faithfulness constraint that would otherwise preserve the dominant value.

Under this definition, dominance reversal is thus defined structurally by its ranking rather than by the presence of any dedicated process that spreads the recessive value. If this is correct, then the data seen in languages like Kinande and Komo would not be indicative of two separate harmony processes, but of a single harmony imperative in which the resolution tending toward either [+ATR] or [−ATR] depends on the input structure and the relative ranking of the constraints that evaluate it. The existing analyses of dominance reversal languages, however, have generally not adopted this framing, instead recasting the pattern as a conflict between two independent harmony processes (see *e.g.,* Otero 2015 for Komo; Meinhardt et al. 2024 for Turkana and the broader observation that previous analyses of Turkana treat the two mappings as separate processes). Since the term has been used in different ways, analyses thus sometimes disagree on whether the patterns they are modelling constitute dominance reversal at all. Descriptively, dominance reversal is defined as a surface pattern where a [+ATR] vowel surfaces as [−ATR] within a system of [+ATR] harmony. Analytically, however, dominance reversal is sometimes analyzed as the interaction of two distinct

processes, one for [+ATR] harmony and the other for [−ATR] harmony (*e.g.,* Leduc and McCollum 2023, Downing and Krämer 2022), or as a ranking outcome in Optimality Theory yielding either full [+ATR] harmony or full [−ATR] harmony (*e.g.,* Baković 2000). For example, while Kenstowicz (2009) and Otero (2015) have argued for Kinande and Komo respectively that the languages exhibit dominance reversal, Downing & Krämer's (2022) analysis of Kinande, and Leduc & McCollum's (2023) analysis of Komo have argued the opposite: that neither language exhibits dominance reversal on the grounds that there is no dedicated mechanism responsible for spreading [−ATR].

And so, the surface data for these languages appear compatible with multiple formal interpretations. The question, then, is whether these all instantiate the same phenomenon, and if so, how to tell. The analyses themselves are not easily comparable. For Kinande alone, for example, Downing & Krämer (2022) use Syntagmatic Correspondence, a framework based on output-output correspondence (Krämer 1998, 2001, 2003), while Kenstowicz (2009) relies on constraint conjunction. The formalisms differ substantially from one another and there is no way to evaluate whether a given language instantiates dominance reversal, or whether a given analysis can model the same phenomenon.

This dissertation argues that dominance reversal in ATR harmony corresponds to a single ranking configuration in Optimality Theory: {BLOCKER, HARMONY-DRIVER} >> IDENT-TRIGGER >> IDENT-TARGET, identified through a factorial typology of nine constraints over a minimal ATR system.

Defining dominance reversal in terms of this ranking makes the question of whether a given language or analysis instantiates it tractable, as it reduces to whether the ranking, once language-specific constraints are abstracted away, instantiates this configuration. This dissertation shows that it does so not only across typologically diverse languages but also across analyses that may initially appear incompatible.

The contribution of this dissertation extends beyond the empirical domain of ATR

harmony. By reducing dominance reversal to a ranking configuration defined over constraint roles, rather than over specific constraints or features, I characterize a type of grammatical resolution that arises whenever a grammar is simultaneously compelled to enforce harmony and prevented from fully doing so. The ingredients that produce this resolution, then, are not specific to the feature ATR, but can recur in any harmony system. More broadly, the results bear on a general question in phonological theory: when a grammar faces competing pressures and has multiple ways to resolve them, what determines which resolution surfaces? I show that the ranking configuration both identifies when dominance reversal is expected and predicts the cross-linguistic variations in its realization, which reduces to principled differences in constraint ranking within the same configuration.

It is common practice in Optimality Theory to pair constraints and their definitions with articulatory or perceptual motivations. For example, a constraint such as *[+LO,+ATR], which prohibits a [+LO,+ATR] segment in the output, is often motivated by the antagonistic articulatory gestures required by the tongue to produce [+LO] and [+ATR]. In this dissertation, however, I adopt the position in Prince (2007:46):

> Since OT is a theory of grammar, the consequences are displayed in the grammars predicted and disallowed — 'typological evidence'. A constraint which cannot be justified on those grounds cannot be justified. Further 'justifying' a constraint functionally (or in any other extrinsic way) can have no effect whatever on its role within the theory. A constraint, viewed locally, can appear wonderfully concordant with some function, but this cannot supplant the theory's logic or compel the global outcome ('efficiency') that is imagined to follow from the constraint's presence, or even make it more likely.

In this dissertation, then, the constraints adopted in this dissertation are therefore

justified by the factorial typology presented in Chapter 2.

The remainder of this dissertation shows that this ranking is recoverable from Komo, Liko, Kinande, Turkana, and Ateso despite the substantial differences in how these languages implement the pattern and in the formalisms that have been used to analyze them.

1.2 How is dominance reversal possible in OT?

What makes the ranking {Blocker, Harmony-Driver} >> IDENT-TRIGGER >> IDENT-TARGET capable of producing dominance reversal is a property of markedness constraints where, while they define a violation profile banning certain structures in the output, they do not, in and of themselves, impose a resolution to any incurred violations (Prince and Smolensky 1993/2004; see also McCarthy 2001 on homogeneity of target/heterogeneity of process). For example, a constraint like AGREE[F] (Lombardi 1996; Beckman 1998; Butska 1998; Baković 2000; Mahanta 2008) does penalize disharmony for the feature [F], but can be fully satisfied by any output that fully agree in [F], regardless of whether the value is [+F] or [−F]. It is the surrounding constraints, then, which select the optimal outcome.

Consider a simple case involving the input /ɛ-i/, a markedness constraint such as AGREE[ATR] (3) and a faithfulness constraint IDENT[ATR] (4).

(3) AGREE[ATR]: Assign one violation mark for each pair of adjacent vowels that disagree in their value for [ATR].

(4) IDENT[ATR]: Assign one violation mark for each output vowel whose value for [ATR] differs from that of its input correspondent.

Under these constraints and with this input, there are three logically possible candidates: the faithful candidate [ɛ-i] (5a), the [+ATR] harmony candidate [e-i]

(5b), and the [−ATR] harmony candidate [ɛ-ɪ] (5c). A fourth candidate ([e-ɪ]), where both vowels surface unfaithfully, is also possible. I do not include this candidate, however, since it is harmonically bounded by the faithful candidate (5a). Of the three candidates in (5a-c), only the faithful candidate violates AGREE[ATR], due to the disagreement between the two vowels. On the other hand, the choice between the two harmony candidates (5b,c) is impossible even though they satisfy the markedness constraint, because both incur a single violation on IDENT[ATR].

(5) Input: /ɛ-i/

/ɛ-i/	AGREE[ATR]	ID[A]	
a. [ɛ-i]	*!		*Faithful Cand.*
b. [e-i]		*	*[+Atr] Harmony*
c. [ɛ-ɪ]		*	*[−Atr] Harmony*

In practice, this indeterminacy does not always surface in published analyses, since the opposing harmony candidate is sometimes omitted from the candidate set. Mahanta (2008:35), for example, shows how [+ATR] harmony in Assamese is derived using AGREE[ATR] and IDENT[ATR], but the tableau includes only the faithful candidate and the [+ATR] harmony candidate, while the [−ATR] harmony candidate is absent.

(6) Input: /√kɔr-i/ (Mahanta 2008:35)

/√kɔr-i/	AGREE[ATR]	ID[A]	
a. [√kɔr-i]	*!		*Faithful Cand.*
☞ b. [√kor-i]		*	*[+Atr] Harmony*

If we add the missing candidate [kɔr-ɪ] (7c), then the current ranking is unable to choose between the [+ATR] and [−ATR] harmony candidates:

(7) Input: /√kɔr-i/

/√kɔr-i/	AGREE[ATR]	ID[A]	
a. [√kɔr-i]	*!		*Faithful Cand.*
b. [√kor-i]		*	*[+Atr] Harmony*
c. [√kɔr-ɪ]		*	*[−Atr] Harmony*

Similarly, in their demonstration of sour grapes, Mullin & Pater (2015) use an AGREE constraint to show harmony in the absence of a blocker. Their candidate set, however, as in Mahanta (2008), excludes the logically possible neutralization candidate [tuuuu].

(8) Input: /Tuuuu/ (Mullin and Pater 2015:5)

/Tuuuu/	*B	AGREE	ID([F])	
a. [Tuuuu]		*!		*Sour Grapes*
b. [TUUuu]		*!	**	*Part. Spreading*
☞ c. [TUUUU]			****	*Full Spreading*

In this specific case, expanding the candidate set to include the neutralization candidate reveals a different outcome. In (9), the candidate set includes [tuuuu] (9d), where the trigger /T/ surfaces without its feature.

(9) Input: /Tuuuu/

/Tuuuu/	*B	AGREE	ID([F])	
a. [Tuuuu]		*!		*Sour Grapes*
b. [TUUuu]		*!	**	*Part. Spreading*
c. [TUUUU]			**!**	*Full Spreading*
☞ d. [tuuuu]			*	*Neutralization*

Like the full spreading candidate (9c), [tuuuu] does not incur violations of AGREE, since all vowels share the same feature value. Unlike (9c), however, [tuuuu] incurs only a single violation of IDENT[F], from the change of /T/ to [t]. By comparison, (9c) incurs four violations, where each /u/ surfaces as [U]. The neutralization candidate (9d) is optimal under this specific ranking, rather than the sour grapes candidate (9a).

In all cases across (7)-(9), then, the markedness constraint does not select a particular feature value for spreading. This outcome occurs because, again, constraints like AGREE do not distinguish between candidates where all the segments are in agreement, regardless of what the value of this agreement is.

Even if we were to specify a constraint like ALIGN-L[+ATR] (10; Kirchner 1993; McCarthy and Prince 1993), for example, which references both [+ATR] and a specific prosodic edge, the same issue would arise.

(10) ALIGN[+ATR], PRWD-L

For every autosegment [+ATR], assign one violation mark for each segment that separates it from the left edge of the Prosodic Word.

As with AGREE above, so long as the faithfulness constraint IDENT[ATR] does not distinguish between [+ATR] and [−ATR], there is no way for the grammar to choose which feature is dominant (11).

(11) Input: /ɛ-i/

/ɛ-i/	ALIGN-L[+ATR]	IDENT[ATR]	
a. [ɛ-i]	*!		*Faithful Cand.*
b. [ɛ-ɪ]		*	*[−Atr] Harmony*
c. [e-i]		*	*[+Atr] Harmony*

Thus, while ALIGN-L[+ATR] evaluates [+ATR] autosegments that are present in the output, the constraint is also satisfied vacuously if the output contains no [+ATR]

specification. As with AGREE, it cannot distinguish between full [+ATR] harmony and [−ATR] harmony.

Specific markedness constraints referencing the desired spreading value cannot establish feature dominance on their own. Breaking the indeterminacy requires splitting the faithfulness constraint into two constraints that target [±ATR] values separately, IDENT[+ATR] (12) and IDENT[−ATR] (13).

(12) IDENT[+ATR] : Assign one violation mark for each output vowel that is [−ATR] whose input correspondent is [+ATR].

(13) IDENT[−ATR] : Assign one violation mark for each output vowel that is [+ATR] whose input correspondent is [−ATR].

By splitting the faithfulness constraint, the two harmony candidates no longer incur identical violation profiles, as seen in (14, 15). Each violates a different faithfulness constraint, and the ranking between IDENT[+ATR] and IDENT[−ATR] is what determines which candidate is selected by the grammar. (14) shows that [+ATR] harmony is derived when IDENT[+ATR] >> IDENT[−ATR], while (15) shows [−ATR] harmony with the opposite ranking.

(14) Input: /ɛ-i/

/ɛ-i/	ALIGN-L[+ATR]	IDENT[+ATR]	IDENT[−ATR]	
a. [ɛ-i]	*!			*Faithful Cand.*
b. [ɛ-ɪ]		*!		*[−Atr] Harmony*
☞ c. [e-i]			*	*[+Atr] Harmony*

(15) Input: /ɛ-i/

/ɛ-i/	Align-L[+Atr]	Ident[−Atr]	Ident[+Atr]	
a. [ɛ-i]	*!			*Faithful Cand.*
b. [e-i]		*!		*[+Atr] Harmony*
☞ c. [ɛ-ɪ]			*	*[−Atr] Harmony*

The resolution is entirely determined by the faithfulness ranking. While markedness constraints like Agree and Align establish that the vowels in each candidate must agree in Atr, it is the ranking of Ident[+Atr] and Ident[−Atr] that determines which value is dominant.

These rankings, however, are mutually exclusive. A single language cannot simultaneously have both Ident[+Atr] >> Ident[−Atr] and Ident[−Atr] >> Ident[+Atr], yet dominance reversal is described precisely as a case where [+Atr] harmony coexists with a more restricted pattern of [−Atr] harmony within the same language.

The question, then, is how a single grammar can produce both [+Atr] and [−Atr] harmony. As established earlier, dominance reversal arises when a blocking constraint and a harmony-driving constraint jointly outrank Ident-Trigger, the faithfulness constraint protecting the dominant value (16).

(16) Dominance Reversal Ranking

{Blocker, Harmony-Driver} >> Ident-Trigger >> Ident-Target

The rankings in (14) and (15) produce full [±Atr] harmony because there is no constraint that prevents the harmony-driving constraint from being fully satisfied. If, however, a constraint ranked above Ident[+Atr] prevents a specific segment from undergoing [+Atr] harmony, such as Ident[−Atr]/[+Lo] (17), then the harmony-driving constraint can no longer be fully satisfied by spreading [+Atr] in a structure that contains a low [−Atr] vowel (18).

(17) IDENT[−ATR]/[+LO] : Assign one violation mark for each output vowel that is [+ATR] whose input correspondent is [+LO,−ATR].

Consider, then, the ranking from (14), where ALIGN-L[+ATR] >> IDENT[+ATR] >> IDENT[−ATR]. Ranking IDENT[−ATR]/[+LO] above IDENT[+ATR] prevents full [+ATR] harmony by protecting the [−ATR] value on low vowels (18a). Partial [+ATR] spreading (18c) and the faithful candidate (18b) incur fatal violations, leaving [−ATR] harmony candidate as optimal (18d).

(18) Input: /a-ɛ-i/

/a-ɛ-i/	IDENT[−ATR]/[+LO]	ALIGN-L[+ATR]	IDENT[+ATR]	IDENT[−ATR]	
a. [ə-e-i]	*!			**	*[+Atr] Harmony*
b. [a-ɛ-i]		*!*			*Faithful Cand.*
c. [a-e-i]		*!		*	*Part. Harmony*
☞ d. [a-ɛ-ɪ]			*		*[−Atr] Harmony*

When the low blocking vowel is absent however, [+ATR] harmony is still considered optimal due to the evaluation falling to the general IDENT[+ATR] >> IDENT[−ATR] ranking (19).

(19) Input: /ɛ-i/

/ɛ-i/	IDENT[−ATR]/[+LO]	ALIGN-L[+ATR]	IDENT[+ATR]	IDENT[−ATR]	
a. [ɛ-i]		*!			*Faithful Cand.*
b. [ɛ-ɪ]			*!		*[−Atr] Harmony*
☞ c. [e-i]				*	*[+Atr] Harmony*

The indifference of the markedness constraint thus allows the *same* language to produce different harmony patterns depending on the structure, allowing patterns like dominance reversal to exist. The logic is as follows. In a structure with no low vowels, the ranking IDENT[+ATR] >> IDENT[−ATR] drives [+ATR] harmony by making changes from [−ATR] to [+ATR] less costly for the grammar than a change

from [+ATR] to [−ATR].[1] In a structure with a low vowel, however, the ranking IDENT[−ATR]/[+LO] >> IDENT[+ATR] creates the opposite scenario, because it becomes less costly for the grammar to change a [+ATR] vowel to [−ATR] than to change a [+LO,−ATR] vowel to [+ATR]. In other words, the pressure to harmonize is so strong that full harmony, regardless of the value, is always preferable to contexts in which preserving a specific feature value, allowing some restricted contexts where [−ATR] harmony is judged to be preferable to [+ATR] harmony. The result is a single grammar with two distinct resolutions of the same harmony imperative: [+ATR] spreading in most cases, and [−ATR] harmony in a more restricted context where [+ATR] harmony cannot fully apply.

The ranking {BLOCKER, HARMONY-DRIVER} >> IDENT[+ATR] >> IDENT[−ATR] thus produces both [+ATR] and [−ATR] harmony within a single grammar, with the resolution ultimately determined by whether the blocking constraint is active.

1.3 Structure and Scope

1.3.1 Structure

This dissertation is organized into six chapters. Chapter 1, the present chapter, introduces the empirical phenomenon, states the central claim, develops the formal intuition behind the analysis, and lays out the predictions that the remaining chapters test.

Chapter 2 establishes the formal foundation. It defines a toy system with a constraint set spanning harmony-driving, blocking, and asymmetric faithfulness constraints, generating the full factorial typology over every logically possible ranking of those constraints. In Chapter 2, I show that every dominance reversal language

[1] This property of markedness constraints connects to Kisseberth's (1970) notion of conspiracy, in which formally unrelated rules work toward the same surface goal. In OT, such a conspiracy reduces to a single markedness constraint whose violations can be resolved in multiple ways depending on the surrounding ranking (*e.g.,* Pater 1999; McCarthy 2001). I return to this connection in Chapter 5.

generated by the 9! typology ultimately instantiates the ranking schema {Blocker, Harmony-Driver} >> Ident-Trigger >> Ident-Target. This constitutes the core theoretical result: dominance reversal corresponds to a specific ranking, unifying languages and analyses that initially appear disparate.

Chapter 3 tests the results from Chapter 2 against data from three languages: Komo, Kinande, and Liko. These languages share little beyond the surface generalization. They belong to different language families, exhibit different types of harmony, directionality and locality, and differ in what counts as a blocker and in what domain blocking applies. Despite these differences, the ranking configuration from Chapter 2 is nonetheless recoverable from all three.

Chapter 4 turns to micro-variation within a single language family. Turkana and Ateso both exhibit Atr harmony and dominance reversal across the same morphophonological domain, with the same bidirectional pattern and the same idiosyncratic morphemes blocking harmony. Dominance reversal, however, surfaces under different conditions across the different systems, coexisting with distinct resolutions that range from partial harmony to full harmony that overrides the blocker's [−Atr] value. This chapter shows that these differences reduce to changes in constraint ranking over the same core configuration.

Chapter 5 extends the argument beyond Atr harmony, showing that the same ranking configuration is recoverable in Stanton's (2020) analysis of nasal harmony in Gurindji, and in Hyman's (1998) analysis of height harmony in Yaka. It also examines two analyses: Wilson's (2011) analysis of nasal place assimilation and Mullin & Pater's (2015) analysis of use it or lose it, in which the same ranking is identified independently and treated as a pathological prediction of OT. I map these analyses to the dominance reversal configuration and show that the ranking that they sought to prevent is the same one that produces dominance reversal in Atr systems. Finally, I show that the configuration maps onto Pater's (1999) analysis of the OshiKwanyama conspiracy,

where two formally distinct repairs to *NC̥ occur in the same language, connecting dominance reversal to the broader idea of conspiracy effects originally identified by Kisseberth (1970).[2]

Chapter 6 concludes by drawing together the implications of the ranking configuration for phonological theory, including the connection between dominance reversal and the Elsewhere Condition.

1.3.2 Predictions

This dissertation generates several predictions. If dominance reversal corresponds to a single ranking configuration, then every language exhibiting dominance reversal should instantiate it, and past analyses should *also* reduce to the same configuration regardless of their formal assumptions. Moreover, by isolating the ranking that defines dominance reversal, surface differences across languages become tractable: differences in the type of harmony-driving constraint, the ranking of the blocker, and the presence of additional constraints produce specific consequences for how dominance reversal surfaces and what resolutions it coexists with.

1.3.3 Scope

The focus of this dissertation is dominance reversal in ATR harmony systems, analyzed within parallel OT. Chapter 5 extends the formal argument beyond ATR and beyond OT, but does not attempt a full empirical investigation of dominance reversal-equivalent effects in other domains. I leave this for future research.

I also limit the empirical investigation to a subset of the languages reported to exhibit dominance reversal in ATR harmony. The languages selected (*i.e.,* Komo,

[2]The term "use it or lose it" comes from Mullin & Pater (2015) and denotes a pattern in which a feature within a domain can spread only if every potential target can receive it. If even one segment cannot bear the spreading feature, the feature is instead deleted from the trigger instead. Use it or lose it, like sour grapes (Wilson 2003, 2006), is considered to be a type of non-myopic pattern (Wilson 2003), where the ability of a feature to spread depends on information beyond the segments immediately adjacent to the trigger (Wilson 2003, 2006).

Kinande, Liko, Turkana, and Ateso) span distinct genetic affiliations, areal contexts, and empirical patterns. Other dominance reversal languages such as Karimojong (Lesley-Neuman 2007, 2012), Toposa (Schröder and Schröder 1984, 1987) or Bondu-So (Hantgan and Davis 2012; Green and Hantgan 2019) are not analyzed here. In some cases, this is because the available documentation does not support the depth of analysis the theory requires; in others, it is because the language instantiates a pattern already represented by an analytically simpler case. Adding languages that do not introduce new empirical distinctions or analytical contributions would not advance the argument.

Finally, the analyses are conducted in parallel OT. The central claim is that a particular ranking configuration is necessary and sufficient for dominance reversal. Showing this requires that the logic of constraint interaction be visible and unobstructed. Introducing additional formal mechanisms, such as harmonic serialism, Stratal OT or weighted constraints would complicate the proof without changing the result. For the same reason, I do not engage with questions about the computational properties of the proposed analyses, or the frequency of dominance reversal as a harmony pattern in ATR systems compared to others. These are productive directions for future research, but fall outside the scope of the present investigation.

CHAPTER 2
THE FORMAL IDENTITY OF DOMINANCE REVERSAL

2.1 Introduction

In typical dominant-recessive systems, one feature value is systematically dominant, spreading and altering the recessive vowels (*e.g.,* Baković 2000; Casali 2003, 2008; Rose and Walker 2011). For example, in Akan (20; Kwa; Berry 1957; Stewart 1967; Clements 1981; Lindau 1987; Dolphyne 1988; Hess 1992; Tiede 1996; Casali 2012), a [+ATR] root vowel triggers harmony across the word, causing [−ATR] vowels to surface as [+ATR] (20c,d), while [−ATR] vowels surface faithfully only when no [+ATR] vowels are present (20a,b).

(20) Akan Vowel Harmony (Casali 2012:36)

	UR	SR	Gloss
	No Harmony		
a.	/mɪ-√da/	[mɪ-√da]	'I sleep'
b.	/ɔ-bɛ-√da/	[ɔ-bɛ-√da]	'he/she will sleep'
	[+Atr] Harmony		
c.	/mɪ-√diʔ/	[mi-√diʔ]	'I eat'
d.	/ɔ-bɛ-√diʔ/	[o-be-√diʔ]	'he/she will eat'

Dominance reversal languages, however, exhibit a pattern in which a normally recessive value exceptionally cause vowels bearing the normally dominant value to harmonize with it. Consider, for example, Turkana (21) and Kinande (22) below. In both languages, [+ATR] harmony occurs in most cases, with a [+ATR] vowel causing [−ATR] vowels to surface as [+ATR] (21b,c; 22b,c). In other, more specific contexts, however, certain [−ATR] vowels not only resist [+ATR] harmony but cause the [+ATR]

vowel to surface as [−Atr] (21d, 22d).[3]

(21) Dominance Reversal in Turkana (Baković 2000:191,209; Noske 2000:777)

	UR	SR	Gloss
No Harmony			
a.	/ɛ-√kɔrɪ/	[ɛ-√kɔrɪ]	'Masc.Sg-ratel'
[+Atr] Harmony			
b.	/ɛ-√kori/	[e-√kori]	'Masc.Sg-giraffe'
c.	/ɛ-√los-ɪ/	[e-√los-i]	's/he will go'
Dominance Reversal			
d.	/a-k-√ido-ʊn-ɛt/	[a-k-√ɪdɔ-ʊn-ɛt]	'ger-k-give.birth-vent-inst.loc'

(22) Dominance Reversal in Kinande (Kenstowicz 2009:23,26)

	UR	SR	Gloss
No Harmony			
a.	/a-ba-√lʊmɛ/	[a-ba-√lʊmɛ]	'man.pl'
[+Atr] Harmony			
b.	/a-ba-√kali/	[ə-bə-√kəli]	'woman/pl'
c.	/ɛ-rɪ-√huk-ɪr-a/	[e-ri-√huk-ir-a]	'to prepare for'
Dominance Reversal			
d.	/ɛ-ri-√hɪrɪ/	[ɛ-rɪ-√hɪrɪ]	'crab'

The surface patterns in Kinande and Turkana are very similar. There is a broad pattern of [+Atr] harmony which is overtaken by a more restricted pattern of [−Atr] harmony. The analyses of these patterns, however, employ vastly different formalisms and assumptions. Kenstowicz (2009), for example, relies on constraint conjunction within a headed-domain framework, while Downing & Krämer (2022) use Syntagmatic Correspondence Theory, evaluating output-output correspondents.

[3]See Chapter 3 for an analysis of Kinande and Chapter 4 for an analysis of Turkana.

Meanwhile, for Turkana, Baković (2000) adopts an OT analysis under Benua's (1995, 1997) Transderivational Correspondence Theory, while Noske (1996, 2000) relies on underspecification.

These analyses are sufficiently different that they cannot be compared directly, making it impossible to determine whether dominance reversal reflects shared grammatical properties or is merely a label for superficial similar patterns. In this chapter, I compute the factorial typology of a minimal ATR harmony system, showing that dominance reversal emerges as a property of a single, unified ranking configuration: {BLOCKER, HARMONY-DRIVER} >> IDENT[+ATR] >> IDENT[−ATR]. I show that every unique grammar yielding dominance reversal, understood as a unique input-output mapping, reduces to that specific ranking. This provides principled formal grounds for unifying dominance reversal systems while abstracting away from the language-specific mechanisms that otherwise make their analyses difficult to compare.

2.2 Method

This section defines the toy system over which the factorial typology is computed, including the input set, constraint set, and the analytical tools used to derive the ranking conditions that characterize each grammar.

2.2.1 Toy System

The system contains three vowel qualities: a high [+ATR] vowel /i/, which serves as the [+ATR] harmony trigger, a mid [−ATR] vowel /ɛ/, which serves as the target of [+ATR] harmony, and a low [−ATR] vowel /a/, which can potentially act as a blocker and whose presence can force the trigger to surface as [−ATR], inducing dominance reversal. Additionally, the purpose of the typology is to isolate the interaction between ATR harmony and blocking. Because the typology is designed to isolate the interaction between ATR harmony and blocking, no candidate differs from its corresponding input

in any feature other than ATR, though constraints reeferring to other feautres are not included. Each input consists of five vowels, varying the position of /i/ and /a/ to test the interaction between harmony, directionality, iterativity, and blocking. The full list of inputs is shown below:

(23) Input set for the factorial typology

Input	Trigger position	Blocker position
ɛɛiɛɛ	medial	none
aɛiɛa	medial	both edges
ɛaiaɛ	medial	both (adjacent)
aɛiɛɛ	medial	left (adjacent)
ɛɛiɛa	medial	right (adjacent)
iɛɛɛɛ	initial	none
ɛɛɛɛi	final	none
aɛɛɛi	final	left
iɛɛɛa	initial	right

The position of /i/ and /a/ varies across inputs to test three properties of the grammar. First, placing /i/ in initial, medial, and final positions distinguishes whether harmony applies leftward, rightward, or bidirectionally. Second, inputs without /a/ establish the baseline harmony pattern, while inputs with /a/ test whether and how the presence of a blocker can change the outcome within the same grammar. Third, mirror-image pairs like /iɛɛɛa/ and /aɛɛɛi/ force the grammar to reveal whether leftward and rightward harmony are treated differently.

2.2.2 Constraints

The nine-constraint set includes two faithfulness constraints of the type IDENT[±ATR], one markedness constraint *[+LO,+ATR] penalizing instances of [ə], two sequential

markedness constraints, and four alignment constraints.

Faithfulness Constraints The faithfulness constraints are split into two value-specific constraints, IDENT[+ATR] and IDENT[−ATR], each penalizing changes to a single ATR value. This split is necessary because a single faithfulness constraint such IDENT-IO[ATR] would assign identical violations to changes affecting either [+ATR] or [−ATR] inputs; both changes involve an unfaithful ATR mapping. The grammar would then be unable to select between [+ATR] harmony and [−ATR] harmony, resulting in a ranking indeterminacy, as was shown in Chapter 1. Splitting faithfulness by value allows the ranking between IDENT[+ATR] and IDENT[−ATR] to encode which value is dominant, as the value protected by the higher-ranked constraint is more costly for the grammar to change. This asymmetry is also what distinguishes dominance reversal from [−ATR] harmony, since both involve [+ATR] vowels surfacing as [−ATR], but only dominance reversal occurs within a broader pattern of [+ATR] harmony. Additionally, since all 9! permutations are computed, the typology does generate languages exhibiting full [−ATR] harmony. However, because the system lacks a counterpart to *[+LO,+ATR] that would block [−ATR] harmony, languages exhibiting both [−ATR] harmony and dominance reversal are not generated (see fn. 6).

(24) Faithfulness Constraints

Name	Definition
IDENT[+ATR]	Assign one violation mark for each output vowel that is [−ATR] whose input correspondent is [+ATR].
IDENT[−ATR]	Assign one violation mark for each output vowel that is [+ATR] whose input correspondent is [−ATR].

Blocking Constraint Modelling dominance reversal requires a context in which [+ATR] harmony cannot fully apply. As shown in Chapter 1, if every vowel were a

licit target of [+ATR] harmony, the grammar would produce full [+ATR] harmony throughout, and there would be no structure in which sacrificing a [+ATR] value would be preferable. The markedness constraint *[+LO,+ATR] provides this context by penalizing [ə], the [+ATR] counterpart of /a/, a common blocking pattern in ATR harmony systems cross-linguistically (Casali 2008).

(25) Blocking Constraint

Name	Definition
*[+LO,+ATR]	Assign one violation mark for each segment in the output that is specified as [+LO] and [+ATR]

Harmony-Driving Constraints The remaining constraints drive harmony. Several constraint families are available for this purpose: AGREE (Lombardi 1996; 1999; Beckman 1998; Butska 1998; Baković 2000), SPREAD (Padgett 1995, 2002), ALIGN (McCarthy and Prince 1993; Kirchner 1993; see also Akinlabi 1994, 1997; Leitch 1996; Pulleyblank 1996; Pulleyblank et al. 1997; Orie 2001), as well as sequential markedness constraints (Archangeli and Pulleyblank 1994; Pulleyblank 2002; Mahanta 2008). Of these, some encode directionality (*e.g.,* ALIGN, sequential markedness constraints) and some do not (*e.g.,* AGREE and some versions of SPREAD). Research on bidirectional harmony, however, has shown that bidirectional patterns most likely consist of two independent unidirectional processes rather than a single bidirectional process (Hyman 2002), and so the typology uses constraints that encode that directionality directly: sequential markedness and ALIGN constraints. The choice to use both ALIGN and sequential markedness constraints is motivated by the fact that these two families ultimately make different predictions when full harmony cannot be obtained. While ALIGN constraints can model partial harmony, sequential markedness constraints cannot, instead predicting a sour grapes effect (Wilson 2003, 2006). Both are used in

the analyses that follow in Chapters 3 and 4.

For both families, the typology includes constraints referencing both [+ATR] and [−ATR]. This is essential since the typology must determine whether the choice of spreading value matters for dominance reversal, and this requires that constraints referencing both values be available to the grammar. For sequential markedness constraints this means defining a constraint such as *[+ATR]C_o[−ATR] instead of *[+ATR]C_oV, while for ALIGN, this means including *both* ALIGN[+ATR] and ALIGN[−ATR].

Of these, sequential markedness constraints penalize a specific ordering of feature disagreement between adjacent vowels. Each constraint targets a specific direction, as can be seen below.

(26) Sequential Markedness Constraints

Name	Definition
*[+ATR]C_o[−ATR]	Assign one violation mark for any sequence of two vowels $V_1C_oV_2$ within a single prosodic word, where V_1 is [+ATR] and V_2 is [−ATR]
*[−ATR]C_o[+Atr]	Assign one violation mark for any sequence of two vowels $V_1C_oV_2$ within a single prosodic word, where V_1 is [−ATR] and V_2 is [+ATR]

Sequential markedness constraints like *[+ATR]C_o[−ATR] cannot model partial harmony. The reason is that while they assign a violation for every instance where a [+ATR] vowel immediately precedes a [−ATR] vowel within the domain (usually the prosodic word), they cannot distinguish between candidates like [iɛɛɛa] and [ieɛɛa] since both contain only one sequence of a [+ATR] vowel immediately followed by a [−ATR] vowel. It does not mean that each candidate can only incur one violation mark, as a candidate like [iɛeɛa] contains two sequences of a [+ATR] vowel followed by a [−ATR] vowel, incurring two violation marks, but it does mean that these constraints

cannot model partial harmony. They produce, instead, sour grapes (Wilson 2003; 2006), where harmony applies either across the whole domain or fails to apply at all, as is shown below.

(27) Input: /iɛɛɛa/

/iɛɛɛa/	*[+Lo,+Atr]	Ident[+Atr]	*[+Atr]C$_o$[−Atr]	Ident[−Atr]	
a.[ieeeə]	*!			****	*[+Atr] Harmony*
b.[iɛɛɛa]		*!			Dominance Reversal
c.[ieeea]			*	*!**	*Part. [+Atr] Harmony Cand.*
d.[ieeɛa]			*	*!*	*Part. [+Atr] Harmony Cand.*
e.[ieɛɛa]			*	*!	*Part. [+Atr] Harmony Cand.*
☞ f.[iɛɛɛa]			*		*Faithful Cand.*

Align constraints, by contrast, penalize the distance between an autosegment, understood here as a span of consecutive vowels sharing the same value for a feature [F], and a prosodic edge (Kirchner 1993). Distance is measured exclusively over intervening vowels.[4] Violations are assessed gradiently (*i.e.,* a single locus of violation can incur multiple violations based on the extent of deviation; McCarthy 2003) by the number of intervening vowels. Additionally, directionality comes from the choice of edge: Align-Right assigns violations based on the distance from the right edge of the specified feature domain to the right edge of the prosodic word, while Align-Left evaluates from the left edge of the specified feature domain to the left edge of the prosodic word. Crossing the two directions with the two Atr values yields four Align constraints total, as shown below:

(28) Align Constraints

[4]Note that alignment is computed exclusively over vowels, while consonants are treated as transparent. There is evidence that vowel harmony can be affected by non-vocalic elements (see *e.g.,* Hansson 2024) that could affect both the outcome and definition of Align, but I leave that to future work.

Name	Definition
Align-L[+Atr]	For every autosegment [+Atr], assign one violation mark for each vowel that separates it from the left edge of the Prosodic Word
Align-L[−Atr]	For every autosegment [−Atr], assign one violation mark for each vowel that separates it from the left edge of the Prosodic Word
Align-R[+Atr]	For every autosegment [+Atr], assign one violation mark for each vowel that separates it from the right edge of the Prosodic Word
Align-R[−Atr]	For every autosegment [−Atr], assign one violation mark for each vowel that separates it from the right edge of the Prosodic Word

A constraint like Align-R[+Atr] assigns one violation for each segment separating a [+Atr] autosegment from the right edge of the prosodic word, so that a [+Atr] autosegment further from the edge incurs more violations than one closer to it. This gradient evaluation is what distinguishes Align from sequential markedness constraints, as Align incurs more or fewer violations depending on how far the autosegment is from any specified edge. In other words, as seen in the tableau below, an output like [ieeea] is preferred by the grammar to an output such as [iɛɛɛa], even though full harmony isn't obtained.

(29) Input: /iɛɛɛa/

/iɛɛɛa/	*[+Lo,+Atr]	Ident[+Atr]	Align-R[+Atr]	Ident[−Atr]	
a.[ieeeə]	*!			****	*Full [+*Atr*] Harmony*
b.[ɪɛɛɛa]		*!			*Dominance Reversal*
c.[iɛɛɛa]			**!**		*Faithful Cand.*
d.[ieɛɛa]			**!*	*	*Part. [+*Atr*]Harmony Cand.*
e.[ieeɛa]			**!	**	*Part. [+*Atr*] Harmony Cand.*
☞ f.[ieeea]			*	***	*Part. [+*Atr*] Harmony Cand.*

As shown above, sequential markedness constraints cannot distinguish between partial harmony and the absence of harmony, while Align constraints can, due to their gradient evaluation. Including both in the typology allows the full range of potential input-output mappings to emerge, and also allows the typology to reveal whether combining a high-ranked Align constraint with a high-ranked sequential markedness constraint yields patterns that neither family produces on its own.

2.2.3 Analytical Tools

The system, then, consists of nine inputs, nine constraints, and a finite set of candidate outputs for each input. For each input, the candidate set includes all typologically distinct resolution types under the assumption that candidates differ from the input only in their Atr specifications across the five-vowel string, without deletion, insertion, or changes in other features. The candidate set covers logically possible outputs that represent distinct harmony patterns (*i.e.,* sour grapes, partial harmony, full [±Atr] harmony, etc.). Because the number of typologically distinct resolutions varies with the position of the trigger and the blocker, the size of the candidate set ranges from three candidates for inputs with triggers at the edge and no blockers to eleven for inputs with a trigger in the middle of the structure flanked by blockers on both sides, yielding fifty input-output pairs in total. The complete list of inputs and candidates can be found in Appendix A.

The factorial typology was computed using OTWorkplace (Prince, Tesar, et al. 2015), which evaluated every candidate against all nine constraints across all

$9! = 362,880$ possible rankings. Rankings that select the same optimal candidate for every input correspond to a single language and the output of this process is the complete set of distinct languages that the system can generate, together with the ranking conditions that characterize each one.

The constraints are encoded as VBA scripts that define how each constraint assigns violations.[5] The scripts for evaluating faithfulness constraints were provided by Eric Baković (personal communication) and adapted for the present analysis. The scripts for evaluating Align constraints were rewritten to correct OTWorkplace's built-in implementation. The original scripts for Align assign no more than one violation mark per misaligned autosegment regardless of its distance from the edge, thus making them unable to model partial harmony. All scripts, including an additional script for IPA support, are available in Appendix D.

The ranking conditions that characterize each language are determined through winner-loser pairs and Recursive Constraint Demotion. A winner-loser pair consists of two candidates competing for the same input: the winner, which is the output selected by the grammar, and the loser, which is a competing candidate that the grammar must reject (Tesar 2014). For each constraint, the pair is annotated with one of three values. W indicates that the constraint prefers the winner, assigning fewer violations to the winner than to the loser. L indicates that the constraint prefers the loser, assigning fewer violations to the loser than to the winner. Finally, a blank cell indicates that the constraint does not distinguish between the two candidates, assigning equal violations to both. The W and L annotations encode the ranking information directly, as at least one constraint marked W must dominate every constraint marked L for the winner to be selected over the loser.

Recursive Constraint Demotion (RCD; Tesar 1995; Tesar and Smolensky 1998; Tesar 2014) is an algorithm that derives a stratified constraint hierarchy from a set

[5] I acknowledge the use of Claude/Anthropic in the writing of these VBA scripts.

of winner-loser pairs. RCD begins with the full set of winner-loser pairs and the full set of constraints, all initially unranked with respect to one another. On the first pass, the algorithm selects any constraint that bears no L annotation (whether blank or W-marked) in any pair, placing them in a stratum that will be the highest ranked of the hierarchy, since ranking those constraints highest can never cause a loser from a winner-loser pair to be selected over a winner. Any pair in which at least one W annotation belongs to a constraint that has just been ranked is then removed from the set. Similarly, constraints that have been ranked in a stratum are also removed from the set. The algorithm then repeats over the remaining set of pairs and constraints, placing the next group of L-free constraints into a second stratum ranked below the first. This continues until every constraint has been assigned to a stratum, resulting in a partial hierarchy such that while the strata are ranked relative to one another, constraints that are *within* a stratum are not. Crucially, because RCD ranks each constraint as high as it can, it is possible for the algorithm to return a ranking where every W-bearing constraint outranks the L-bearing constraints. This ranking is consistent with the winner-loser pairs, but does not represent a complete characterization of the rankings that those pairs permit since each pair requires at least one W-bearing constraint to outrank all L-bearing constraints. The stratified hierarchy therefore represents a simplified summary of the ranking information encoded by the winner-loser pairs. This compressed representation is sufficient for the purpose of identifying the language generated by a grammar, but it does not preserve the complete ranking requirements established by the winner-loser pairs themselves.

2.3 Results

The factorial typology over this system generates twenty-six distinct languages, all listed in Appendix C. Of these 26 languages, exactly 7 exhibit dominance reversal: languages in which [+ATR] harmony applies in the absence of a blocker, but the

[+ATR] trigger surfaces as [−ATR] when harmony is blocked.[6] The remaining 19 languages show input-output mappings consistent with other phonological patterns (*e.g.,* full [±ATR] harmony, partial [±ATR] harmony, sour grapes and a fully faithful mapping), which I do not focus on in this dissertation. Therefore, the rest of this section focuses exclusively on the 7 languages exhibiting an input-output mapping consistent with dominance reversal (as shown in 30).

(30) Input-Output Table for Dominance Reversal Languages[7]

Languages / Inputs :	a. /ɛɛiɛɛ/	b. /aɛiɛa/	c. /ɛaiaɛ/	d. /aɛiɛɛ/	e. /ɛɛiɛa/	f. /iɛɛɛɛ/	g. /ɛɛɛɛi/	h. /aɛɛɛi/	i. /iɛɛɛa/
L#10	eeiee	aɛɪɛa	ɛaɪaɛ	aeiee	ɛɛɪɛa	ieeee	eeeei	aeeei	ɪɛɛɛa
L#11	eeiee	aɛɪɛa	ɛaɪaɛ	aɛiee	ɛɛɪɛa	ieeee	eeeei	aɛɛɛi	ɪɛɛɛa
L#12	eeiee	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	eeiɛa	ieeee	eeeei	aɛɛɛɪ	ieeea
L#13	eeiee	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	eeiɛa	ieeee	eeeei	aɛɛɛɪ	iɛɛɛa
L#14	eeiee	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	ɛɛɪɛa	ieeee	eeeei	aɛɛɛɪ	ɪɛɛɛa
L#21	eeiɛɛ	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	eeiɛa	iɛɛɛɛ	eeeei	aɛɛɛɪ	iɛɛɛa
L#26	ɛɛiee	aɛɪɛa	ɛaɪaɛ	aɛiee	ɛɛɪɛa	ieeee	ɛɛɛɛi	aɛɛɛi	ɪɛɛɛa

Consider, specifically, form (30a) for the pattern. In all cases, the input /ɛɛiɛɛ/ induces [+ATR] harmony, with the directionality varying depending on the language (bidirectional, progressive or regressive). In the presence of a blocker, however, the outcome depends on the language. With an input like (30b), where /a/ flanks the trigger on both sides, all 7 languages exhibit dominance reversal, where the [+ATR] vowel surfaces as [−ATR]. In cases where /a/ occurs only on one side of the trigger, the outcome depends on the language. In L#14, for example, dominance reversal always applies. In L#21 and L#26, on the other hand, harmony applies exclusively in one direction, as shown in outputs (L#21a,f) and (L#26a,g), and dominance reversal

[6]The typology as constructed models dominance reversal in systems where [+ATR] is the dominant value, which reflects the attested empirical domain investigated in this dissertation. The case where dominance reversal would emerge in a broad system of [−ATR] harmony is logically possible, though modeling it would require an additional blocking constraint that prevents full [−ATR] harmony from the ranking IDENT[−ATR] >> IDENT[+ATR]. No such constraint is included in the current system, though I predict that the outcome would be identical to the present results. I leave empirical investigation of this prediction for future work.

[7]Cells are referred to by language and input. For example, L#10d designates the output of L#10 for input d.

always occurs whenever /a/ is present within the domain (*e.g.,*L#21b, L#26b). The remaining four languages, however, exhibit either partial harmony (L#10, L#12) or sour grapes (L#11, L#13). Outputs like (L#10d) and (L#12e) show partial harmony, /a/ remains [−ATR] in the output, and harmony proceeds up to that vowel, while outputs like (L#11d) and (L#13e) show sour grapes instead: in the presence of an /a/, harmony fails to apply leftward and rightward respectively even though (L#11) and (L#13) do exhibit bidirectional harmony in the absence of a low vowel (L#11,13a). Despite these differences, all 7 languages exhibit input-output mappings consistent with dominance reversal.

The variation between the languages also pertains to the directionality of the pattern. While all 7 languages instantiate some form of dominance reversal, one language exhibits dominance reversal when an /a/ is on either side (L#14), three languages exhibit dominance reversal exclusively when an /a/ is to the right (L#10,11,26) and three languages exhibit dominance reversal exclusively when an /a/ is to the left (L#12,13,21). These differences will be shown to follow from which harmony-driving constraints occupy the first stratum alongside the blocking constraint in the ranking hierarchies derived below.

The outputs in the table above are considered *winners* from the concept of winner-loser pairs. As mentioned in the previous section, a winner-loser pair consists of two candidates that are in competition for the same input, specifying what ranking is necessary for the winner to be selected, thus providing crucial information regarding which constraints must outrank others (Tesar 2014).

Consider one of these winner-loser pairs below from Language 10.

(31) Winner-Loser Pair Example

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C$_o$[−A]	A-L[−A]	A-R[+A]	ID[+A]	*[−A]C$_o$[+A]	A-L[+A]	ID[−A]	A-R[−A]
a.	aɛiɛɛ	aeiee	əeiɛɛ	W	W	W	W		L	L	L	L

In (31a), the input is /aɛiɛɛ/, and the winner-loser pair for this input is [aeiee] and [əeiee] respectively. For this pair, four constraints prefer the winner (W): *[+Lo,+Atr], *[+Atr]C$_o$[−Atr], Align-Left[−Atr] and Align-Right[+Atr] and at least one must dominate every L-bearing constraint for the winner to be selected, though it does not determine which constraint dominates the L-bearing constraints, or how the W-bearing constraints are ranked relative to one another. Four constraints prefer the loser (L): *[−Atr]C$_o$[+Atr], Align-Left[+Atr], Ident[−Atr] and Align-Right[−Atr], while only one constraint has no preference: Ident[+Atr].

The evaluation of a single winner-loser pair, however, is insufficient to determine the constraint hierarchy for a given grammar. The ranking must satisfy the complete set of winner-loser pairs simultaneously. As an example, consider the following winner-loser pairs for L#10, analyzed using RCD.[8]

(32) List of Winner-Loser Pairs Before First RCD Pass

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C$_o$[−A]	A-L[−A]	A-R[+A]	Id[+A]	*[−A]C$_o$[+A]	A-L[+A]	Id[−A]	A-R[−A]
a.	aɛiɛɛ	aeiee	əeiɛɛ	W	W	W	W		L	L	L	L
b.	aɛiɛa	aɛiɛa	əeiɛa	W	W	W	W	L			W	
c.	aɛiɛɛ	aeiee	əeiee	W					L	L	W	L
d.	aɛiɛa	aɛiɛa	aeieə	W				L	W	W	W	W
e.	aɛiɛa	aɛiɛa	əeieə	W				L		W		
f.	ɛaiaɛ	ɛaiaɛ	eəiəe	W				L			W	
g.	iɛɛɛa	iɛɛɛa	ieeeə	W				L		W		
h.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ		W	W	W		W	W	L	W
i.	aɛiɛɛ	aeiee	aɛiɛɛ		W	W	W			W	L	L
j.	ɛɛiɛɛ	eeiee	eeiɛɛ		W	W	W				L	
k.	aɛiɛa	aɛiɛa	aeiea		W	W	W	L	W	W	W	W
l.	aɛiɛa	aɛiɛa	aɛiɛa		W	W	W	L	W	W		W
m.	iɛɛɛa	iɛɛɛa	ieeea		W	W	W	L			W	
n.	iɛɛɛa	iɛɛɛa	iɛɛɛa		W	W	W	L				
o.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ					W			L	
p.	aɛɛɛi	aeeei	aɛɛɛi					W	L	L	L	L
q.	ɛɛiɛɛ	eeiee	ɛɛiee						W	W	L	W
r.	aɛiɛɛ	aeiee	aɛiee							W	L	L

[8]This list is non-exhaustive but sufficient for expository purposes; see Appendix C for the full list of winner-loser pairs.

This list contains eighteen winner-loser pairs and nine constraints. Four constraints bear no L annotations in any of the pairs: *[+Lo,+Atr], *[+Atr]C_o[−Atr], Align-Left[−Atr] and Align-Right[+Atr]. Following RCD, these four constraints are placed in the first stratum, the resulting partial hierarchy after the first RCD pass is shown in (33).

(33) Constraint Hierarchy After First RCD Pass

{ *[+L,+A], *[+A]C_o[−A], Align-L[−Atr], Align-R[+Atr]}

Because those constraints and pairs a. through n. have been accounted for by the top stratum constraints dominating the rest of the constraints, they can be removed from the set. The following table shows the necessary winner-loser pairs for the next pass of RCD over the remaining five constraints:

(34) List of Winner-Loser Pairs Before Second RCD Pass

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	Id[+A]	*[−A]C_o[+A]	A-L[+A]	Id[−A]	A-R[−A]
o.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ					W			L	
p.	aɛɛɛi	aeeei	aɛɛɛɪ					W	L	L	L	L
q.	ɛɛiɛɛ	eeiee	ɛɛiee						W	W	L	W
r.	aɛiɛɛ	aeiee	aɛiee							W	L	L

Once again, the algorithm attempts to rank constraints as high as possible consistent with the set of winner-loser pairs. In the case of (34), the only non L-bearing constraint is Ident[+Atr]. This constraint will be placed in a second stratum *below* the first one and taken out of the list of constraints for the third pass of the RCD (35).

(35) Constraint Hierarchy After Second RCD Pass

{ *[+L,+A], *[+A]C_o[−A], Align-L[−Atr], Align-R[+Atr]} >> {Ident[+Atr]}

The RCD algorithm then continues with a third pass, taking the remaining four constraints and evaluating them against the remaining winner-loser pairs (36). Since

both *[−Atr]C$_o$[+Atr] and Align-Left[+Atr] do not favor any losers, they are put in a third stratum of the hierarchy, below Ident[+Atr] (37)

(36) List of Winner-Loser Pairs Before Third RCD Pass

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C$_o$[−A]	A-L[−A]	A-R[+A]	Id[+A]	*[−A]C$_o$[+A]	A-L[+A]	Id[−A]	A-R[−A]
q.	ɛɛiɛɛ	eeiee	ɛɛiee						W	W	L	W
r.	aɛiɛɛ	aeiee	aɛiee							W	L	L

(37) Constraint Hierarchy After Third RCD Pass

{ *[+L,+A], *[+A]C$_o$[−A], Align-L[−Atr], Align-R[+Atr]} >> {Ident[+Atr]} >> {*[−A]C$_o$[+A], Align-L[+Atr]}

And so at this point, because no winner-loser pairs remain, both Ident[−Atr] and Align-Right[−Atr] can be ranked in the lowest stratum with the fourth RCD pass, giving the final constraint hierarchy for Language 10 in (38)

(38) Constraint Hierarchy After Fourth RCD Pass

{ *[+L,+A], *[+A]C$_o$[−A], Align-L[−Atr], Align-R[+Atr]} >> {Ident[+Atr]} >> {*[−A]C$_o$[+A], Align-L[+Atr]} >> {Ident[−Atr], Align-Right[−Atr]}

This hierarchy is one of the rankings consistent with L#10 in the sense discussed above. The top stratum contains both the blocking constraint *[+Lo,+Atr], and harmony-driving constraints, which jointly prevent /a/ from surfacing as [+Atr], while enforcing agreement between a [+Atr] vowel and any [−Atr] vowel to its right. Ident[+Atr], being alone in the second stratum, means that the imperative to harmonize and the imperative to prevent [ə] both take priority over preserving the identity of a [+Atr] vowel in the input, but preserving a [+Atr] vowel is generally prioritized over preserving a [−Atr] vowel, since Ident[−Atr] is ranked in a lower stratum. Thus, in inputs without a low vowel, the ranking of Ident[+Atr] >> Ident[−Atr] enforces [+Atr] harmony, while in inputs that *do* contain a low vowel

to the right, full rightward [+ATR] harmony becomes impossible and the grammar resolves this conflict by causing the [+ATR] vowel to surface as [−ATR].

And so, running RCD through all seven dominance reversal languages shows that they differ only in which harmony-driving constraints occupy the first stratum versus the third or fourth and whether IDENT[−ATR] occupies the third or fourth stratum. These differences are summarized in (39).

Since languages have difference in directionality of dominance reversal, but also resolutions which coexists with dominance reversal (*e.g.,* sour grapes, partial harmony, *etc.*), the specific constraints occupying each stratum vary across languages.

(39) Differences in ranking between the languages

Language	First Stratum	Third Stratum	Fourth Stratum
L#10	*[+ATR]C_o[−ATR], ALIGN-L[−ATR], ALIGN-R[+ATR]	*[−ATR]C_o[+ATR], ALIGN-L[+ATR]	ALIGN-R[−ATR], IDENT[−ATR]
L#11	*[+ATR]C_o[−ATR], ALIGN-L[−ATR], ALIGN-R[+ATR]	*[−ATR]C_o[+ATR], ALIGN-R[−ATR]	ALIGN-L[+ATR], IDENT[−ATR]
L#12	*[−ATR]C_o[+ATR], ALIGN-L[+ATR], ALIGN-R[−ATR]	*[+ATR]C_o[−ATR], ALIGN-R[+ATR]	ALIGN-L[−ATR], IDENT[−ATR]
L#13	*[−ATR]C_o[+ATR], ALIGN-L[+ATR], ALIGN-R[−ATR]	*[+ATR]C_o[−ATR], ALIGN-L[−ATR]	ALIGN-R[+ATR], IDENT[−ATR]
L#14	*[+ATR]C_o[−ATR], ALIGN-L[−ATR], ALIGN-R[+ATR] *[−ATR]C_o[+ATR], ALIGN-L[+ATR], ALIGN-R[−ATR]	ID[−A]	
L#21	*[−ATR]C_o[+ATR], ALIGN-L[+ATR], ALIGN-R[−ATR]	ID[−A]	*[+ATR]C_o[−ATR], ALIGN-L[−ATR], ALIGN-R[+ATR]
L#26	*[+ATR]C_o[−ATR], ALIGN-L[−ATR], ALIGN-R[+ATR]	ID[−A]	*[−ATR]C_o[+ATR], ALIGN-L[+ATR], ALIGN-R[−ATR]

Three properties, however, are constant across all seven languages. First, *[+LO,+ATR] is always in the first stratum. Second, IDENT[+ATR] is always in the second stratum, immediately below the harmony-driving and blocking constraints. Third, IDENT[−ATR] is always below IDENT[+ATR].

(40) Similarities in ranking between the languages

Language	First Stratum	Second Stratum	Third/Fourth Stratum
L#10	*[+Lo,+Atr]	Ident[+A]	Ident[−A]
L#11	*[+Lo,+Atr]	Ident[+A]	Ident[−A]
L#12	*[+Lo,+Atr]	Ident[+A]	Ident[−A]
L#13	*[+Lo,+Atr]	Ident[+A]	Ident[−A]
L#14	*[+Lo,+Atr]	Ident[+A]	Ident[−A]
L#21	*[+Lo,+Atr]	Ident[+A]	Ident[−A]
L#26	*[+Lo,+Atr]	Ident[+A]	Ident[−A]

Crucially, dominance reversal occurs independently of which harmony-driving constraint occupies the first stratum. Dominance reversal can be said to occur, then, whenever a harmony-driving constraint and *[+Lo,+Atr] >> Ident[+Atr] >> Ident[−Atr], as shown in (41) and (42); and the differences between those two tableaux ultimately pertains to distinctions in directionality of evaluation. In this sense, a harmony-driving constraint is better understood as a role where any constraint in that position penalize disharmony.

(41) Input: /ai/

/ai/	*[+Lo,+Atr]	Harmony-Driver	Id[+A]	Id[−A]	
a. [əi]	*!			*	*[+Atr] Harmony*
b. [ai]		*!			*Faithful cand.*
☞ c. [aɪ]			*		*Dominance Reversal*

(42) Input: /ia/

/ia/	*[+Lo,+Atr]	Harmony-Driver	Id[+A]	Id[−A]	
a. [iə]	*!			*	*[+Atr] Harmony*
b. [ia]		*!			*Faithful Cand.*
☞ c. [ɪa]			*		*Dominance Reversal*

Every dominance reversal language in the typology, and no other, necessarily satisfies this ranking schema, where *[+Lo,+Atr] and at least one harmony-driving constraint jointly dominate Ident[+Atr] >> Ident[−Atr].

In (43) I show, then, the comparison between the possible resolution based on ranking differences, abstracting away from certain ranking distinction that would not change the type of harmony being modeled. For example, the ranking between *[+Lo,+Atr] and Ident[+Atr] determines whether an input /ə/ (a [+Lo,+Atr] vowel) is preserved in the output (Ident[+Atr] >> *[+Lo,+Atr]) or not (*[+Lo,+Atr] >> Ident[+Atr]) in context where the harmony-driver would not bear on the outcome. The rankings in (43) correspond, then, to a characterization of how disharmony is resolved, rather than the exhaustive rankings that define each harmony type.

(43) Ranking Comparison between the Types of Harmony

Harmony Type	First Stratum	Second Stratum	Third Stratum	Fourth Stratum
Dominance Reversal	*[+Lo,+Atr], Harmony-Driver	Ident[+Atr]	Ident[−Atr]	
Full [+Atr] Harmony	Ident[+Atr] , Harmony-Driver	Ident[−Atr]	*[+Lo,+Atr]	
Partial [+Atr] Harmony	*[+Lo,+Atr]	Ident[+Atr]	Align[+Atr]	Ident[−Atr]
Full [−Atr] Harmony	Ident[−Atr] , Harmony-Driver	*[+Lo,+Atr]	Ident[+Atr]	

(43) thus represents different resolutions of the conflicts for the same set of grammatical pressures. In other words, the constraints remained identical throughout the typology and it is the ranking differences between them which ultimately resulted in the different input-output mappings. Full [+Atr] harmony, for example, prioritizes preserving input [+Atr] vowels and Atr agreement in the output, yielding [+Atr] spreading as the optimal outcome, while full [−Atr] harmony also prioritizes Atr agreement in the output, but values preserving input [−Atr] vowels, thus yielding full [−Atr] spreading throughout the domain. Partial harmony also emerges as a result of a specific ranking where, for example, *[+Lo,+Atr] >> Ident[+Atr] >> Align[+Atr] penalizes both a [+Atr] [ə] and changing an input [+Atr] vowel.

Dominance reversal, then, occupies a specific position in this space of resolutions

where IDENT[+ATR] is ranked high enough to induce [+ATR] harmony where it can, but not high enough to protect [+ATR] vowels in a structure that contains a blocking segment, leading to the loss of [+ATR] in order to satisfy the harmony-driving constraint.

To conclude, I began this chapter by comparing Kinande's and Turkana's ATR harmony systems, noting that the two languages exhibit surface patterns that are strikingly similar to one another, raising the question of whether these similarities were merely coincidental or reflected a shared identity. The results from Chapter 2 provide a tractable way of answering that question. If the only ranking that can yield dominance-reversal-like effects is the ranking of {BLOCKER, HARMONY-DRIVER} >> IDENT[+ATR] >> IDENT[−ATR], then necessarily any OT analysis of dominance reversal should reduce to that configuration, making languages like Kinande and Turkana now comparable at a level that goes beyond surface data. Additionally, as will be seen in the following chapters, dominance reversal exhibits variation in how it occurs, varying from the domain of application (Chapter 3) to microvariation in how different resolutions can co-occur with dominance reversal for the same structure (Chapter 4). The configuration accommodates this variation because dominance reversal is not produced by any specific constraint, but by the position each type of constraint occupies in the hierarchy. The factorial typology showed this directly, where every language that modeled dominance reversal placed the blocking constraint and at least one harmony-driving constraint in the first stratum, IDENT[+ATR] alone in the second and IDENT[−ATR] below it. Differences between the 7 languages consisted of, for example, which harmony-driving constraint belonged to the first, third or fourth stratum. These differences then, distinguish languages with regards to directionality and resolutions that could co-occur with dominance reversal whenever an /a/ is present in the structure. Thus, so long as a blocking constraint, which should be understood as any constraint capable of blocking the repair strategy induced by

HARMONY-DRIVER >> IDENT[+ATR] >> IDENT[−ATR], and a harmony-driving constraint jointly dominate IDENT[+ATR], which in turn dominates IDENT[−ATR], the grammar produces dominance reversal. In chapters 3 and 4 I thus show that the configuration is recoverable not only across five different languages, but also across past analyses.

CHAPTER 3
MACROVARIATIONS

3.1 Introduction

Chapter 2 established that dominance reversal maps to a single ranking configuration in OT: Blocker, Harmony-Driver >> Ident[+Atr] >> Ident[−Atr], which produces two distinct outcomes depending on the input structure. In a structure where no blocker is present, the ranking Harmony-Driver >> Ident[+Atr] >> Ident[−Atr] derives [+Atr] harmony, since avoiding violations of the harmony-driving constraint through a [−Atr] → [+Atr] input-output mapping is considered less costly to the grammar than the opposite. However, in a structure which contains a blocking segment that prevents full [+Atr] harmony, the grammar evaluates that a [+Atr] → [−Atr] input-output mapping is preferred to full [+Atr] harmony, since it avoids violations both of the Blocker and the Harmony-Driver, which are higher ranked than Ident[+Atr].

The purpose of this chapter is to test whether this configuration is recoverable from typologically diverse Atr harmony systems: Komo (Koman; Otero 2015; 2018; Olejarczuk et al. 2019; Leduc and McCollum 2023), Kinande (Bantu; Schlindwein 1987; Mutaka 1994, 1995, 2007; Archangeli and Pulleyblank 1994, 2002; Hyman 2002; Downing and Krämer 2022), and Liko (Bantu; Kutsch Lojenga 2008; Wit 2015a, 2015b, 2016; McCollum et al. 2020). These languages differ in language family, in directionality, in what constitutes the blocker and in the morphological or phonological domain in which blocking applies. Despite this level of variation, I show that the ranking from Chapter 2 is recoverable from all three systems, using the same type of harmony-driving constraint: a sequential markedness constraint requiring adjacent

segments to agree for [ATR] (*cf.* Pulleyblank 2002). The differences among the three languages reduce to differences in the blocking condition and in the domain in which it applies, as well as the vowels involved in the harmony-driving constraint.

In Komo (§3.2), [+ATR] high vowels trigger spreading to [−HI] vowels, but when spreading is blocked by a high [−ATR] vowel, those same triggers surface as [−ATR]. Kinande (§3.3) extends this to the morphological domain. In this language, a [+ATR] prefix vowel surfaces as [−ATR] when it precedes a high [−ATR] root, demonstrating that high-ranked faithfulness to a morphological domain (the root) can function as the blocking condition. Finally, in Liko (§3.4), high enclitic vowels act as blockers, never surfacing as [+ATR], resulting in non-high vowels surfacing as [−ATR], showing that the blocking condition can require a morphophonological property, since only [+HI,−ATR] vowels in clitic position can condition dominance reversal.

3.2 Komo

Komo exhibits what Otero (2015) identifies as a type of phonologically conditioned dominance reversal, necessitating "two distinct and highly productive" (Otero 2015:212) spreading processes for the feature [ATR]: a canonical regressive [+ATR] harmony, and a secondary progressive [−ATR] harmony.

3.2.1 Basic Harmony

Komo possesses 7 contrastive vowels /i ɪ u ʊ ɛ ɔ a/ and three non-contrastive vowels [e o ə] surfacing exclusively as the output of vowel harmony on /ɛ ɔ a/ (Otero 2015:213). The language exhibits leftward [+ATR] harmony, which is triggered by high vowels only and targets [−HI] vowels only (44). Harmony is evident in alternations across morpheme boundaries, and can spread from a suffix, causing the [+ATR] value of the root vowel to change (44d-f).

(44) Suffix-triggered harmony (Otero 2015:215)

	UR	SR	Gloss
a.	/√ham-a/	[√ham-a]	'yawn-1SG'
b.	/√kɛʃ-a/	[√kɛʃ-a]	'thresh-1SG'
c.	/√dɔt'-an/	[√dɔt'-an]	'squat-1PL'
d.	/√ham-uk/	[√həm-uk]	'yawn-PFV'
e.	/√kɛʃ-i/	[√keʃ-i]	'thresh-2SG'
f.	/√dɔt-uk/	[√dot'-uk]	'squat-PFV'

Harmony also occurs within roots, where mid and low vowels in roots will surface as [+ATR] when followed by a high [+ATR] root vowel (45a-c). Due to the strict directionality of the harmony pattern, if a [−ATR] vowel follows a [+ATR] vowel within the root, then the disharmonic root will surface faithfully (45d-f).

(45) Root-internal harmony (Otero 2015:215,216)

	UR	SR	Gloss
a.	/√bɛzi/	[√bezi]	'be thin'
b.	/√gɔdum/	[√godum]	'sow'
c.	/√ami/	[√əmi]	'shake'
d.	/√durɛ/	[√durɛ]	'louse'
e.	/√ʃigɔʃ/	[√ʃigɔʃ]	'insect.sp'
f.	/√dibal/	[√dibal]	'grasshopper'

Further evidence for strictly regressive [+ATR] harmony is the non-alternation of [−ATR] suffixes that are attached to [+ATR] roots (46).

(46) No progressive harmony (Otero 2015:215-216)

	UR	SR	Gloss
a.	/√tus-ɛ/	[√tus-ɛ]	'be pushed-2SG'
b.	/√kiʃ-a/	[√kiʃ-a]	'harvest-1SG'
c.	/√sas-i-p'-ak/	[√səs-i-pak]	'to greet-ITV-3SG.F-1SG.O'

Additionally, the application of harmony in Komo is limited to the vowel closest to the trigger. This creates what Jurgec (2011) refers to as an icy target effect, where a segment that undergoes harmony then blocks the feature from propagating further. This effect is clearly illustrated in (47d-f), with forms like /zɛbɛ-i/ → [zɛbe-i], where the /ɛ/ closest to the trigger surfaces as [+ATR] as a result of harmony, becoming [e], but blocks harmony from continuing, leaving the leftmost vowel unaffected as [ɛ].

(47) Icy Target effect (Otero 2015:217,218)

	UR	SR	Gloss
a.	/√maza-na/	[√maza-na]	'be tired-1SG'
b.	/√k'ɔnzɔ-na/	[√k'ɔnzɔ-na]	'to prepare-1SG'
c.	/√zɛbɛ-na/	[√zɛbɛ-na]	'to enjoy-1SG'
d.	/√maza-i/	[√mazə-i]	'be tired-2SG'
e.	/√k'ɔnzɔ-i/	[√kɔnzo-i]	'to prepare-2SG'
f.	/√zɛbɛ-i/	[√zɛbe-i]	'to enjoy-2SG'

Regressive harmony in Komo can be modeled with the following sequential markedness constraint (Archangeli and Pulleyblank 1994; Pulleyblank 2002; Mahanta 2008) (48).

(48) *[−ATR]C_o[+HI, +ATR]: Assign one violation mark for any sequence of two vowels $V_1C_oV_2$ within a single prosodic word, where V_1 is [−ATR] and V_2 is [+HI,+ATR].

The [+Hi] specification on the trigger is crucial for the analysis of Komo. A more general constraint like *[−Atr]C$_o$[+Atr] would incorrectly force derived [−Hi, +Atr] vowels to also trigger harmony, leading to unattested iterative spreading. Thus, the explicit [+Hi] specification is what creates the icy target effect observed in (47) by correctly restricting the harmony trigger to high vowels only, preventing non-high [+Atr] vowels from initiating harmony themselves. Using a sequential markedness constraint like *[−Atr]C$_o$[+Hi,+Atr] to drive harmony creates an icy target effect, where a non-high vowel is simultaneously a target of harmony without subsequently acting as a trigger. Using this constraint also creates a prediction where harmony is expected to spread iteratively if the [−Atr] vowel is also [+Hi], though that prediction cannot be tested in Komo since, as I will show below, the presence of a high [−Atr] vowel within the domain yields dominance reversal.

Furthermore, I assume general faithfulness constraints for [±Atr] preventing gratuitous changes to the [Atr] value of the input.[9] These constraints are defined below:

(49) Ident[+Atr]: Assign one violation mark for each output vowel that is [−Atr] whose input correspondent is [+Atr].

(50) Ident[−Atr]: Assign one violation mark for each output vowel that is [+Atr] whose input correspondent is [−Atr].

As shown in Chapter 1, because the harmony-driving constraint can be satisfied both by leftward [+Atr] harmony or rightward [−Atr] harmony, Ident[+Atr] must outrank Ident[−Atr] to preserve the [+Atr] value on the harmony triggers. As a result of this ranking, the harmony candidate in (51c) is preferred over the faithful candidate in a context where a [−Hi, −Atr] vowel precedes a high [+Atr] vowel.

[9]Violations on the sequential markedness constraint could, in principle, be avoided by changing a [+Hi,+Atr] to [−Hi]. In order to prevent such an outcome, I assume a high-ranked constraint such as Ident[Hi], though I do not explicitly model it.

(51) Input: /√ham-uk/

/√ham-uk/	*[−ATR]C$_0$[+HI, +ATR]	ID[+A]	ID[−A]	
a. [√ham-uk]	*!			*Faithful Cand.*
b. [√ham-ʊk]		*!		*Dominance Reversal*
☞ c. [√həm-uk]			*	*[+Atr] Harmony*

This ranking extends straightforwardly to cases involving derived mid vowels, where the [+HI] specification in the harmony-driving constraint correctly derives the form in (52d).

(52) Input: /√zɛbɛ-i/

/√zɛbɛ-i/	*[−ATR]C$_0$[+HI, +ATR]	ID[+A]	ID[−A]	
a. [√zɛbɛ-i]	*!			*Faithful Cand.*
b. [√zɛbɛ-ɪ]		*!		*Dominance Reversal*
c. [√zebe-i]			**!	*[+Atr] Harmony*
☞ d. [√zɛbe-i]			*	*Part. [+Atr] Harmony*

Since the constraint refers specifically to [+HI, +ATR] vowels, a sequence where a [−ATR] vowel precedes a [−HI,+ATR] vowel does not incur violations of the harmony-driving constraint. Thus although both candidates (52c,d) incur no violations of the harmony-driving constraint, full harmony throughout the word incurs gratuitous violations of IDENT[−ATR].

In sum, Komo's basic pattern can be captured with a sequential markedness constraint that enforces regressive spreading from high [+ATR] vowels, with derived [−HI,+ATR] vowels correctly excluded as triggers. This establishes the core mechanics of the pattern where high vowels are the only active triggers and harmony is strictly regressive.

3.2.2 Dominance Reversal

While the above data show successful [+ATR] spreading onto [−HI] vowels, Komo also exhibits a phonologically-conditioned pattern of dominance reversal. Specifically, when a high [+ATR] trigger is directly preceded by another high vowel that is [−ATR], spreading is blocked, resulting in a repair where the high underlying [+ATR] vowel surfaces as its [−ATR] counterpart (53a-c).

(53) Dominance reversal (Otero 2015:216-217)

	UR	SR	Gloss
a.	/√jɪl-i-p/	[√jɪl-ɪ-p]	'to see-ITV-3SG.F'
b.	/√dɪl-uk/	[√dɪl-ʊk]	'to stamp.PFV'
c.	/√ɓʊg-uk/	[√ɓʊg-ʊk]	'to wait-PFV'

Much like regressive [+ATR] harmony shown above, dominance reversal is also strictly directional. A high [+ATR] vowel surfaces as [−ATR] only when it follows a high [−ATR] vowel. The data in (54) confirm that the reverse is not true: a high [+ATR] vowel that precedes a high [−ATR] vowel surfaces faithfully with its [+ATR] feature intact.

(54) No dominance reversal before blockers (Otero 2015:217)

	UR	SR	Gloss
a.	/√kiʃ-ʊ-i/	[√kiʃ-ʊ-ɪ]	'to harvest-VENT-2sg'
b.	/√sun-ʊ-i/	[√sun-ʊ-ɪ]	'to point-VENT-2sg'
c.	/√jil-i-r-ɪ/	[√jil-ir-ɪ]	'to see-ITV-3SG.MASC-3SG.NEUT.O'

However, unlike regressive [+ATR] spreading, all high [+ATR] vowels following a high [−ATR] vowel will lose their [+ATR] feature (Olejarczuk et al. 2019). This is shown in (55).

(55) Iterative dominance reversal (Olejarczuk et al. 2019:38)

	UR	SR	Gloss
a.	/√ʃɪt-uk-un/	[√ʃɪt-ʊk-ʊn]	'to whistle-PFV-3PL'
b.	/√pʊk-uk-i/	[√pʊk-ʊk-ɪ]	'to cross-PFV-2SG'

Thus, because *[−ATR]C$_o$[+HI,+ATR] penalizes any sequence of a [−ATR] vowel preceding a [+HI,+ATR] vowel, /ɪC$_o$i/ sequences, whether underlying (/ʃɪt-uk-un/) or derived ([ʃɪt-ʊk-un]), generate the same violations.

Accounting for this blocking effect via a markedness constraint is problematic. A general constraint like *[+HI,+ATR] is empirically falsified by the presence of /i u/ in Komo's surface forms. On the other hand, since the grammar has access to faithfulness constraints governing feature values (*i.e.,* IDENT[±ATR]), a minimal and natural extension is to permit faithfulness to a more restricted natural class of segments. The blocking effect of Komo's high [−ATR] vowels can therefore be captured directly by a high-ranking faithfulness constraint IDENT[−ATR]/[+HI] (56) ranked above the more general IDENT[+ATR] to prevent /ɪ/→[i].[10]

(56) IDENT[−ATR]/[+HI]: Assign one violation mark for each output vowel that is [+ATR] whose input correspondent is [+HI,−ATR].

The high rank of this specific faithfulness constraint is what induces dominance reversal. It creates a conflict, where the harmony-driving constraint, *[−ATR]C$_o$[+HI, +ATR], penalizes the faithful candidate (57b) [jɪl-ir], but IDENT[−ATR]/[+HI] prohibits the expected harmonic repair of changing the high [−ATR] vowel to [+ATR] (Candidate 57a; [jilir]). Because both of these constraints outrank IDENT[+ATR], dominance reversal, which incurs a violation of this lower-ranked constraint, emerges as the optimal outcome (candidate 57c).

[10]The reason why [+HI,−ATR] vowels in Komo act as blockers instead of being licit targets of harmony is not immediately clear and warrants further research.

(57) Input: /√jɪl-ir/

/√jɪl-ir/	ID[−A]/[+H]	*[−ATR]C$_o$[+HI, +ATR]	ID[+A]	ID[−A]	
a. [√jil-ir]	*!			*	*[+Atr] Harmony*
b. [√jɪl-ir]		*!			*Faithful Cand.*
☞ c. [√jɪl-ɪr]			*		*[−Atr] Harmony*

In (58), because *[−ATR]C$_o$[+HI,+ATR] is violated only when a [−ATR] vowel *precedes* a high [+ATR] vowel, the sequence /i…ʊ/ in (58c) does not incur any violation on the harmony-driving constraint, leading the leftward [−ATR] harmony (58c) to incur gratuitous violations of IDENT[+ATR].

(58) Input: /√kiʃ-ʊ-i/

/√kiʃʊi/	ID[−A]/[+H]	*[−ATR]C$_o$[+HI, +ATR]	ID[+A]	ID[−A]	
a. [√kiʃ-u-i]	*!			*	*[+Atr] Harmony*
b. [√kiʃ-ʊ-i]		*!			*Faithful Cand.*
c. [√kɪʃ-ʊ-ɪ]			**!		*Bidir. Dominance Reversal*
☞ d. [√kiʃ-ʊ-ɪ]			*		*Dominance Reversal*

Additionally, any sequence of multiple [+HI, +ATR] vowels after [+HI, −ATR] vowels will violate *[−ATR]C$_o$[+HI,+ATR] (59). As a result, the candidate featuring iterative dominance reversal (59d) is preferred to other candidates (59b,c).

(59) Input: /√ʃɪt-uk-un/

/√ʃɪt-uk-un/	ID[−A]/[+H]	*[−ATR]C$_o$[+HI, +ATR]	ID[+A]	ID[−A]	
a. [√ʃit-uk-un]	*!			*	*[+Atr] Harmony*
b. [√ʃɪt-uk-un]		*!			*Faithful Cand.*
c. [√ʃɪt-ʊk-un]		*!	*		*Part. Dominance Reversal*
☞ d. [√ʃɪt-ʊk-ʊn]			**		*Dominance Reversal*

Finally, when a [−HI] vowel intervenes between a sequence of /ɪ…i/, regressive [+ATR] harmony spreads onto the [−HI] vowel, and the high [+ATR] vowel retains its [+ATR] feature (60).

(60) Harmony in the presence of /ɪ/ (Otero 2015:218)

a. /√tɪt'a-i/ [√tɪt'ə-i] 'be thin-2SG'

This partial harmony pattern is correctly predicted by the analysis without any modification to the ranking. Changing the intervening [−HI, −ATR] vowel to [+ATR] avoids violations on the harmony-driver (61d), while both the harmony candidate (61a) and dominance reversal (61c) incur gratuitous violations of faithfulness constraints.

(61) Input: /√tɪt'a-i/

/√tɪt'a-i/	ID[−A]/[+H]	*[−ATR]C$_o$[+HI, +ATR]	ID[+A]	ID[−A]	
a. [√tit'ə-i]	*!			**	*[+Atr] Harmony*
b. [√tɪt'a-i]		*!			*Faithful Cand.*
c. [√tɪt'a-ɪ]			*!		*Dominance Reversal*
☞ d. [√tɪt'ə-i]				*	*Part. [+Atr] Harmony*

To summarize, vowel harmony in Komo is driven by a sequential markedness constraint *[−ATR]C$_o$[+HI,+ATR]. This constraint assigns violations for every sequence in which a [−ATR] vowel is immediately followed by a [+HI,+ATR] vowel. The restriction that the [+ATR] vowel must also be [+HI] yields the icy target effect, since a sequence of [−ATR]C$_o$[−HI,+ATR] would not incur any violations and changing an input [−ATR] vowel to [+ATR] in such a sequence would incur a gratuitous violation on IDENT[−ATR]. When a [+HI,−ATR] vowel precedes a high [+ATR] vowel, the high-ranked constraint IDENT[−ATR]/[+HI] prevents [+ATR] spreading as a repair strategy to avoid violations of *[−ATR]C$_o$[+HI,+ATR], thus inducing dominance reversal.

This analysis, then, maps directly onto the ranking configuration from Chapter 2, inducing a general repair strategy where changing a vowel from [−ATR] to [+ATR] is considered less costly to the grammar as a way of avoiding violations of the harmony-driving constraint *[−ATR]C$_o$[+HI,+ATR] than the opposite. High-ranked

IDENT[−ATR]/[+HI] acts as the BLOCKING constraint preventing the general repair strategy from occurring in a structure that contains a high [−ATR] vowel which immediately precedes a high [+ATR] vowel. In that specific structure, the repair strategy is going to prioritize changing a [+ATR] vowel to [−ATR], therefore avoiding violations of the two high-ranked constraints IDENT[−ATR]/[+HI] and *[−ATR]C_o[+HI,+ATR], even though it incurs violations of IDENT[+ATR].

The next section turns to Kinande, which presents a different instantiation of the same configuration. Where in Komo the blocker is a phonologically predictable high [−ATR] vowel, in Kinande the blocker is the morphological root, preventing parasitic harmony from applying to the root from a [+HI,+ATR] prefix.

3.3 Kinande

Kinande features two distinct harmony processes: a dominant, regressive [+ATR] harmony that spreads leftward from roots and suffixes, and a more restricted, progressive and parasitic harmony, where a high [+ATR] vowel causes a following high [−ATR] vowel to assimilate. A central challenge has been the pattern of dominance reversal associated with class 5 nominals, in which a prefix appears to lose its [+ATR] value when followed by a [−ATR] root, a phenomenon which has been addressed through various formalisms (Kenstowicz 2009; Downing and Krämer 2022). Building on these prior analyses and descriptions, I argue that the pattern is derived from the ranking established in Chapter 2: the [+ATR] prefix vowel surfaces as [−ATR] due to a high-ranked faithfulness constraint protecting the root vowel (IDENT-ROOT[−ATR]) from becoming [+ATR].

3.3.1 Basic Harmony

Kinande possesses a vowel inventory of seven contrastive vowels /i ɪ u ʊ ɛ ɔ a/ and three non-contrastive vowels [e o ə] (Downing and Krämer 2022:3-5). The language

exhibits a basic [+ATR] harmony pattern where a [+ATR] vowel spreads its feature leftward onto preceding vowels (62c,d).

(62) Regressive Harmony (Downing and Krämer 2022:11)

	UR	SR	Gloss
a.	/a-ba-√lʊmɛ/	[a-ba-√lʊmɛ]	'man.PL'
b.	/ɔ-mʊ-√lʊmɛ/	[ɔ-mʊ-√lʊmɛ]	'man.SG'
c.	/a-ba-√kali/	[ə-bə-√kəli]	'woman/PL'
d.	/ɔ-mʊ-√kali/	[o-mu-√kəli]	'woman.SG'

Harmony does not proceed rightward onto a following [−HI] vowel, a restriction that holds even within a root (63b,c).

(63) Directional Asymmetry (Downing and Krämer 2022:4, Kenstowicz 2009:5)

	UR	SR	Gloss
a.	/ɛ-rɪ-√hum-a/	[e-ri-√hum-a]	'move.INF'
b.	/ɔ-mʊ-√sisa/	[o-mu-√sisa]	'to defeat'
c.	/ɛ-rɪ-√liban-ɪr-a/	[e-ri-√liban-ɪr-a]	'to disappear for'

The data presented above show the directional asymmetry where [+ATR] harmony in Kinande proceeds uninhibited leftward regardless of the height of the target, but does not spread to [−HI] vowels progressively. This leftward-spreading pattern can be modeled through a sequential markedness constraint *[−ATR]C_o[+ATR], preventing a [−ATR] vowel from preceding a [+ATR] vowel (64).

(64) *[−ATR]C_o[+ATR]: Assign one violation mark for any sequence of two vowels $V_1C_oV_2$ within a single prosodic word, where V_1 is [−ATR] and V_2 is [+ATR].

Harmony in Kinande is modeled with the ranking *[−ATR]C_o[+ATR] >> IDENT[+ATR] >> IDENT[−ATR], which favors [+ATR] harmony as the optimal repair strategy. As

explained previously, because *[−Atr]C$_o$[+Atr] evaluates each instance of potential disagreement in the output, iterativity is enforced without additional mechanisms. This is shown below:

(65) Input: /ɛ-rɪ-√hum-a/

/ɛ-rɪ-√hum-a/	*[−Atr]C$_o$[+Atr]	Id[+A]	Id[−A]	
a. [ɛ-rɪ-√hum-a]	*!			*Faithful Cand.*
b. [ɛ-ri-√hum-a]	*!		*	*partial [+Atr] Harmony*
c. [ɛ-rɪ-√hʊm-a]		*!		*Dominance Reversal*
d. [e-ri-√hum-ə]			***!	*[+Atr] Harmony*
☞ e. [e-ri-√hum-a]			**	*Regr. [+Atr] Harmony*

This process, however, is not limited to triggers in roots. As the data below show, the same pervasive leftward harmony occurs when the [+Atr] trigger originates in a suffix (66), inducing harmony across all preceding vowels, causing both the root and prefix vowels to surface as [+Atr]. Notably, even a [+Atr] vowel in prefix position is able to trigger [+Atr] harmony on a preceding prefix (66d).

(66) Affix-triggered Harmony (Downing and Krämer 2022:5, 10, 15)

	UR	SR	Gloss
a.	/ɛ-rɪ-√hɛk-ɛs-i-a/	[e-ri-√hek-es-i-a]	'carry.caus.'
b.	/ɛ-rɪ-√lɔg-ɛs-i-a/	[e-ri-√log-es-i-a]	'bewitch.caus.'
c.	/ɔ-mʊ-√hak-i/	[o-mu-√hək-i]	'smear.agent.'
d.	/ɛ-ri-√hɛmbɛ/	[e-ri-√hɛmbɛ]	'5.horn'

In Kinande, a [+Atr] suffix vowel can change the Atr value of a preceding root vowel (66), a systematic property that challenges the proposed universal Ident-Root >> Ident-Affix (McCarthy and Prince 1995; Urbanczyk 2006). Kinande is not unique in exhibiting this pattern of suffix dominance over roots. Noske (2000), for

instance, makes a parallel argument for Turkana and proposes a high-ranked IDENT-IO suffix[ATR] (67) and IDENT-IO root[ATR] (68), where IDENT-IO suffix[ATR] is ranked above IDENT-IO root[ATR].

(67) IDENT-IO suffix[ATR]: "Let β be an output segment in a suffix and α the input correspondent of β. If β is [αATR], then α has to be [αATR]" (Noske 2000:795)

(68) IDENT-IO root[ATR]: "Let β be an output segment in a root and α the input correspondent of β. If β is [αATR], then α has to be [αATR]" (Noske 2000:791)

Regressive harmony in Kinande, then, mirrors the pattern seen in Turkana, where a suffix vowel is able to override the [ATR] value of the root vowel. However, a general symmetric constraint like IDENT-IO[ATR] is insufficient since, as the later data will show, the grammar must be able to treat changes *to* [+ATR] on a suffix differently from changes *from* [+ATR].

For the regressive pattern currently under discussion, the crucial interaction is between the preservation of the [+ATR] trigger on the suffix and the preservation of the [−ATR] vowels of the root. Therefore, I adapt Noske's constraints above, positing two constraints that are empirically motivated by this pattern:

(69) IDENT-SUFFIX[+ATR] : Assign one violation mark for each output vowel in a suffix that is [−ATR] whose input correspondent is [+ATR].

(70) IDENT-ROOT[−ATR]: Assign one violation mark for each output vowel in a root that is [+ATR] whose input correspondent is [−ATR].

Thus, the ranking of *[−ATR]C_o[+ATR], IDENT-SUFFIX[+ATR] >> IDENT-ROOT[−ATR] properly captures the dominance of suffix vowels over root vowels *contra* McCarthy & Prince's (1995) assumptions, ensuring that harmony from a suffix proceeds even at the cost of violating faithfulness to the root's [−ATR] feature (71).

(71) Input: /ɔ-mʊ-√hak-i/

/ɔ-mʊ-√hak-i/	*[−Atr]C$_o$[+Atr]	Id-Suff[+A]	Id-Rt[−A]	Id[+A]	Ident[−Atr]	
a. [ɔ-mʊ-√hak-i]	*!					*Faithful Cand*
b. [ɔ-mʊ-√həkɨ-i]	*!				*	*partial [+Atr] Harmony*
c. [ɔ-mʊ-√hak-ɪ]		*!		*		*Dominance Reversal*
☞ d. [o-mu-√həkɨ-i]			*		***	*Regr. [+Atr] Harmony*

This ranking establishes the basic grammar for regressive harmony which interacts with a more restricted process of progressive parasitic harmony, where [+Atr] spreads rightward between adjacent high vowels.

3.3.2 Parasitic Harmony

Parasitic harmony in Kinande occurs when a [+Hi,+Atr] vowel can cause a following [+Hi,−Atr] vowel to surface as [+Atr] (72a,b). This pattern applies only within a sequence of adjacent high vowels, and cannot cross non-high vowels (72c).

(72) Parasitic Harmony (Kenstowicz 2009:23; Downing and Krämer 2022:4)

	UR	SR	Gloss
a.	/ɛ-rɪ-√huk-ɪr-a/	[e-ri-√huk-ir-a]	‘to prepare for’
b.	/ɛ-rɪ-√huk-ʊl-ɪr-a/	[e-ri-√huk-ul-ir-a]	‘to dish out for’
c.	/ɛ-rɪ-√liban-ɪr-a/	[e-ri-√liban-ɪr-a]	‘disappear’

The data in (72) show that progressive [+Atr] harmony applies between adjacent high vowels, causing suffix high vowels to surface as [+Atr] if the preceding vowel is also high. The current constraint ranking, however, only handles regressive harmony. Thus, because no constraint is currently ranked high enough to penalize sequences of [+Hi,+Atr]C$_o$[+Hi,−Atr], there is no mechanism that would force progressive spreading of the [+Atr] feature, as shown in (73), causing the grammar to incorrectly select the regressive harmony candidate (73d) instead of the progressive harmony candidate (73c).

(73) Input: /ɛ-rɪ-√huk-ɪr-a/

/ɛ-rɪ-√huk-ɪr-a/	*[−Atr]Co[+Atr]	Id-Suff[+A]	Id-Rt[−A]	Id[+A]	Ident[−Atr]	
a. [ɛ-rɪ-√huk-ɪr-a]	*!					*Faithful Cand.*
b. [ɛ-rɪ-√hʊk-ɪr-a]				*!		*Dominance Reversal*
c. [e-ri-√huk-ir-a]					***!	*Parasitic Harmony*
☞ d. [e-ri-√huk-ɪr-a]					**	*Regr. [+Atr] Harmony*

To account for parasitic harmony, then, the grammar requires a dedicated markedness constraint that targets only disharmonic sequences of adjacent high vowels. To model this process, I posit the constraint *[+Hi, +Atr]Co[+Hi, −Atr], defined below:

(74) *[+Hi, +Atr]Co[+Hi, −Atr] : Assign one violation mark for any sequence of two vowels $V_1C_oV_2$ within a single prosodic word, where V_1 is [+Hi, +Atr] and V_2 is [+Hi,−Atr].

This markedness constraint must be ranked above the general Ident[+Atr] >> Ident[−Atr] to force assimilation to the [+Atr] vowel. Because no high-ranked positional faithfulness constraint protects [−Atr] vowels in suffixes, this ranking is sufficient to derive the parasitic harmony pattern, as shown in (75).

(75) Input: /ɛ-rɪ-√huk-ɪr-a/

/ɛ-rɪ-√huk-ɪr-a/	*[−Atr]Co[+Atr]	Id-Suff[+A]	*[+Hi, +Atr]Co[+Hi, −Atr]	Id[+A]	Id[−A]	
a. [ɛ-rɪ-√huk-ɪr-a]	*!		*			*Faithful Cand.*
b. [e-ri-√huk-ɪr-a]			*!		**	*Regr. [+Atr] harmony*
c. [ɛ-rɪ-√hʊk-ɪr-a]				*!		*Dominance Reversal*
d. [e-ri-√huk-ir-ə]					****!	*[+Atr] Harmony*
☞ e. [e-ri-√huk-ir-a]					***	*Part. [+Atr] Harmony*

(75) shows that progressive parasitic harmony is successfully modeled through *[+Hi, +Atr]Co[+Hi, −Atr], penalizing sequences in which a high [+Atr] vowel is followed by a high [−Atr] vowel (75a,b). This constraint operates alongside the higher-ranked *[−Atr]Co[+ATR] constraint, which drives regressive spreading across

vowels of all heights (75a). Candidates that spread [+ATR] rightward across a sequence of high vowels (*e.g.,* candidate 75d,e), then successfully satisfy *[+HI, +ATR]C_o[+HI, −ATR], while candidates where [+ATR] spreads onto following [−HI] vowels (75d) incur gratuitous violations of IDENT[−ATR]. This ranking effectively captures parasitic progressive [+ATR] harmony only occurring between high vowels.

As previously shown, high [+ATR] prefix vowels are capable of triggering regressive harmony on a preceding prefix vowel (66). However, when such a prefix is followed by a high [−ATR] root vowel, the expected parasitic progressive harmony modeled above is blocked. Instead, the [+ATR] prefix vowel surfaces as [−ATR] (76c,d).

(76) Prefix Harmony (Kenstowicz 2009:26)

	UR	SR	Gloss
a.	/ɛ-ri-√hɛmbɛ/	[e-ri-√hɛmbɛ]	'horn'
b.	/ɛ-ri-√lɔlɔ/	[e-ri-√lɔlɔ]	'sin'
c.	/ɛ-ri-√bʊgʊ/	[ɛ-rɪ-√bʊgʊ]	'plantain'
d.	/ɛ-ri-√hɪrɪ/	[ɛ-rɪ-√hɪrɪ]	'crab'

The data in (76a,b) confirm the prefix is underlyingly specified as [+ATR], allowing it to spread [+ATR] to a preceding vowel. This contrasts with (76c,d), where the high [−ATR] vowel of the root blocks progressive harmony. Consequently, the prefix surfaces as [−ATR], resulting in dominance reversal.

No other constraint is necessary to model dominance reversal. The position of a high [+ATR] vowel in prefix position preceding a root composed of high [−ATR] vowels does incur violations of *[+HI,+ATR]C_o[+HI, −ATR] (77d), while IDENT-ROOT[−ATR] prevents changes to the root (77b,c). The optimal repair strategy under this circumstance is dominance reversal causing the prefix vowel /i/ to surface as [ɪ] (77e).

(77) Input: /ɛ-ri-√hɪrɪ/

/ɛ-ri-√hɪrɪ/	*[−Atr]C$_o$[+Atr]	Id-Rt[−A]	*[+Hi, +Atr]C$_o$[+Hi, −Atr]	Id[+A]	Id[−A]	
a. [ɛ-ri-√hɪrɪ]	*!		*			*Faithful Cand.*
b. [e-ri-√hiri]		*!*			***	*[+Atr] Harmony*
c. [e-ri-√hirɪ]		*!	*		**	*partial [+Atr] Harmony*
d. [e-ri-√hɪrɪ]			*!		*	*Regr. [+Atr] Harmony*
☞ e. [ɛ-rɪ-√hɪrɪ]				*		*Dominance Reversal*

In this ranking, Ident-Root[−Atr] thus functions as the blocking constraint, eliminating any candidate that satisfies the markedness constraint *[+Hi, +Atr]C$_o$[+Hi, −Atr] by incurring a change in the root (77b,c), while ruling out the candidates that preserve the disharmony between high vowels (77a,d). Thus, the optimal candidate (77e) is selected, with its single violation of low-ranked Ident[+Atr], yielding dominance reversal.

Kinande's analysis, then, provides a clear instantiation of the canonical ranking for dominance reversal. In this case, Ident-Root[−Atr] acts as the blocking constraint by protecting the [−Atr] value of root vowels, preventing [+Atr] harmony as a resolution to avoid violations of the high-ranked harmony-driving constraint *[+Hi,+Atr]C$_o$[+Hi,−Atr]. Because both of these constraints outrank Ident[+Atr], the grammar selects the candidate that violates this lower-ranked general faithfulness constraint, resulting in dominance reversal. Kinande therefore confirms that the ranking configuration also extends to the morphological domain.

3.3.3 On Previous Analyses

The results from Chapter 2 predict that any analysis which derives dominance reversal should instantiate the same ranking configuration, abstracting away from any particular formalism or underlying assumptions. In this section, I turn to past analyses of Kinande: Kenstowicz (2009) and Downing & Krämer (2022). While the constraints and ranking in Downing & Krämer can successfully be mapped to the ranking configuration, Kenstowicz's analysis, by contrast, cannot since it makes use exclusively of markedness constraints, making it impossible to fill certain positions such as Ident-Trigger and

IDENT-TARGET.

3.3.3.1 Kenstowicz (2009)

Kenstowicz (2009) adopts headed harmonic domains in the sense of Cole & Kisseberth (1993) and McCarthy (2004), formalizing the blocking effect that induces dominance reversal through a constraint conjunction *Affix&*LR. He identifies dominance reversal as a ganging up effect (Kenstowicz 2009:33), where two independently lower-ranked constraints, *LR (penalizing rightward harmony) and *AFFIX (penalizing affixal triggers), combine to block rightward [+ATR] harmony from an affix. The tableau in (78) shows parasitic harmony, where the conjunction *AFFIX&*LR is vacuously satisfied since spreading is initiated from the root, while *(ɪ) eliminates the faithful candidate (78a) and the dominance reversal candidate (78b), both of which contain [−ATR] high-vowel spans. Lastly, *ADJSPAN penalizes any adjacent feature spans.

(78) Input: /√sim-ɪr/ (*cf.* Kenstowicz 2009:32)

/√sim-ɪr/	*AFFIX & LR	*(ɪ)	*ADJ SP	*LR	*AFFIX	
a. [√sim-ɪr]		*!	*			*Faithful Cand.*
b. [√sɪm-ɪr]		*!			*	*Dominance Reversal*
☞ c. [√sim-ir]				*		*Progr. [+Atr] Harmony*

For dominance reversal (79), however, progressive harmony (79a) now violates *AFFIX&*LR, because the [+ATR] vowel is in a prefix position with rightward spreading, thus violating both constraints simultaneously. The remaining candidates, then, both incur a violation on *(ɪ) (79b,c) leading the evaluation to fall to the lower ranking *ADJ SP >> *LR >> *AFFIX. The faithful candidate (79b) violates the higher ranked *ADJ SP, yielding dominance reversal (79c) as the optimal candidate.

(79) Input: /ri-√hɪrɪ/ (*cf.* Kenstowicz 2009:33)

/ri-√hɪrɪ/	*Affix & LR	*(ɪ)	*Adj Sp	*LR	*Affix	
a. [ri-√hiri]	*!			*	*	*Progr. [+Atr] harmony*
b. [ri-√hɪrɪ]		*	*!			*Faithful Cand.*
☞ c. [rɪ-√hɪrɪ]		*				*Dominance Reversal*

And so, while it is straightforward to claim that *Affix&*LR acts as the blocking constraint, preventing the more general repair strategy (rightward [+Atr] harmony), it is less obvious which role the other constraints occupy in the hierarchy. The reason is that Kenstowicz's analysis uses exclusively markedness constraints, without any explicit faithfulness constraints to protect input segments. For example, *(ɪ) appears to occupy the Ident-Trigger position in some sense, by preventing a repair from /i/ to [ɪ]. However, this constraint fundamentally discourages *any* [ɪ] in the output, regardless of its input correspondent. It would equally drive a change from an input /ɪ/, rather than protecting an input /i/ specifically. Similarly, there is no constraint that unambiguously functions as Ident-Target. There have to be some implicit faithfulness constraints that are active, otherwise any input structure would always be changed to some unmarked value (see Prince & Smolensky 1993/2004:27 on the Economy Property). What these faithfulness constraints are and how they are ranked are left undefined, making the mapping to Chapter 2's ranking difficult to properly establish.

3.3.3.2 Downing & Krämer (2022)

Downing & Krämer (2022) use Syntagmatic Correspondence (Krämer 1998, 2001, 2003) to show that the loss of the prefix vowel [+Atr] value in Kinande can be accounted for without requiring a dedicated mechanism to induce [−Atr] harmony, an argument which Downing & Krämer use to argue that Kinande does not exhibit dominance reversal (Downing and Krämer 2022:18).

To model parasitic harmony (80), Downing & Krämer use a high-ranked S-

IDENT^{HI}(±ATR) to drive harmony between high vowels, eliminating the faithful candidate (80a), while the ranking of MAX^{HI}(+ATR) >> DEP(+ATR) drives a general repair in which [+ATR] harmony (80c) is considered a better outcome to avoid violations of S-IDENT^{HI}(+ATR) than [−ATR] harmony (80b).

(80) Input: /...√himb-ır-ə/ (*cf.* Downing & Krämer 2022:8)

/...√himb-ır-ə/	S-IDENT^{HI}(±ATR)	DEP$^{\text{HI, STEM}}$(+ATR)	MAX^{HI}(+ATR)	DEP(+ATR)	
a. [...√himb-ır-ə]	*!				*Faithful Cand.*
b. [...√hımb-ır-a]			*!		*Dominance Reversal*
☞ c. [...√himb-ir-a]				*	*Part. [+Atr] Harmony*

On the other hand, the ranking of DEP$^{\text{HI, STEM}}$(+ATR) >> MAX^{HI}(+ATR) models dominance reversal in a sequence of a prefix [+ATR] vowel preceding a [−ATR] root, since DEP$^{\text{HI, STEM}}$(+ATR) prevents stem vowels from surfacing as [+ATR] (81b). Dominance reversal (81c) is thus preferred over the faithful candidate (81a) due to violations of the harmony-driving constraint S-IDENT^{HI}(±ATR).

(81) Input: /...-ri-√hırı/ (*cf.* Downing and Krämer 2022:16)

/...-ri-√hırı/	S-IDENT^{HI}(±ATR)	DEP$^{\text{HI, STEM}}$(+ATR)	MAX^{HI}(+ATR)	DEP(+ATR)	
a. [...-ri-√hırı]	*!				*Faithful Cand.*
b. [...-ri-√hiri]		*!*		**	*Parasitic Harmony*
☞ c. [...-rı-√hırı]			*		*Dominance Reversal*

In Downing & Krämer's analysis, then, DEP$^{\text{HI, STEM}}$(+ATR) functions as the blocking constraint, preventing the more general rightward [+ATR] harmony resolution, while S-IDENT^{HI}(±ATR) functions as the harmony-driver by demanding agreement between high vowels for ATR. And so, while Max^{Hi}(+Atr) >> DEP(+ATR) normally drives [+ATR] harmony by making it more costly for the grammar to change the [+ATR] value of a high vowel over preserving the [−ATR] value of any vowels. When the [−ATR] vowel is in a stem, the ranking of DEP$^{\text{HI, STEM}}$(+ATR) >> Max^{Hi}(+Atr) creates the opposite scenario, making it more costly to the grammar to change a [−ATR] high

stem vowel to [+ATR]. The ranking, then, instantiates the same dominance reversal configuration seen in this chapter and Chapter 2: BLOCKER, HARMONY-DRIVER >> IDENT[+ATR] >> IDENT[−ATR].

As discussed in Chapter 1, markedness constraints driving harmony are indifferent to resolution and [±ATR] dominance emerges from a specific ranking that determines that one value is more costly for the grammar to change than the other to avoid violations of the harmony-driver. Dominance reversal, however, isn't derived by a specific constraint, but by a specific ranking configuration where a blocker prevents full [+ATR] harmony, forcing the grammar to sacrifice the trigger's specification to satisfy the harmony-driving constraint.

Thus, despite the fact that Downing & Krämer explicitly argue that their analysis does not constitute dominance reversal (Downing and Krämer 2022:15-18), on the grounds that [−ATR] spreading emerges without a dedicated constraint spreading [−ATR], the fact that their ranking maps to the dominance reversal configuration makes it clear that they do instantiate the pattern.

Kinande's analyses therefore confirm that the ranking configuration from Chapter 2 extends to the morphological domain. In my analysis from this chapter, the pattern is derived through the interaction of IDENT-ROOT[−ATR] and *[+HI,+ATR]C_o[+HI,−ATR], both ranked above IDENT[+ATR], creating a ranking where a prefix [+ATR] vowel will surface as [−ATR] when it is followed by a [−ATR] root. This is also confirmed by Downing & Krämer's (2022) analysis since, despite using substantially different assumptions and constraint families, their ranking maps to the ranking configuration, even though they argued that Kinande is not a case of dominance reversal. This has the additional benefit of expanding the results of Chapter 2 where the constraints occupying each position in the ranking are defined by their functions. By contrast, Kenstowicz's analysis highlight a clear limitation of the current configuration. Since it does not include any faithfulness constraints, some of the

positions like IDENT-TRIGGER and IDENT-TARGET can't be filled, a limitation I leave for further research.

I now turn to Liko, where I show how the combination of morphological position and phonological specification also instantiates dominance reversal.

3.4 Liko

Liko extends the empirical scope in two ways. First, it provides a third independent attestation of dominance reversal, where the blocking constraint is a positional markedness constraint, showing that the configuration is not restricted to the domain or the type of constraint that occupies the BLOCKER position. Second, the same ranking that produces dominance reversal in Liko also produces sour grapes, with the choice of resolution determined by the height of the [+ATR] vowel. This co-occurrence allows for a direct comparison of the two patterns under identical morphophonological conditions, showing that they can be derived from a single constraint system rather than treated as typologically distinct processes.

The descriptive work on this system by de Wit (2015a, 2015b) analyzes Liko's harmony as an interaction between two competing domains. He posits a primary [+ATR] domain where [+ATR] is the active spreading feature accompanied by a second [−ATR] domain created by [−ATR] enclitics. He proposes that these enclitics can cause the [+ATR] feature of a preceding non-high vowel to be "delinked" under an autosegmental framework, which accounts for non-high [+ATR] vowels surfacing as [−ATR]. By contrast, high vowels exhibit greater "[+ATR] strength" (de Wit 2015a:38), which protects them from this delinking process.

Building on de Wit's descriptive insights, I reanalyze the Liko pattern as a single system of [+ATR] harmony in which all surface alternations for the dominance reversal pattern follow from the configuration established in Chapter 2. [+ATR] spreads to licit targets except when a high [−ATR] enclitic vowel occurs in the structure, preventing

any [+ATR] spreading from occurring. The choice of repair, then, is conditioned by the height of the [+ATR] trigger: high [+ATR] vowels are preserved, yielding a sour grapes effect, while non-high [+ATR] vowels surface as [−ATR], resulting in dominance reversal. The analysis proceeds in three stages: basic progressive spreading, sour grapes, and dominance reversal, showing how all three surface patterns emerge from the same constraint system.

3.4.1 Basic Harmony

Liko possesses nine contrastive vowels /i ɪ u ʊ e ɛ o ɔ a/ and exhibits a bidirectional [+ATR] harmony system in which the feature [+ATR] spreads from any [+ATR] vowel onto adjacent [−ATR] vowels (82). Although harmony applies both progressively and regressively, the formal analysis in this chapter will focus specifically on progressive spreading. This focus is motivated by the fact that the pattern central to this dissertation, dominance reversal, is conditioned exclusively by following vowels. A full and complete analysis of Liko would, however, need to handle the regressive pattern as well.

The data in (82) show a bidirectional harmony pattern where [+ATR] is the active, spreading feature. A [+ATR] vowel, whether in a root (82d,e,g) or a suffix (82f), is able to trigger [+ATR] harmony. For instance, in /ta-gum-ag-a/ → [to-gum-og-o] (82f), the feature of the root /gum/ and the derived [+ATR] vowel [o] spread to both prefix and suffix vowels.

(82) Bidirectional ATR harmony (de Wit 2015a:9; de Wit 2015b:73, 283, 352)

	UR	SR	Gloss
a.	/ta-√gʊm-ag-a/	[ta-√gʊm-ag-a]	'1PL-iron-PLUR-FV'
b.	/ka-√pɪk-a/	[ka-√pɪk-a]	'9b-prepare.for.sowing-FV'
c.	/wa-√lal-a/	[wa-√lal-a]	'2SG-sleep-FV.INST'
d.	/ta-√lik-a/	[to-√lik-o]	'1PL-dry-FV'
e.	/ta-√gum-ag-a/	[to-√gum-og-o]	'1PL-crawl-PLUR-FV'
f.	/√pɪk-a-ni/	[√pik-o-ni]	'prepare.for.sowing-FV-IMP.ADDR'
g.	/wa-√kos-an-a/	[wo-√kos-on-o]	"2SG-pour-ASS-FV.INST"

Harmony also extends across clitic boundaries, where a high [+ATR] vowel can trigger harmony on certain, adjacent clitic vowels, as shown in (83).[11]

(83) ATR harmony on enclitics (de Wit 2015b:102)

	UR	SR	Gloss
a.	/sɪ-√bʊkʊ=sɔ/	[sɪ-√bʊkʊ=sɔ]	'19-shrub, drug=19'
b.	/√kʊnzɪmgɪ=kɔ/	[√kʊnzɪmgɪ=kɔ]	'15.sweet potato=15'
c.	/√suku=sɔ/	[√suku=so]	'7.burning piece of wood=7'
d.	/kʊ-√tili=kɔ/	[ku-√tili=ko]	'15.ear=15'

Progressive harmony can be induced in Liko through a sequential markedness constraint defined in (84).

(84) *[+ATR]C_o[−ATR]: Assign one violation mark for two vowels $V_1C_oV_2$ within a single prosodic word, where V_1 is [+ATR] and V_2 is [−ATR]

[11]There is one [−HI,−ATR] clitic /=nɔ/ mentioned in de Wit (2015b) which does not undergo [+ATR] harmony. This behavior could be attributed to an idiosyncratic property of that specific morpheme. Baković (2000), for instance, provides a direct precedent with his analysis of "inalterably recessive" vowels in other harmony systems like Kalenjin, Turkana and Turkish. This pattern could be formally captured by positing a high-ranked positional faithfulness constraint, such as REC-IDENT[ATR] (Baković 2000) that targets this specific morphological class.

All vowels are therefore expected to participate in the process. The ranking *[+ATR]C_o[−ATR] >> IDENT[+ATR] >> IDENT[−ATR] captures this behavior (85), favoring [+ATR] harmony (85d).

(85) Input: /...-√gum-ag-a/

/...-√gum-ag-a/	*[+ATR]C_o[−ATR]	ID[+A]	ID[−A]	
a. [...-√gum-ag-a]	*!			*Faithful Cand.*
b. [...-√gum-og-a]	*!		*	*Part. [+Atr] Harmony*
c. [...-√gʊm-ag-a]		*!		*Dominance Reversal*
☞ d. [...-√gum-og-o]			**	*[+Atr] Harmony*

This constraint ranking also trivially handles attested cases of harmony across an enclitic boundary (86).

(86) Input: /√suku=sɔ/

/√suku=sɔ/	*[+ATR]C_o[−ATR]	ID[+A]	ID[−A]	
a. [√suku=sɔ]	*!			*Faithful Cand.*
b. [√sʊkʊ=sɔ]		*!*		*Dominance Reversal*
☞ c. [√suku=so]			*	*[+Atr] Harmony*

In sum, Liko's basic harmony pattern is one in which the feature spreads iteratively from any [+ATR] vowel onto following [−ATR] vowels across roots, suffixes and [−HI] clitics. [+HI] clitics, however, do not participate in this process, consistently surfacing with their underlying [−ATR] value. As mentioned above, when a [+HI,−ATR] clitic blocks harmony from a [−HI,+ATR] vowel, the latter surfaces as [−ATR] (dominance reversal); when the trigger vowel is [+HI,+ATR], however, no harmony applies though the [+ATR] specification is preserved (sour grapes). The choice of resolution is determined entirely by the height of the [+ATR] vowel as the next two sections will show.

3.4.2 Dominance Reversal

Dominance reversal occurs when a non-high [+ATR] vowel occurs in a domain with a high [−ATR] clitic vowel.[12] (87a,c,e) show that the vowel of the root is underlyingly [+ATR] since it causes the underlyingly [−ATR] vowel of the prefix to surface as [+ATR], while (87b,d,f) show dominance reversal. In Liko, whenever a non-high [+ATR] vowel and a high [−ATR] enclitic occur in the same domain, the underlyingly [+ATR] vowel surfaces as [−ATR] (87b,d,f).[13] These forms also interact with an independent process of height dissimilation (see fn. 14), which is not formalized in this dissertation since it is not consequential to the analysis of dominance reversal.

(87) Dominance Reversal (de Wit 2015b:105)

	UR	SR	Gloss
a.	/ma-√wo/	[mo-√wo]	'6-thorn'
b.	/kʊ-√wo=kɔ/	[ku-√wa=kʊ][14]	'15-thorn=15'
c.	/ma-√kwo/	[mo-√kwo]	'6-death'
d.	/kʊ-√kwo=kɔ/	[ku-√kwa=kʊ]	'15-death=15'
e.	/ɓi-√ngo/	[ɓi-√ngo]	'8-neck'
f.	/sɪ-√ngo=sɔ/	[si-√nga=sʊ]	'7-neck=7'

The resistance of these high [−ATR] clitics to [+ATR] harmony cannot be attributed to a lexically restricted class of morphemes that idiosyncratically fail to harmonize, an approach taken for other languages (see Baković 2000 on inalterably recessive vowels). It also cannot be conditioned phonologically, as high [−ATR] vowels in other

[12] All available examples of dominance reversal in Liko involve nominal constructions. An extensive search of the primary descriptive source did not reveal any verbal constructions that unambiguously create the context for dominance reversal.

[13] This does raise the question of why a language like Liko, which does not prohibit [ɔ], would have /o/ surfacing as [a] as a result of dominance reversal instead of [ɔ]. It is not clear at first glance why that would be the case, though I leave this question open.

[14] Liko also exhibits height dissimilation: when a [−HI] vowel precedes a [−HI] enclitic vowel, the enclitic surfaces as high. This could be formalized with a high-ranked constraint like *[−HI]=[−HI], penalizing adjacent [−HI] vowels across a clitic boundary.

morphological positions (*i.e.,* suffixes) are licit targets. Instead, the data from Liko show that the conditioning is truly morphophonological, where high [−ATR] vowels in clitic position do not undergo harmony.

The behavior of the class 15 clitic, /=kɔ/, shows this perfectly: it acts as a target of harmony (88a) when its vowel remains [−HI], but blocks harmony when it raises to [+HI] (88b; also see fn.14 on clitic raising). As mentioned above, the blocking effect in Liko is determined by the interaction of a specific morphological domain (the clitic) with specific phonological features ([+HI] and [−ATR]).

(88) Morphophonological Blocking Effect (de Wit 2015b:105)

	UR	SR	Gloss
a.	/kʊ-√tili=kɔ/	[ku-√tili=ko]	'15.ear=15'
b.	/kʊ-√kwo=kɔ/	[ku-√kwa=kʊ]	'15.death=15'

The generalization is that [+HI, −ATR] vowels in clitic positions are invariably opaque to harmony. Clitics are prosodically weak elements, making them a canonical site for positional neutralization, where phonological contrasts are resolved in favor of unmarked values (*e.g.,* Casali 1997; Beckman 1997, 1998; Lacy 2006; Kiparsky 2006; Casali 2014). Additionally, research on /2IU/ systems reveals a pattern where [−ATR] vowels in these inventories consistently function as the unmarked value in non-prominent contexts (Casali 2014).[15, 16] This behavior is captured formally by a high-ranked positional markedness constraint *=C_o[+HI,+ATR] (89) which penalizes high [+ATR] vowels in clitic position.

(89) *=C_o[+HI,+ATR]: Assign one violation mark for each vowel within a clitic that is [+HI,+ATR].

[15] /2IU/ refers to languages with a phonemic ATR contrast in the high vowels (Casali 2008).

[16] This raises the question of why [−HI,−ATR] clitics undergo harmony rather than remaining in their unmarked, [−ATR] state. This asymmetry potentially suggests that the grammar's imperative to enforce this unmarkedness is stronger for high vowels than for mid vowels (a point that is parallel to Komo's [+HI,−ATR] vowel's invariability). Once again, the explanation of this counter-intuitive behavior in the high [−ATR] vowels is not immediately clear and warrants further research.

Since both *[+Atr]C_o[−Atr] and *=C_o[+Hi,+Atr] outrank Ident[+Atr], the only constraint protecting the input [+Atr] value of non-high vowels, the ranking is best satisfied by the dominance reversal candidate (90c) as [+Atr] harmony would incur violations on the high ranked *=C_o[+Hi,+Atr] constraint and partial harmony or sour grapes would incur violations on *[+Atr]C_o[−Atr]. Crucially, no additional constraint is necessary to induce dominance reversal (90).[17, 18]

(90) Input: /...-√kwo=kʊ/

/...-√kwo=kʊ/	*=C_o[+Hi,+Atr]	*[+Atr]C_o[−Atr]	Id[+A]	Id[−A]	
a. [...-√kwo=ku]	*!			*	*[+Atr] Harmony*
b. [...-√kwo=kʊ]		*!			*Faithful Cand.*
☞ c. [...-√kwa=kʊ]			*		*Dominance Reversal*

Liko thus provides a third, independent attestation of the ranking that yields dominance reversal shown in Chapter 2. The blocking constraint in Liko is a positional

[17]Derived high vowels are also opaque to [+Atr] harmony and can therefore contribute to both sour grapes and use it or lose it. However, because this process is ultimately orthogonal to the analysis of non-myopia itself, since it does not cause or condition the effacement of the trigger's [+Atr] feature, I assume the raised vowel as part of the input in the tableaux that follow.

[18]As seen in the data, Liko also exhibits a regressive harmony process, where [+Atr] spreads on the preceding prefix. In cases with dominance reversal (*e.g.,* /kʊ-√kwo=kɔ/ → [ku-√kwa=kʊ]), this gives the impression of a redocking effect, where [+Atr] shifts from the root to the prefix vowel. The analysis in this chapter focuses exclusively on Liko's progressive harmony and thus does not provide a complete account of the pattern, which would include an account of regressive harmony. This is an important limitation since the current analysis would not be able to account for this pattern. For example, if I take the input /kʊ-√kwo=kɔ/, the relevant candidate set includes the height dissimilation candidate [kʊ-√kwo=kʊ], the dissimilated and partial harmony candidate [ku-√kwo=kʊ], the full dominance reversal [kʊ-√kwa=kʊ] and the redocked candidate [ku-√kwa=kʊ]. As it stands, the current ranking would select the incorrect dominance reversal candidate *[kʊ-√kwa=kʊ], and not the redocked candidate [ku-√kwa=kʊ]. A complete analysis of Liko [+Atr] harmony could potentially address this issue by replacing the sequential markedness constraint with Align-L[−Atr] which would assign violations based on the distance of a [−Atr] autosegment from the left edge of the word. By using this constraint with the ranking Ident[+Atr] >> Ident[−Atr], full rightward [+Atr] harmony is still preferred as a resolution to avoid violations on the harmony-driving constraints, while in cases where full harmony is impossible, partial leftward [−Atr] harmony would be preferable, as it would align the [−Atr] autosegment closer to the left edge of the word. This constraint in combination with a high-ranked constraint such as *Float which would prevent an unassociated [+Atr] autosegment would make the redocked candidate [ku-√kwa=kʊ] optimal since it minimizes violations on Align-L[−Atr] by having [−Atr] closer to the left edge while preserving at least one association line between a segment and [+Atr]. The definition of Align-L[−Atr] also prevents it from modelling partial harmony and, instead, predicts sour grapes, because the sour grape candidate incurs fewer violations by having the [−Atr] autosegment closer to the left edge compared to the partial [+Atr] harmony candidate. The full details of such an analysis are left for future work.

markedness constraint, *=C$_o$[+Hi,+Atr], rather than the faithfulness constraints seen in Komo and Kinande, showing that whether the blocking constraint is a markedness or faithfulness constraint isn't as important as its position in the overall hierarchy. So long as this blocking constraint and the harmony-driver outrank Ident[+Atr], the dominance reversal effect can be modeled.

3.4.3 Sour Grapes

When the trigger is [+Hi], the same blocking configuration that yields dominance reversal instead produces sour grapes, where the [+Atr] vowel surfaces in the output, but no harmony occurs, as shown in (91).

(91) Sour grapes effect (de Wit 2015a:29; 2015b:95)

	UR	SR	Gloss
a.	/...-√ɓin-a=gʊ/	[...-√ɓin-a=gʊ]	'...-dance-FV=NEG'
b.	/...-√kumb-ag-a=gʊ/	[...-√kumb-ag-a=gʊ]	'...-carry-PLUR-FV=NEG'
c.	/...-√dim-is-a=ndɪ/	[...-√dim-is-a=ndɪ]	'...-CAUSE-FV=P'

In (91a-c), the trigger's [+Atr] feature is preserved and no harmony spreads rightward at all, leaving even the vowels that would otherwise be licit targets between the [+Atr] vowel and the clitic high [−Atr] vowel as [−Atr]. This behavior can be captured formally by the high-ranking faithfulness constraint:

(92) Ident[+Atr]/[+Hi]: Assign one violation mark for each output vowel that is [−Atr] whose input correspondent is [+Hi, +Atr].

The sour grapes pattern emerges, then, from the interaction between the positional markedness constraint protecting high [−Atr] clitic vowels (*=C$_o$[+Hi,+Atr]) and the high-ranked faithfulness constraint Ident[+Hi,+Atr], where *=C$_o$[+Hi,+Atr] and Ident[+Hi,+Atr] outrank the harmony-driving constraint *[+Atr]C$_o$[−Atr]

(93). As a result, candidates with harmony (93b,c) or dominance reversal (93a) are eliminated, leaving the faithful candidate (93e) as optimal, resulting in sour grapes.

(93) Input: /...-√kumb-ag-a=gʊ/

/...-√kumb-ag-a=gʊ/	Id[+A]/[+H]	*=C_o[+Hi,+Atr]	*[+Atr]C_o[−Atr]	Id[+A]	Id[−A]	
a. [...-√kʊmb-ɔg-a=gʊ]	*!					*Dominance Reversal*
b. [...-√kumb-og-o=gu]		*!			***	*Full [+Atr] Harmony*
c. [...-√kumb-og-o=gʊ]			*		*!*	*Part. [+Atr] Harmony*
d. [...-√kumb-og-a=gʊ]			*		*!	*Part. [+Atr] Harmony*
☞ e. [...-√kumb-ag-a=gʊ]			*			*Faithful Cand.*

In contrast, when a [−Atr] enclitic is [−Hi], changing its Atr value does not incur violations of *=C_o[+Hi,+Atr], making full harmony possible (94).

(94) Input: /√suku=sɔ/

/√suku=sɔ/	Id[+A]/[+H]	*=C_o[+Hi,+Atr]	*[+Atr]C_o[−Atr]	Id[+A]	Id[−A]	
a. [√suku=sɔ]			*!			*Faithful Cand.*
b. [√sʊkʊ=sɔ]				*!*		*Dominance Reversal*
☞ c. [√suku=so]					*	*[+Atr] Harmony*

Taken together, these results show that a single constraint ranking accounts for both dominance reversal and sour grapes in Liko. The distinction between the two reduces to a ranking relationship between Ident[+Atr]/[+Hi], *=C_o[+Hi,+Atr] and Ident[+Atr]. In dominance reversal, since the [+Atr] vowel is non-high, it is only protected by Ident[+Atr], ranked below the harmony-driver and *=C_o[+Hi,+Atr], thus mandating a repair strategy where violations of Ident[+Atr] are preferred over violations of either *=C_o[+Hi,+Atr] or *[+Atr]C_o[−Atr], making the mapping [+Atr] → [−Atr] optimal. Conversely, since high [+Atr] vowels are protected by the higher ranked constraint Ident[+Hi,+Atr], the ranking prioritizes protecting *both* the [+Atr] vowel and the clitic vowel. Avoiding violations of *[+Atr]C_o[−Atr], then, becomes impossible, and so the evaluation of the remaining candidates falls to Ident[−Atr], resulting in the grammar selecting the faithful candidate since it

incurs the fewest violations of that constraint. In both cases, the blocking constraint is the same, though the grammar will prioritize a different resolution depending on the height of the trigger vowel.

3.5 Conclusion

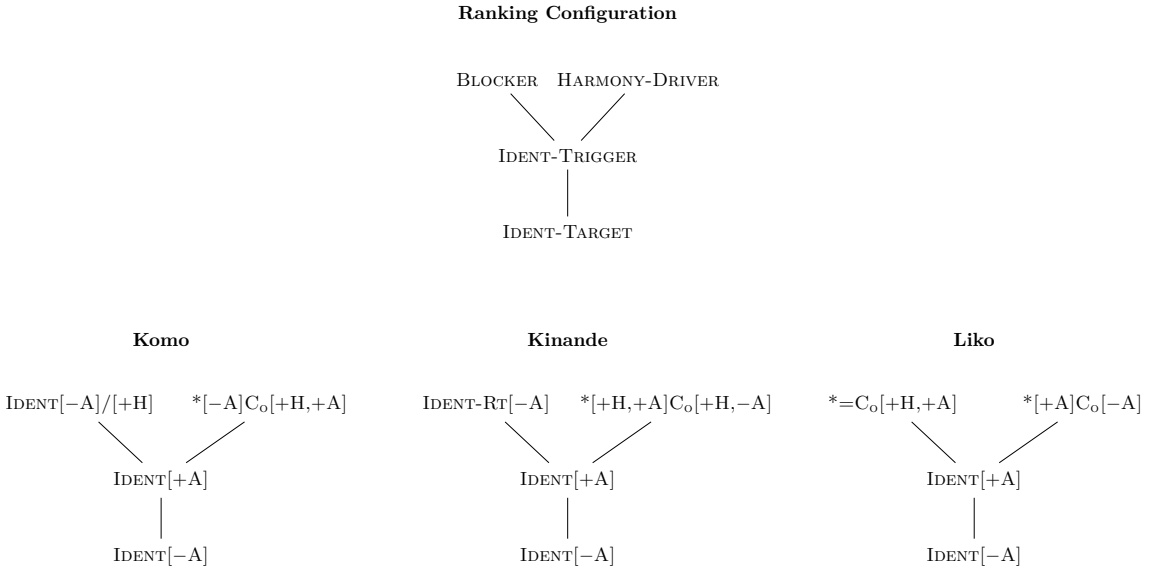
The purpose of this chapter was to determine whether the ranking configuration for dominance reversal identified in Chapter 2 could be empirically instantiated in attested languages. The analyses of Komo, Kinande and Liko provide three independent confirmations that it can be, despite the substantial differences across the three systems.

The three languages were selected to show how robust the ranking was across different morphophonological conditions. Komo offers the most transparent instantiation, with the faithfulness constraint (IDENT[−ATR]/[+HI]) and harmony-driving constraint (*[−ATR]C_o[+HI,+ATR]) ranked above IDENT[+ATR], causing underlying [+ATR] vowels to surface as [−ATR] in a structure containing a high [−ATR] vowel. Kinande further shows that this effect can be extended to the morphological domain, with IDENT-ROOT[−ATR] occupying the position of the blocking constraint, causing prefixes that are [+HI,+ATR] to surface as [−ATR] when preceding a high [−ATR] root vowel. Finally, Liko shows that the blocking constraint can also be a morphophonological markedness constraint, preventing high [+ATR] vowels in clitic positions, thus causing non-high [+ATR] vowels to lose their [+ATR] value in a structure containing a high [−ATR] clitic vowel.

Figure (3.1) summarizes the ranking for each language, comparing it to the abstract configuration from Chapter 2. Across all three languages, the ranking relationship is identical: a blocking constraint and a harmony-driving constraint outrank IDENT[+ATR], which in turn outranks IDENT[−ATR]. Thus, despite significant differences in the nature of their constraints, they exhibit the same ranking hierarchy,

showing that dominance reversal emerges from the ranking relationships rather than any language-specific property.

Figure 3.1: Hasse diagrams comparing the abstract ranking configuration with Komo, Kinande, and Liko. Lines indicate strict domination.



The analysis of Liko additionally shows that the same ranking can also derive what have often been treated as distinct. Sour grapes has typically been analyzed as separate from dominance reversal, often treated as a pathological prediction of certain markedness constraints (*e.g.,* Wilson 2003, 2006). In Liko, however, both effects ultimately emerge from the same ranking, with the choice between them determined by the height of the [+Atr] vowel in the structure under evaluation. A markedness constraint, then, can be said to define a space of possible resolutions, with the interaction between the constraints and the input structure determining which resolution surfaces, a point which is expanded on further in the following chapter.

To conclude, this chapter expanded on the results of Chapter 2, showing that the configuration corresponding to dominance reversal is recoverable from analyses of attested languages, across different families, harmony types, directionalities and domains. Chapter 4 turns from this cross-linguistic validation to intralinguistic

variation, explaining microvariations between the closely related languages Turkana and Ateso.

CHAPTER 4
MICROVARIATIONS

4.1 Introduction

Chapters 2 and 3 established a simple but consequential point: dominance reversal is a resolution that becomes available when a grammar is simultaneously compelled to enforce harmony while prevented from realizing full [+ATR] harmony. In Chapter 2, I identified the specific ranking architecture BLOCKER, HARMONY-DRIVER >> IDENT-TRIGGER >> IDENT-TARGET that produces this pattern, while Chapter 3 validated that configuration across three typologically different systems: Komo, Kinande, and Liko.

This chapter shifts from external to internal. If the ranking is the configuration that makes dominance reversal possible, what does it predict about the space of potential resolutions once harmony is prevented? The core idea is that the ranking that yields dominance reversal does not mandate it. Rather, given a pressure to harmonize and a pressure for some segment to remain invariant, the grammar must choose which part of the structure to sacrifice. The options are limited and interpretable: a grammar can sacrifice full harmony, the blocker's specification or the trigger's specification. And the resolution that is ultimately favored depends on different parameters such as the type of harmony-driving constraint, how the grammar partitions the set of potential triggers, and the strength of the blocker itself.

The empirical focus of the chapter is Turkana (*e.g.,* Dimmendaal 1983; Hulst and Smith 1986; Vago and Leder 1987; Noske 1990, 1996, 2000; Baković 2000; McCollum et al. 2020; Meinhardt et al. 2024) and Ateso (Kitching 1915; Kiggen 1953a, 1953b; Hilders and Lawrance 1956, 1958; Otaala 1981; Noske and Otaala 2006; Barasa 2012,

2015, 2016, 2017, 2018), two closely related Teso-Turkana languages with near-identical harmony systems defined through a bidirectional pattern of [+ATR] harmony with idiosyncratic suffixes that retain their [−ATR] values. I show that three distinct instantiations of dominance reversal emerge as microvariations of the same underlying architecture. Turkana illustrates a "Strong" system in which dominance reversal is the uniform repair whenever spreading is blocked, and a "Medium" system in which dominance reversal coexists with partial harmony. Ateso provides a "Weak" pattern, where dominance reversal coexists with full harmony as resolutions of the same structure.

The theoretical contribution is that once the four constraints forming the basic dominance reversal configuration are fixed, small modifications yield predictable shifts in where dominance reversal applies and what it can coexist with. The ranking, ultimately, defines what is possible, and the surface variation reduces to principled and explainable differences in constraint configuration within the same architecture. §4.2 focuses on Turkana, §4.4 focuses on Ateso, and §4.5 concludes.

4.2 Turkana

Turkana is an Eastern Nilotic language spoken in northwestern Kenya (Dimmendaal 1983) featuring a nine-vowel inventory (/i ɪ u ʊ e ɛ o ɔ a/; *e.g.,* Baković 2000:187). The harmony system is bidirectional, with [+ATR] spreading from any [+ATR] vowel in both directions, targeting all vowels within the prosodic word, with the exception of certain dominant suffixes which resist harmony despite containing phonologically licit targets (Dimmendaal 1983; Noske 2000; Baković 2000). In addition, Turkana also exhibits an independent raising process in which /a/ raises to a mid vowel when immediately following a [+ATR] vowel (Noske 1996, 2000).

This complexity has produced competing analyses in the literature. Baković (2000) describes a system in which all [+ATR] vowels surface as [−ATR] in the presence of

an inalterably recessive suffix (following Baković's (2000) use of the term). Noske (1996, 2000), on the other hand, describes a system in which only mid [+ATR] vowels surface as [−ATR], while high [+ATR] vowels trigger partial [+ATR] harmony. While it is unclear whether these reflect geographically distinct varieties or different stages of documentation, I treat them here as dialects of the same language. The goal of this section is twofold: first, to show that these competing analyses can be unified as dialectal microvariations of a single system, and second, to show that the differences between them emerge from a distinction in the harmony-driving constraint involved in dominance reversal.

The following sections establish each component of the grammar independently.

4.2.1 Basic Harmony

Before examining how Turkana instantiates the dominance reversal ranking established in Chapter 2, I need to first establish the basic harmony pattern which is shared across both varieties described in the literature. This baseline grammar will serve as the foundation for dominance reversal, which will be addressed in subsequent sections.

The data in (95) show the pattern. In (95a,b), all vowels surface as [−ATR] in the absence of [+ATR] vowels. In (95c,d), the presence of a [+ATR] vowel in either suffix position (95d) or root position (95c) causes surrounding vowels to surface as [+ATR].

(95) Bidirectional Harmony (both varieties; Noske 2000:777, 781)

	UR	SR	Gloss
a.	/ŋɪ-√kɔrɪ-a/	[ŋɪ-√kɔrɪ-a]	'ratel.PL'
b.	/a-√gyɛl-ʊn/	[a-√gyɛl-ʊn]	'to buy'
c.	/ŋɪ-√kori-a/	[ŋi-√kory-o]	'leopard.PL'
d.	/ɛ-√gyɛl-ʊn-e/	[e-√gyel-un-e]	'way of buying'

To model this bidirectional pattern, I adopt separate directional ALIGN constraints (Kirchner 1993; McCarthy and Prince 1993), rather than a symmetrical constraint like SPREAD or AGREE. This choice is motivated both empirically and typologically. Empirically, the harmony behavior in Turkana is not perfectly symmetrical with regards to the behavior of the low vowels. Additionally, cross-linguistic typology supports the independence of directional spreading (Hyman 2002), with harmony systems frequently exhibiting asymmetries in scope and blocking, often displaying a bias towards regressive assimilation (Hyman 2002). Separate alignment constraints allow this asymmetry to be derived from the ranking of the harmony drivers themselves. These constraints are defined below.

(96) ALIGN[+ATR], PRWD-L

For every feature [+ATR], assign one violation mark for each segment that separates it from the left edge of the Prosodic Word.

(97) ALIGN[+ATR], PRWD-R

For every feature [+ATR], assign one violation mark for each segment that separates it from the right edge of the Prosodic Word.

For harmony to occur in either direction, the relevant ALIGN constraint must outrank the faithfulness constraint that penalizes changing a [−ATR] vowel to [+ATR]. This establishes a provisional ranking where the harmony drivers and IDENT[+ATR] outrank IDENT[−ATR]. Note that, at this stage of the analysis, the constraints ALIGN-R[+ATR], IDENT[+ATR] and ALIGN-L[+ATR] are not yet ranked with respect to one another.

The tableau in (98) shows how this interaction models a basic case of progressive harmony.

(98) Input: /ŋɪ-√bur-ɪn/

/ŋɪ-√bur-ɪm/	Align-R[+A]	Id[+A]	Align-L[+A]	Id[−A]	
a. [ŋɪ-√bur-ɪm]	*!		*		*Faithful Cand.*
b. [ŋi-√bur-ɪm]	*!			*	*Part. [+Atr] Harmony*
c. [ŋɪ-√bʊr-ɪm]		*!			*Dominance Reversal*
☞ d. [ŋi-√bur-in]				**	*[+Atr] Harmony*

In this tableau, candidates (98a,b) fatally violate the Align constraints, because the feature [+Atr] is not aligned to either edge of the word. Candidate (98c) is ruled out by its violation of Ident[+Atr]. And so, the optimal candidate is (98d), where [+Atr] harmony spreads across the entire word, showing that spreading [+Atr] is preferred in this basic context, thus establishing the basic grammar for Turkana.

4.2.2 Beyond Basic Harmony

The basic grammar established in the previous section predicts total, bidirectional harmony: any [+Atr] vowel, whether in a root, prefix or suffix, should spread its feature to every [−Atr] vowel in the prosodic word.

Turkana, however, exhibits two different blocking effects that prevent full spreading: morphological (§4.2.2.1) and phonological (§4.2.2.2).

4.2.2.1 Morphological Dominance

In Turkana, certain suffixes contain vowels that should undergo harmony but systematically fail to. The data in (99) show that this blocking effect cannot be attributed to vowel quality, as /ɛ/ harmonizes in prefixes (99a), yet the same vowel in the suffix /-ɛt/ resists harmony (99b). This also cannot be attributed to a general property of suffixes, since the high vowel in the suffix /-ʊn/ does undergo harmony without issue (99c).[19]

[19]Looking across the different data sources for Turkana, there is no evidence of an underlying mid suffix vowel that alternates under harmony, other than the raised low vowels. This warrants further research.

(99) Lexical Dominance (both varieties; Noske 2000:777,780)

	UR	SR	Gloss
a.	/ɛ-√risik/	[e-√risik]	'anti-witchcraft.SG'
b.	/a-√tiŋ-ɛt/	[a-√tiŋ-ɛt]	'handle.SG'
c.	/a-√lim-ʊn/	[a-√lim-un]	'to pray this way'

I assume, following Baković (2000), that the blocking effect must be a lexical property of specific morphemes. This type of morphologically specific behavior is relatively well-documented in the literature on vowel harmony. For example, Baković analyzes similar behavior in Kalenjin, Turkana, and Turkish, arguing that certain 'inalterably recessive' morphemes resist harmony due to a high-ranked positional faithfulness constraint he labels REC-IDENT[ATR], which I formalize as follows (100):

(100) REC-IDENT[ATR]: Let β be an output segment belonging to a morpheme marked as [RECESSIVE], and let α be the input correspondent of β. The specification for the feature [ATR] for β must be identical to the specification for the feature [ATR] for α.

While I adopt Baković's assumption that blocking is conditioned morphologically, I depart from his implementation. Baković's model uses a symmetrical AGREE[F] constraint that requires adjacent vowels to match in ATR value, without distinguishing the value. I assume, however, that [+ATR] harmony systems are fundamentally asymmetrical, and thus split REC-IDENT[ATR] into value-specific constraints (101, 102), where the specification REC- restricts evaluation to segments belonging to morphemes marked as $_R$ ([RECESSIVE]), distinguishing it from general IDENT[−ATR] which applies to all segments.

(101) REC-IDENT[−ATR]: Assign one violation mark for each output vowel V_R that is [+ATR] whose input correspondent is [−ATR] and is part of a morpheme

specified as [RECESSIVE].

(102) REC-IDENT[+ATR]: Assign one violation mark for each output vowel V_R that is [−ATR] whose input correspondent is [+ATR] and is part of a morpheme specified as [RECESSIVE].

Of these, REC-IDENT[−ATR] is the relevant constraint for Turkana, since all known recessive morphemes are underlyingly [−ATR]. To derive the blocking effect, this constraint must be ranked above the harmony-driving constraints, resulting in the ranking REC-IDENT[−ATR] >> ALIGN-R[+ATR], IDENT[+ATR], ALIGN-L[+ATR] >> IDENT[−ATR].

The following tableaux demonstrate this interaction by comparing a dominant suffix (103) with a non-dominant suffix (104). Note that in (103), the grammar cannot yet choose between the faithful candidate and the dominance reversal candidate. This is expected, since the point here is to show that REC-IDENT[−ATR] eliminates the harmony candidate only for inalterably recessive suffixes. The resolution of this indeterminacy is the subject of §4.2.3.

(103) Input: /...√bu-ɛt$_R$/

/...√bu-ɛt$_R$/	REC-ID[−A]	ALIGN-R[+A]	ID[+A]	ALIGN-L[+A]	ID[−A]	
a. [...√bu-et$_R$]	*!				*	*[+Atr] Harmony*
b. [...√bu-ɛt$_R$]		*				*Faithful Cand.*
c. [...√bʊ-ɛt$_R$]			*			*Dominance Reversal*

In (103), the recessive suffix /-ɛt$_R$/ is protected by the high-ranked REC-IDENT[−ATR], which prohibits any change to its underlying [−ATR] specification (103a).

By comparison, when a suffix is not marked as recessive, there is no constraint that protects the underlying feature value, as shown in (104). Harmony therefore applies, satisfying the high-ranked alignment constraints (104c).

(104) Input: /a-√buk-ʊn/

/a-√buk-ʊn/	Rec-Id[−A]	Align-R[+A]	Id[+A]	Align-L[+A]	Id[−A]	
a. [a-√buk-ʊn]		*!				*Faithful Cand.*
b. [a-√bʊk-ʊn]			*!			*Dominance Reversal*
☞ c. [a-√buk-un]					*	*[+Atr] Harmony*

Together, these tableaux show that vowels in these inalterably recessive suffixes behave as morphologically conditioned blockers whose opacity cannot be reduced to phonological features.

4.2.2.2 Phonological Blocking

The low vowel /a/ in Turkana interacts with harmony in two distinct ways. First, /a/ acts as a blocker, resisting [+Atr] harmony and consistently surfacing as [−Atr], a behavior which is phonologically predictable. Second, /a/ raises to a mid vowel when it directly follows a [+Atr] vowel, a behavior I show, following Noske (1996, 2000), is independent from vowel harmony proper.

Blocking Effect Unlike the recessive suffixes shown in the previous section, /a/ blocks harmony in all contexts. This behavior is shown in (105), where harmony proceeds bidirectionally to licit targets, except for /a/, which remains [−Atr]. Although this behavior occurs uniformly throughout the language, the clearest evidence comes from sequences where /a/ precedes the [+Atr] vowel, as when /a/ follows a [+Atr] vowel, raising applies and obscures the pattern (§4.2.2.2).

(105) Dominance of /a/ (both varieties; Noske 2000:777—781)

	UR	SR	Gloss
a.	/a-√lim-ʊn/	[a-√lim-un]	'to pray this way'
b.	/ɛ-√ram-e/	[ɛ-√ram-e]	'way of beating'
c.	/a-k-√mɔk-a-kɪn-e/	[a-k-√mɔk-a-kin-e]	'way of lighting'

/a/ acting as a blocker of [+ATR] harmony is quite common in ATR systems (Casali 2008), particularly in languages with nine-vowel inventories where /a/ lacks a direct [+ATR] counterpart (Stewart 1970; Casali 2008), as is the case for Turkana, and can be modeled by the constraint *[+LO,+ATR] defined below:[20]

(106) *[+LO, +ATR]: Assign one violation mark for each segment in the output that is specified as [+LO] and [+ATR].

*[+LO, +ATR] penalizes the presence of [ə] anywhere in the output. The vowel /ə/ is recognized cross-linguistically as highly unstable and prone to diachronic loss within African vowel inventories (Rottland and Otaala 1983; Dimmendaal 2002; Casali 2016). By ranking *[+LO,+ATR] above the ALIGN constraints, any candidate that attempts to satisfy harmony by changing /a/ to [ə] will incur a fatal violation, effectively preventing full harmony as a possible repair strategy for low vowels.

However, while ranking *[+LO,+ATR] above the harmony-driving constraint eliminates the harmonic candidate, it is not sufficient to select the optimal candidate (107), due to the lack of ranking between ALIGN-L[+ATR] and IDENT[+ATR], resulting in the wrong candidate (107d) to be selected.

(107) Input: /a-√gol-ʊn/

/a-√gol-ʊn/	*[+LO,+ATR]	ALIGN-R[+A]	ID[+A]	ALIGN-L[+A]	ID[−A]	
a. [ə-√gol-un]	*!				**	*[+Atr] Harmony*
b. [a-√gol-ʊn]		*		*!		*Faithful Cand.*
c. [a-√gol-un]				*	*!	*Part. [+Atr] Harmony*
☞ d. [a-√gɔl-ʊn]			*			*Dominance Reversal*

The grammar thus requires a ranking between ALIGN-L[+ATR] and IDENT[+ATR] to select the partial harmony candidate (107c). IDENT[+ATR] must outrank ALIGN-

[20]While this constraint is sufficient to capture the Turkana pattern, Ozburn (2019, 2022) shows that low vowels can remain neutral to [+ATR] harmony even in languages where /a/ does have a phonemic [+ATR] counterpart.

L[+ATR], leading the grammar to favor partial harmony and tolerate the resulting disharmony (108).

(108) Input: /a-√gol-ʊn/

/a-√gol-ʊn/	*[+LO,+ATR]	ALIGN-R[+A]	ID[+A]	ALIGN-L[+A]	ID[−A]	
a. [ə-√gol-un]	*!				**	*[+Atr] Harmony*
b. [a-√gol-ʊn]		*!		*		*Faithful Cand.*
c. [a-√gɔl-ʊn]			*!			*Dominance Reversal*
☞ d. [a-√gol-un]				*	*	*Part. [+Atr] Harmony*

In (108), high-ranked *[+LO,+ATR] eliminates the harmonic candidate (108a) and the new ranking correctly chooses the partial harmony candidate (108d) over dominance reversal (108c).

This establishes that /a/ resists [+ATR] harmony due to *[+LO,+ATR]. The low vowel, however, also undergoes an independent raising process when it directly follows a [+ATR] vowel.

Low Vowel Raising When /a/ directly follows a [+ATR] vowel, it surfaces as either [ɔ] (109b) or [o] (109c), with the distinction between the two being that the itive suffix /-ar$_R$/ is an idiosyncratic suffix that never undergoes [+ATR] harmony.

(109) /a/∼[o] alternation (both varieties; Baković 2000:210, Noske 2000:777,780)

	UR	SR	Gloss
a.	/ŋɪ-√kɔrɪ-a/	[ŋɪ-√kɔrɪ-a]	'ratel.PL'
b.	/a-√rip-ar$_R$/	[a-√rip-ɔr$_R$][21]	'to skim off'
c.	/ɛ-√ibus-a-kɪn/	[e-√ibus-o-kin]	'it has fallen down'

This outcome is problematic for any analysis that treats this behavior solely as a property of harmony. If raising /a/ to a mid vowel occurred as a repair strategy to

[21]Note that /a-rip-ar$_R$/ → [a-rip-ɔr$_R$] is specific to the Medium variant. The Strong variant would predict [a-rɪp-ɔr$_R$]; though I have not found this particular form in Dimmendaal's (1983) or Baković's (2000) data. Both variants do, however, exhibit the /a/ → [ɔ] raising.

vowel harmony, then the outcome should necessarily be either [+ATR] [o] or [a] itself in cases where [+ATR] harmony doesn't apply. The appearance of [ɔ] represents a gratuitous change under such an account: both /a/ and [ɔ] are [−ATR], and /a/ is cross-linguistically one of the most stable vowels in any inventory (Casali 2014, 2016). Positing a change from /a/ to [ɔ] in this context lacks phonological justification.

Noske (1996, 2000) accounts for this pattern by positing that raising and harmony are distinct processes, with raising applying first and then feeding harmony, although she does not formalize it in OT.[22].

A single level of evaluation would not successfully account for the combination of raising and dominance reversal because they are in counterbleeding order with each other, as exemplified with /a-√pet-ar$_R$/ → [a-√pɛt-ɔr$_R$]. The context that causes /a/ → [ɔ] is not present at the surface because the presence of the inalterably recessive suffix also causes the [+ATR] vowel that would trigger raising to surface as [−ATR] (see §4.4 for more details on dominance reversal). Counterbleeding effects are notoriously difficult to do in a single parallel evaluation because the opaque candidate [a-√pɛt-ɔr$_R$] contains an unfaithful mapping with no apparent surface motivation, causing a seemingly gratuitous faithfulness violation compared to a more faithful candidate such as *[a-√pɛt-ar$_R$] (see also McCarthy 1999; Baković 2007). In order to account for this interaction, I assume a Stratal OT implementation (*e.g.,* Bermúdez-Otero 1999, 2003; Kiparsky 2000, 2003, 2015) where phonological evaluation is interleaved with morphological structure across a series of lexical domains which conventionally include stem and word strata. Under this implementation, parallel evaluation occurs in steps, where an output of an evaluation at one stratum becomes input to the next. Crucially, each stratum consists of its own independent evaluation system, with different constraints and rankings being possible.

[22]van der Hulst & Smith (1986) also posit raising, though they refer to it as a post-lexical phonetic rule occurring after harmony (Hulst and Smith 1986:269). The raising of /a/, however, *is* phonologically predictable. It also wouldn't explain why /a/ is a target of harmony in some contexts (/ibus-a-kɪn/ → [ibus-o-kin]) and not others (/ibus-a-kɪn-a/ → [ɪbʊs-ɔ-kɪn-a])

Under this account, then, I assume the ranking of *[+ATR]C_o[+LO] (110) >> IDENT[+ATR] >> IDENT[+LO] (111) at the lower stratum in order to avoid violations on *[+ATR]C_o[+LO] by changing the height of a low vowel rather than the ATR value of a [+ATR] vowel. At this lower level, furthermore, I also assume that the harmony-driving constraints ALIGN-R[+ATR] and ALIGN-L[+ATR] are ranked below IDENT[±ATR] to prevent harmony altogether.

(110) *[+ATR]C_o[+LO]: Assign one violation mark for any sequence of two vowels $V_1C_oV_2$ within a single prosodic word, where V_1 is [+ATR] and V_2 is [+LO].

(111) IDENT[+LO]: Assign one violation mark for each output vowel that is [−LO] whose input correspondent is [+LO].

The ranking *[+ATR]C_o[+LO] >> IDENT[+ATR] >> IDENT[+LO] thus causes /a/ to raise to [−ATR] [ɔ] when it follows a [+ATR] vowel (112).

(112) Input: /a-√pet-ar$_R$/

/a-√pet-ar$_R$/	REC-ID[−A]	*[+ATR]C_o[+LO]	ID[+A]	ID[−A]	ID[+LO]	
a. [a-√pet-ər$_R$]	*!	*		*		*[+Atr] Harmony*
b. [a-√pet-or$_R$]	*!			*	*	*Raising and [+Atr] Harmony*
c. [a-√pet-ar$_R$]		*!				*Faithful Cand.*
d. [a-√pɛt-ar$_R$]			*!			*Dominance Reversal*
☞ e. [a-√pet-ɔr$_R$]					*	*Raising*

The winning candidate (112e) *a-√pet-ɔr$_R$* then acts as the input to harmony (113), where the high-ranked REC-IDENT[−ATR] prevents the inalterably recessive suffix from surfacing as [+ATR], though the details of how dominance reversal is modeled in Turkana are developed in §4.4.

(113) Input: /a-√pet-ɔr$_R$/

/a-√pet-ɔr$_R$/	Rec-Id[−A]	*[+Lo,+Atr]	Align-R[+A]	Id[+A]	Align-L[+A]	Id[−A]	
a. [ə-√pet-or$_R$]	*!	*				**	[+Atr] Harmony
b. [a-√pet-or$_R$]	*!				*	*	*Part. Harmony*
c. [a-√pet-ɔr$_R$]			*		*!		*Faithful Cand.*
☞ d. [a-√pɛt-ɔr$_R$]				*			*Dominance Reversal*

Thus separating raising from harmony by using Stratal OT successfully models the relevant mappings.

In forms that contain a low vowel which is not inalterably recessive, however, the interaction between raising and harmony produces a feeding order where /a/ → [ɔ] → [o], as seen in forms like /√ibus-a-kɪn/ → [√ibus-o-kin]. This is shown in (114) and (115).

(114) Input: /√ibus-a-kɪn/

/√ibus-a-kɪn/	Rec-Id[−A]	*[+Atr]C$_o$[+Lo]	Id[+A]	Id[−A]	Id[+Lo]	
a. [√ibus-ə-kin]		*!		**		*[+Atr] Harmony*
b. [√ibus-a-kɪn]		*!				*Faithful Cand.*
c. [√ɪbʊs-a-kɪn]			*!			*Dominance Reversal*
d. [√ibus-o-kin]				*!*	*	*Raising and [+Atr] Harmony*
☞ e. [√ibus-ɔ-kɪn]					*	*Raising*

(115) Input: /√ibus-ɔ-kɪn/

/√ibus-ɔ-kɪn/	Rec-Id[−A]	*[+Lo,+Atr]	Align-R[+A]	Id[+A]	Align-L[+A]	Id[−A]	
a. [√ibus-ɔ-kɪn]			*!*				*Faithful Cand.*
b. [√ɪbʊs-ɔ-kɪn]				*!*			*Dominance Reversal*
☞ c. [√ibus-o-kin]						**	*[+Atr] Harmony*

Under this ranking, candidate (115c) wins because it satisfies the harmony-driving constraint without violating either Rec-Ident[−Atr] or *[+Lo,+Atr], since neither the suffix /-a/ nor /-kɪn/ is dominant, and the blocking vowel /-a/ rose to mid [ɔ] before harmony applied at the higher evaluation and is therefore a licit target by the time harmony applies.

A stratal implementation of the analysis leaving raising at a lower level of the evaluation which then feeds into a higher level where harmony can be modeled is thus capable of modeling the counterbleeding order between raising and dominance reversal. This explains the different surface realizations of the low vowel. When /a/ is not inalterably recessive, the raised [ɔ] is submitted to the higher level and can subsequently harmonize with the [+ATR] value of the trigger and surface as [o]. In cases where /a/ is inalterably recessive, raising does occur, causing the inalterably recessive /a/ to surface as [ɔ] since it does not require a change in ATR value, though it does not harmonize due to high-ranked REC-IDENT[−ATR].

This section has established three mechanisms that interact with the basic harmony from §4.2.1: morphological blocking from inalterably recessive suffixes, phonological blocking from the low vowel /a/, and low vowel raising. While the ranking of *[+LO,+ATR], IDENT[+A] >> ALIGN-L[+ATR] governs leftward partial spreading, the ranking indeterminacy between ALIGN-R[+A] and IDENT[+ATR] governing rightward spreading remains to be determined. The following section shows how ranking these constraints with respect to one another can produce dominance reversal.

4.2.3 Dominance Reversal

Turkana provides two distinct instantiations of this pattern: a Strong and a Medium variant, with data taken from (Dimmendaal 1983; Baković 2000) and (Noske 1990, 1996, 2000) respectively. In the Strong variant, all [+ATR] vowels surface as [−ATR] in the presence of an inalterably recessive suffix to their right, while in the Medium variant, dominance reversal is confined to a specific sequence of a non-high [+ATR] vowel immediately followed by an inalterably recessive non-high [−ATR] vowel. As will be shown below, both patterns instantiate the same ranking developed in Chapter 2, differing only in the harmony-driving constraint. §4.2.3.1 focuses on the Strong pattern, while §4.2.3.2 focuses on the Medium pattern.

4.2.3.1 Strong Dominance Reversal

The analysis begins with the system described in Dimmendaal (1983) and Baković (2000). In this dialect, when progressive harmony is blocked, the [+ATR] vowels lose their feature specification and surface as [−ATR] regardless of their height. This is shown in (116b), where high [+ATR] vowels and mid [+ATR] vowels surface as [−ATR] in the presence of the idiosyncratic suffix /-a$_R$/, contrasting with (116a), where the same root /ibus/ surfaces as [+ATR] and causes following suffixes to harmonize.

(116) Strong dominance reversal (Baković variety; Dimmendaal 1983:285; Baković 2000:210; Meinhardt et al. 2024:1207)

	UR	SR	Gloss
a.	/ɛ-√ibus-a-kɪn-ɪ/	[e-√ibus-o-kin-i]	'3-drop-EPI-DAT-V'
b.	/ɛ-√ibus-a-kɪn-a$_R$/	[ɛ-√ɪbʊs-ɔ-kɪn-a$_R$]	'GER-drop-EPI-DAT-VOI'
c.	/a-√pol-a-ʊn-ɛt$_R$/	[a-√pɔl-ɔ-ʊn-ɛt$_R$]	'growth.SG'

While the analysis so far has established that REC-IDENT[−ATR] must be highly ranked to function as a blocker, this alone is insufficient to derive the correct output. As shown in (117), once the full harmony candidate (117a) is eliminated, the current ranking selects partial harmony (117c) over the faithful candidate (117b), since (117c) incurs fewer violations of ALIGN-R[+ATR]. Dominance reversal (117d) is not selected because ALIGN-R[+ATR] and IDENT[+ATR] are not yet ranked with respect to one another.

(117) Input: /. . .-√ibus-a-kɪn-a$_R$/

/. . .-√ibus-ɔ-kɪn-a$_R$/	REC-ID[−A]	*[+LO,+ATR]	ALIGN-R[+A]	ID[+A]	ID[−A]	
a. [. . .-√ibus-o-kin-ə$_R$]	*!	*			***	*[+Atr] Harmony*
b. [. . .-√ibus-ɔ-kɪn-a$_R$]			***!			*Faithful Cand.*
☞ c. [. . .-√ibus-o-kin-a$_R$]			*		**	*Part. [+Atr] Harmony*
d. [. . .-√ɪbʊs-ɔ-kɪn-a$_R$]				**!		*Dominance Reversal*

To derive dominance reversal, the grammar must prefer violations of IDENT[+ATR] over violations of the harmony driver. By ranking ALIGN-R[+ATR] *above* IDENT[+ATR], the grammar favors dominance reversal over partial harmony, which yields the attested pattern as shown in (118).

(118) Input: /...-√ibus-a-kɪn-a$_R$/.[23]

/...-√ibus-ɔ-kɪn-a$_R$/	REC-ID[−A]	*[+LO,+ATR]	ALIGN-R[+A]	ID[+A]	ID[−A]	
a. [...-√ibus-o-kin-ə$_R$]	*!	*			***	*[+Atr] Harmony*
b. [...-√ibus-ɔ-kɪn-a$_R$]			*!**			*Faithful Cand.*
c. [...-√ibus-o-kin-a$_R$]			*!		**	*Part. [+Atr] Harmony*
☞ d. [...-√ɪbʊs-ɔ-kɪn-a$_R$]				**		*Dominance Reversal*

This Strong variant represents the minimal instantiation of dominance reversal, comparable to the systems seen in Chapter 3: a general harmony-driving constraint applies uniform pressure on all vowels to harmonize, and is ranked alongside REC-IDENT[−ATR] which protects the [−ATR] value of inalterably recessive suffixes. These two constraints are jointly ranked above IDENT[+ATR], resulting in an outcome where *any* [+ATR] vowels surface as [−ATR] in the presence of an inalterably recessive suffix. This ranking will serve as the starting point for the Medium variant that I now turn to, showing how the same structural configuration can yield a different outcome merely by changing the harmony-driving constraint responsible for dominance reversal.

4.2.3.2 Medium Dominance Reversal

In the system described by Noske (1996, 2000), when full harmony is impossible, mid [+ATR] vowels immediately preceding the dominant vowels undergo dominance reversal (119a), while high [+ATR] and non-high vowels preceding non-dominant

[23]The inalterably recessive suffix /-a$_R$/ in candidate (118c) is not eliminated by *[+ATR]C$_o$[+LO] because it immediately follows a [−ATR] vowel underlyingly (thus as the input to raising). The current analysis does create a prediction that an inalterably recessive /a/ would successfully undergo raising to [ɔ] when immediately following a [+ATR] vowel. I am not able to verify such a prediction, however, as I have not seen such a form in the available data and leave its investigation for future research.

vowels trigger partial harmony (119b-d).

(119) Medium Dominance Reversal (Noske variety; Noske 2000:779,780,782)

	UR	SR	Gloss
a.	/a-√lep-ar$_R$/	[a-√lɛp-ɔr$_R$]	'to milk out'
b.	/a-√bu-ɛt$_R$/	[a-√bu-ɛt$_R$]	'swelling.SG'
c.	/a-k-√ido-ʊn-ɛt$_R$/	[a-k-√ido-un-ɛt$_R$]	'birth.SG'
d.	/a-√pol-o-ʊn-ɛt$_R$/	[a-√pol-o-un-ɛt$_R$]	'growth.SG'

The data in (119) show that when a mid [+ATR] vowel like /e/ is immediately followed by an inalterably recessive blocker, it surfaces as [ɛ] (119a), mirroring the Strong variant pattern.[24]

However, when a high [+ATR] vowel like /i/ immediately precedes the blocker (119b), or when the non-high [+ATR] vowel is followed by a [−ATR] vowel that is not dominant (119c), the result is partial harmony. In all cases, the inalterably recessive suffix remains [−ATR].

The ranking for the Strong variant would not accurately model this type of dominance reversal due to the dependence on the feature [±HI]. Because of the ranking ALIGN-R[+ATR] >> IDENT[+ATR], the grammar will prioritize full harmony over preserving the identity of any [+ATR] vowel. Because high [+ATR] vowels never undergo dominance reversal, as (120) shows for /a-k-√ido-ʊn-ɛt$_R$/ and as was shown in (118), this ranking does not accurately model the Medium variant of Turkana.

(120) Input: /a-k-√ido-ʊn-ɛt$_R$/

[24]Across the available Turkana descriptions (*e.g.,* Dimmendaal 1983; Noske 1996, 2000; Baković 2000), all inalterably recessive suffixes appear to contain exclusively [−HI] vowels. This raises the question of whether the lexical specification as [RECESSIVE] is restricted to non-high morphemes, or whether this is an accidental gap in the data. If the restriction is systematic, it may reflect a broader tendency for high vowels to behave differently with respect to morphological specification in ATR systems, parallel to the asymmetry noted for Liko in Chapter 3 (fn. 16). This question warrants further research.

/...k-√ido-ʊn-ɛt$_R$/	Rec-Id[−A]	Align-R[+A]	Id[+A]	Id[−A]	
a. [...k-√ido-un-et$_R$]	*!			***	*[+Atr] Harmony*
b. [...k-√ido-ʊn-ɛt$_R$]		*!*			*Faithful Cand.*
c. [...k-√ido-un-ɛt$_R$]		*!		*	*Part. [+Atr] Harmony*
☞ d. [...k-√ɪdɔ-ʊn-ɛt$_R$]			**		*Dominance Reversal*

To derive the attested partial harmony for high vowels, the grammar must prioritize satisfying Ident[+Atr] over Align-R[+Atr], which preserves the trigger's feature, yielding partial harmony when spreading is blocked as was shown in (118).

However, ranking Ident[+Atr] >> Align-R[+Atr] is not sufficient since it also protects the [+Atr] value of mid vowels, therefore never inducing dominance reversal (121).

(121) Input: /a-√lep-ar$_R$/

/a-√lep-ɔr$_R$/	Rec-Id[−A]	Id[+A]	Align-R[+A]	Id[−A]	
a. [a-√lep-or$_R$]	*!			*	*[+Atr] Harmony*
b. [a-√lɛp-ɔr$_R$]		*!			*Dominance Reversal*
☞ c. [a-√lep-ɔr$_R$]			*		*Faithful Cand.*

This ranking paradox between (120)-(121) occurs because constraints like Align-R[+Atr] cannot distinguish the height of the trigger in a way that would allow the grammar to preserve the [+Atr] value of high vowels, but not mid vowels. This motivates the need for a markedness constraint that selectively penalizes non-high [+Atr] vowels when followed by a [−Hi,−Atr] vowel (122).

(122) *[−Hi,+Atr]C$_o$[−Hi,−Atr]: Assign one violation mark for any sequence of two vowels V$_1$C$_o$V$_2$ within a single prosodic word, where V$_1$ is [−Hi,+Atr] and V$_2$ is [−Hi,−Atr].

This constraint mirrors the *[+Hi,+Atr]C$_o$[+Hi,−Atr] constraint motivated for Kinande in Chapter 3, targeting high vowels. Both are instances of parasitic harmony,

where agreement of one feature is conditioned by another feature (Cole and Trigo 1988; Kaun 1995).

To model dominance reversal, *[−Hi,+Atr]C$_o$[−Hi,−Atr] must outrank Ident[+Atr], alongside Rec-Ident[−Atr], establishing the ranking Rec-Ident[−Atr], *[−Hi,+Atr]C$_o$[−Hi,−Atr] >> Ident[+Atr].

Thus, in cases where a [−Hi,+Atr] vowel directly precedes a [−Hi,−Atr] dominant recessive vowel, the combination of Rec-Ident[−Atr] and *[−Hi,+Atr]C$_o$ [−Hi,−Atr] leaves only dominance reversal as the available repair (123).

(123) Input: /a-√lep-ar$_R$/

/a-√lep-ɔr$_R$/	*[−Hi,+Atr]C$_o$[−Hi,−Atr]	Rec-Id[−Atr]	Id[+A]	Align-R[+A]	Id[−A]	
a. [a-√lep-ɔr$_R$]	*!			*		*Faithful Cand.*
b. [a-√lep-or$_R$]		*!			*	*[+Atr] Harmony*
☞ c. [a-√lɛp-ɔr$_R$]			*			*Dominance Reversal*

And, in the absence of a sequence that would violate *[−Hi,+Atr]C$_o$[−Hi,−Atr] and Rec-Ident[−Atr], the evaluation is passed down to the lower-ranked constraints, where Ident[+Atr] >> Align-R[+Atr] correctly selects partial harmony as the optimal outcome, as is shown below:

(124) Input: /...-√pol-o-ʊn-ɛt$_R$

/...-√pol-o-ʊn-ɛt$_R$/	*[−Hi,+Atr]C$_o$[−Hi,−Atr]	Rec-Id[−Atr]	Id[+A]	Align-R[+A]	Id[−A]	
a. [...-√pol-o-un-et$_R$]		*!			**	*[+Atr] Harmony*
b. [...-√pɔl-ɔ-ʊn-ɛt$_R$]			*!*			*Dominance Reversal*
c. [...-√pol-o-ʊn-ɛt$_R$]				**!		*Faithful Cand.*
☞ d. [...-√pol-o-un-ɛt$_R$]				*	*	*Part. [+Atr] Harmony*

This form also rules out an alternative ranking where no sequential markedness constraint is invoked. This alternative would have a ranking where Ident[+Atr] is demoted *below* Align-R[+Atr] and an additional faithfulness constraint Ident[+Atr]/[+Hi] protecting only [+Hi,+Atr] vowels would be ranked alongside Rec-Id[−Atr] and Align-R[+Atr]. As shown below in (125), this ranking would cause dominance

reversal to be the optimal resolution on any structure that contains both an inalterably recessive suffix and a non-high [+ATR] vowel, leading to the selection of the wrong optimal candidate (125d) *[a-√pɔl-ɔ-ʊn-ɛt$_R$].

(125) Input: /...-√pol-o-ʊn-ɛt$_R$/

/...-√pol-o-ʊn-ɛt$_R$/	ID[+A]/[+H]	REC-ID[−ATR]	ALIGN-R[+A]	ID[+A]	ID[−A]	
a. [...-√pol-o-un-et$_R$]		*!			**	*[+Atr] Harmony*
b. [...-√pol-o-ʊn-ɛt$_R$]			*!*			*Faithful Cand.*
c. [...-√pol-o-un-ɛt$_R$]			*!		*	*Part. [+Atr] Harmony*
☞ d. [...-√pɔl-ɔ-ʊn-ɛt$_R$]				**		*Dominance Reversal*

In (125), no candidates incur any violations of IDENT[+ATR]/[+HI] since none of the [+ATR] vowels are [+HI]. Subsequently, this leaves the evaluation to the ranking of REC-IDENT[−ATR], ALIGN-R[+ATR] >> IDENT[+ATR] which results in dominance reversal for that specific structure. The mapping of /a-√pol-o-ʊn-ɛt$_R$/ → [a-√pol-o-un-ɛt$_R$] thus shows that Noske's data instantiate a different variant of dominance reversal which applies only to sequences of non-high vowels that disagree in ATR value.

Overall, /a-√pol-o-ʊn-ɛt$_R$/ → [a-√pol-o-un-ɛt$_R$] shows that the data from Noske corresponds to a distinct subtype of dominance reversal that applies exclusively to a specific sequence of non-high vowels disagreeing in ATR value due to the harmony-driving constraint *[−HI,+ATR]C$_o$[−HI,−ATR].

4.3 Interim Discussion

The Strong and Medium variants thus share the same ranking architecture for inducing dominance reversal, but differ in the scope of the harmony-driving constraint. This same ranking architecture also underlies the two major prior treatments of Turkana (Baković 2000; Noske 2000) despite their substantial differences in formal assumptions.

Baković (2000) draws on Benua's (1995, 1997) Transderivational Correspondence

Theory, in which output-to-output identity constraints evaluate faithfulness to the stem at each cycle. The analysis treats [+ATR] and [−ATR] harmony as separate processes, where [+ATR] spreads cyclically outward from roots to affixes, while [−ATR] harmony operates "anti-cyclically," proceeding inward when dominant suffixes are present. The constraint REC-IDENT[ATR], as mentioned in §4.2.1, prevents idiosyncratic morphemes from changing [ATR] value, while AGREE[ATR] drives bidirectional harmony. The local conjunction *[−LO,−ATR] &, IO-IDENT[ATR] penalizes underlyingly [+ATR] non-low vowels that surface as [−ATR].

The tableau in (126) illustrates the interaction. Candidate (126a), with [+ATR] harmony, fatally violates REC-IDENT[ATR] by altering the recessive suffix, while candidate (126b) fatally violates AGREE[ATR] due to disharmonic sequences. Candidate (126c) is the optimal candidate, yielding dominance reversal. Baković posits two faithfulness constraints protecting the input's ATR value: IO-IDENT[ATR] and SA-IDENT[ATR]. The main difference between the two is that while IO-IDENT[ATR] licenses every [ATR] feature, SA-IDENT[ATR], on the other hand, compares a segment in an affixed output form with its correspondent in the stem of affixation, requiring them to agree for ATR (Baković 2000:23).

(126) Input: /k-√ido-ʊn-ɛt$_R$/ (Baković 2000:213)

/k-√ido-ʊn-ɛt$_R$/	REC-ID[ATR]	AGREE[ATR]	*[−LO,−ATR] &IO-ID[ATR]	SA-ID[ATR]	IO-ID[ATR]	
a. [kidounet$_R$]	*!				**	*[+Atr] Harmony*
b. [kidounɛt$_R$]		*!			*	*Faithful Cand.*
☞ c. [kɪdɔʊnɛt$_R$]			*	***	**	*Dominance Reversal*

In Baković's analysis, then, REC-IDENT[ATR] functions as the blocking constraint and, alongside AGREE[ATR], outranks the faithfulness constraints that penalize a change in the ATR value of the trigger.

Comparatively, Noske's analysis (1996, 2000) relies on underspecification rather than morphological idiosyncrasies. Target vowels are lexically unspecified for [ATR] and receive their value through spreading (Noske 2000:773). Dominance reversal

effects are derived, then, from the ranking of IDENT-IO suffix[ATR], ALIGN-R[ATR], ALIGN-L[ATR] >> IDENT-IO Root[ATR] (IDENT-TRIGGER) (127).

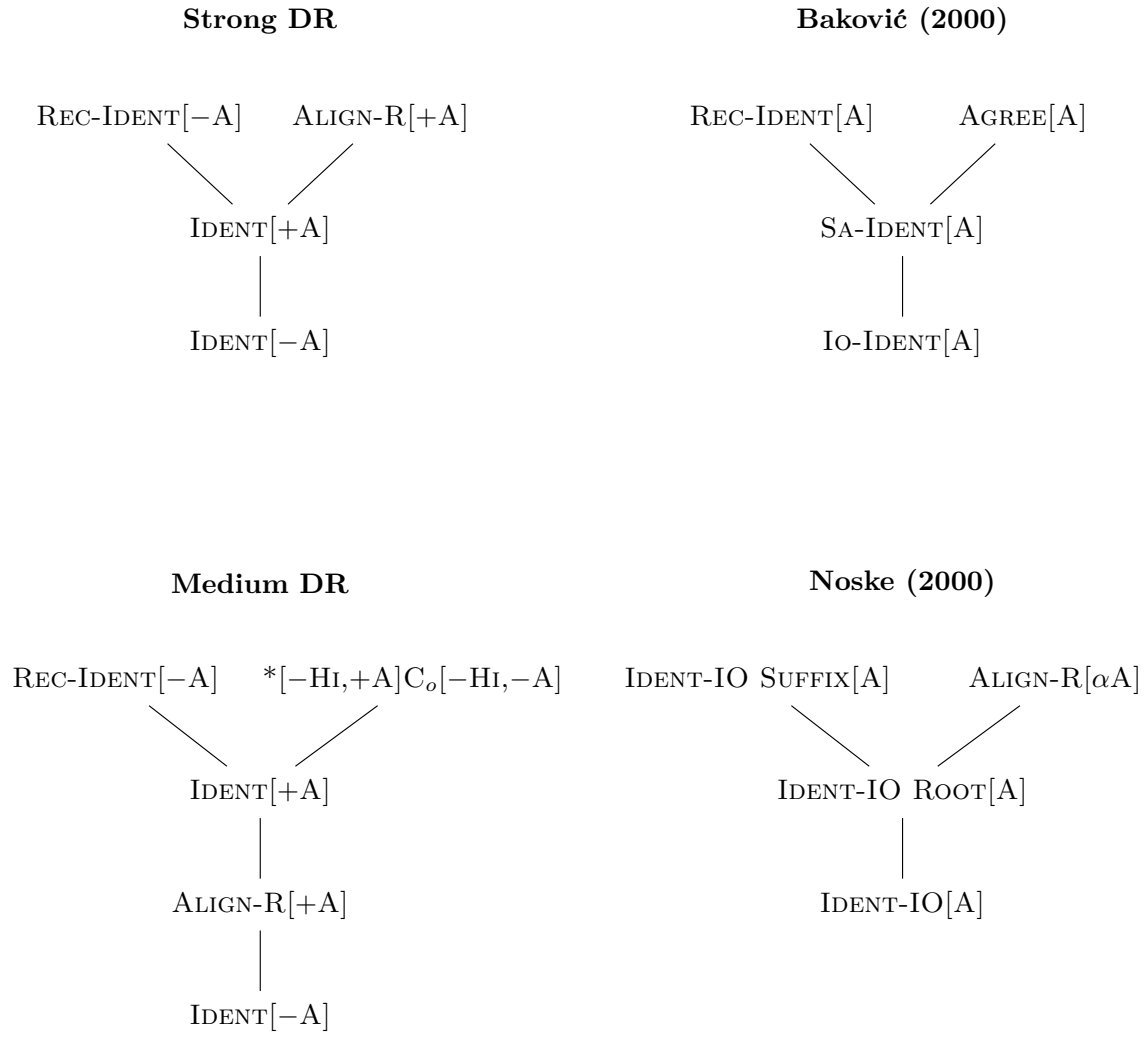
(127) Input: /a-√gol-ɔr/ (Noske 2000:798)

/a-√gol-ɔr/	ID-IOsuff[A]	ALIGN-R[αATR]	ALIGN-L[αATR]	ID-IOrt[A]	ID-IO[A]	
a. [a-√gɔl-or]	*!	*	*	*	**	*Redocking*
b. [ə-√gol-or]	*!				*	*[+Atr] Harmony*
c. [a-√gol-ɔr]		*!	*			*Faithful Cand.*
☞ d. [a-√gɔl-ɔr]				*	*	*Dominance Reversal*

Candidate (127d) wins since it satisfies both the blocking constraint IDENT-IO suffix[ATR] and the harmony-driving constraint ALIGN-R[αATR], resulting in dominance reversal in the presence of a suffix that is specified for [−ATR] in the input.

Thus, Noske's IDENT-IO suffix[ATR] has the same function as Baković's (2000) REC-IDENT[ATR] and the value-specific REC-IDENT[−ATR] that I use in this dissertation. Their position in the ranking protects a subset of ATR vowels, thereby preventing the primary resolution of [+ATR] harmony and instead favoring dominance reversal as the alternative repair strategy. And so, all four analyses arrive at the same organization, where a constraint that prevents full [+ATR] harmony and a harmony-driving constraint dominate IDENT-TRIGGER.

Figure 4.1: Hasse diagrams comparing the Strong and Medium rankings with Baković's (2000) and Noske's (2000) analyses. Lines indicate strict domination.



Thus, across all four analyses, the blocking constraint occupies the same stratum as the harmony-driving constraint or outranks it, leading the grammar to prioritize protecting the [−Atr] value of those vowels above all else. However, it is logically possible to rank the blocking constraint *below* the harmony-driving constraint, which could theoretically make full harmony possible as a resolution for a structure that contains a blocker. The following section instantiates this ranking.

4.4 Ateso

I now turn to Ateso, a Teso-Turkana language closely related to Turkana, exhibiting the same bidirectional [+ATR] harmony system (128b,e), including low vowel raising (128e,f) and morphologically idiosyncratic suffixes (128f).

(128) Ateso Harmony

	UR	SR	Gloss
Turkana (both varieties)			
(Noske 2000:777, 779; Meinhardt et al. 2024:1207)			
a.	/ɛ-√kɔrɪ/	[ɛ-√kɔrɪ]	'RATEL.SG'
b.	/ɛ-√ibus-a-kɪn-ɪ/	[e-√ibus-o-kin-i]	'3-drop-EPI-DAT-V'
c.	/a-√lep-ar$_R$/	[a-√lɛp-ɔr$_R$]	'to milk out'
Ateso			
(Barasa 2018:65,66,68)			
d.	/ɛ-√mɔn/	[ɛ-√mɔn]	'bull'
e.	/ɛ-√duk-akɪn/	[e-√duk-okin]	'(s)he built for'
f.	/a-√pet-ar$_R$/	[a-√pɛt-ɔr$_R$]	'kick away from the speaker'

The first difference between Ateso and Turkana, however, is that in Ateso /a/ does undergo [+ATR] harmony, surfacing as [ə] (129), causing /a/ to have four surface alternants: [a] (129b), [ə] (129a,e), [ɔ] (129c), [o] (129d). I assume that the mid vowels [ɔ] and [o] (129c,d) surface due to the same raising process as in Turkana.[25] The /a/ → [ə] mapping (129a,e), on the other hand, shows that /a/, contrary to Turkana, is a licit target of harmony.

[25]There is one potential root /boja/ → [bojə] in Barasa (2018:63) that appears to be an exception to raising. A possible explanation is that raising does not occur in roots, which can be accounted for by ranking a constraint like IDENT-ROOT[+LO] alongside IDENT[+ATR].

(129) /a/ as a target of harmony (Barasa 2017:34; Barasa 2018:66-68)

	UR	SR	Gloss
a.	/a-√bila/	[ə-√bilə]	'F-shrine'
b.	/a-√kɔkɔ-ar$_R$/	[a-√kɔkɔ-ar$_R$]	'to steal from'
c.	/a-√pet-ar$_R$/	[a-√pɛt-ɔr$_R$]	'kick away from the speaker'
d.	/ɛ-√duk-akɪn/	[e-√duk-okin]	'(s)he built for'
e.	/ɛ-√mɪn-akɪn-o-si/	[e-√min-əkin-o-si]	'they love each other'

Barasa's (2017, 2018) description of the low vowel's behavior is somewhat contradictory. He generally attributes raising to harmony, while also claiming that /a/ is both neutral and transparent to vowel harmony, and that [ə] is a phonologically predictable allophone (Barasa 2018:64). This tension most likely stems from attempting to account for how harmony can simultaneously cause /a/ to surface as both [o] and [ə] since, as discussed in the previous section, attributing the raising of /a/ to [+ATR] harmony is difficult to justify, especially in a context where the [+ATR] counterpart of /a/ is available to the grammar.

The Turkana and Ateso forms [a-lɛp-ɔr$_R$] and [a-pɛt-ɔr$_R$], where /a/ raises to [−ATR] [ɔ], already show this difficulty, as /a/ raising to [−ATR] [ɔ] as a result of [+ATR] harmony represents a gratuitous change under such an account. Ateso presents an additional problem in that /a/ would have two [+ATR] counterparts ([o] and [ə]). If the grammar permits the direct [+ATR] counterpart of /a/, [ə], then the existence of [o] as a harmonic repair is unmotivated.

Ateso, then, provides further support for the separation of raising from harmony. /a/ raises to [ɔ] when directly following a [+ATR] vowel; [o] surfaces when harmony subsequently applies to that raised vowel; and /a/ → [ə] occurs when harmony, but not raising, applies. As shown in (130), modeling this behavior merely requires demoting the constraint *[+LO,+ATR] to allow [ə] as a viable repair strategy (130c).

(130) Input: /ɛ-√mɪn-akɪn-o-si/

/ɛ-√mɪn-akɪn-o-si/	Align-R[+A]	Id[+A]	Align-L[+A]	Id[−A]	*[+Lo,+Atr]	
a. [ɛ-√mɪn-akɪn-ɔ-sɪ]		*!*				*Dominance Reversal*
b. [ɛ-√mɪn-akɪn-o-si]			*!***			*Faithful Cand.*
☞ c. [e-√min-əkin-o-si]				****	*	*[+Atr] Harmony*

And since raising occurs *prior* to harmony, this ranking can also account for /a/ → [o], as [ɔ] can act as an input to the harmony process (131c).

(131) Input: /ɛ-√duk-akɪn/

/ɛ-√duk-ɔkɪn/	Align-R[+A]	Id[+A]	Align-L[+A]	Id[−A]	*[+Lo,+Atr]	
a. [ɛ-√duk-ɔkɪn]	*!*		*			*Faithful Cand.*
b. [ɛ-√dʊk-ɔkɪn]		*!				*Dominance Reversal*
☞ c. [e-√duk-okin]				***		*[+Atr] Harmony*

Thus, by demoting *[+Lo,+Atr] and maintaining low vowel raising as an independent process, Ateso's four-way alternation for the low vowel /a/ is trivially accounted for.

4.4.1 Dominance Reversal

Ateso also exhibits two distinct resolutions for inalterably recessive suffixes (132). In one case, dominance reversal occurs when sequences of non-high [+Atr] vowels directly precede inalterably recessive suffixes (132a). In the other, [+Atr] harmony overtakes the inalterably recessive vowel, yielding full harmony (132b,c).

(132) Dominance Reversal (Barasa 2017:34,163; 2018:63,68)

	UR	SR	Gloss
a.	/a-√pet-ar$_R$/	[a-√pɛt-ɔr$_R$]	'kick away from the speaker'
b.	/a-√buk-ar$_R$/	[ə-√buk-or$_R$]	'pour away from the speaker'
c.	/a-√twan-ar(ɪ)$_R$/	[ə-√twən-ər(i)$_R$]	'to stretch oneself'

Additionally, (132c) does not result in dominance reversal because of the glide /w/. There is evidence in Ateso that glides participate in [+ATR] harmony. This can be seen in mappings like /a-√twan-ar(ɪ)$_R$/ → [ə-√twən-ər(i)$_R$]. In this form, every vowel surfaces as [+ATR]. However, in Ateso, the infinitive prefix and the itive suffix are both underlyingly [−ATR] /a-/ and /-ar(ɪ)/ respectively, making them unable to act as triggers of [+ATR] harmony. Thus, the source of agreement must come from the root. The root, however, cannot contain an underlyingly low [+ATR] vowel /twən/ since it would have resulted in a sequence of [+ATR]C$_o$[+LO,+ATR], a context that would have caused the itive suffix to raise to a mid vowel. The fact that the itive suffix surfaces, instead, with a low [+ATR] vowel [ər(i)] indicates that the preceding low vowel in the root has to be [−ATR]. This would mean that there is no vowel in the form /a-√twan-ar(ɪ)$_R$/ that bears a [+ATR] feature capable of triggering [+ATR] harmony. The logical assumption is, therefore, that the glide /w/ does bear a [+ATR] feature that is capable of spreading to [−ATR] vowels. [26].

Overall, the behavior of dominance reversal in Ateso is similar to the Medium variant of Turkana, targeting only non-high vowels, with the distinction that high [+ATR] vowels trigger full harmony instead of partial harmony. However, while the Medium variant ranking *does* successfully induce dominance reversal when the [+ATR] vowel is [−HI], as in /a-√pet-ar$_R$/ (133), it fails to model full harmony when the [+ATR] vowel is [+HI] (134).

(133) Input: /a-√pet-ar$_R$/

/a-√pet-ɔr$_R$/	*[−H,+A]C$_o$[−H,−A]	REC-ID[−A]	ID[+A]	ALIGN-R[+A]	ALIGN-L[+ATR]	ID[−A]	
a. [a-√pet-ɔr$_R$]	*!			*!	*		*Faithful Cand.*
b. [ə-√pet-or$_R$]		*!				**	*[+Atr] Harmony*
☞ c. [a-√pɛt-ɔr$_R$]			*				*Dominance Reversal*

[26]Although glides triggering [+ATR] harmony is not a common property of ATR systems cross-linguistically, it is a recurring behavior in East African languages. Casali (2008) notes that in several Nilo-Saharan languages semivowels /j/ or /w/ interact with ATR values of adjacent vowels. More broadly, it is generally not surprising for glides to participate in vowel harmony across harmony types given their affinity with high vowels (Hansson 2024:21).

(134) Input: /a-√buk-ar$_R$/

/a-√buk-ɔr$_R$/	*[−H,+A]C$_o$[−H,−A]	Rec-Id[−A]	Id[+A]	Align-R[+A]	Align-L[+Atr]	Id[−A]	
a. [ə-√buk-or$_R$]		*!				**	*[+Atr] Harmony*
b. [a-√bʊk-ɔr$_R$]			*!				*Dominance Reversal*
☞ c. [a-√buk-ɔr$_R$]				*	*		*Faithful Cand.*

The Medium variant ranking thus incorrectly selects the faithful candidate (134c) because Rec-Ident[−Atr] outranks both Ident[+Atr] and Align-R[+Atr], blocking harmony from overriding the suffix. In order to generate the correct output, Ident[+Atr] and Align-R[+Atr] must be reranked above Rec-Ident[−Atr] (135). This change allows full [+Atr] even in structures containing an inalterably recessive suffix, effectively making the inalterably recessive suffixes susceptible to change from [−Atr] to [+Atr] in certain structures. On the other hand, while generalizing the sequential markedness constraint from the Medium variant to *[+Atr]C$_o$[−Atr] such that high vowels can now also be licensed by this constraint would also work, this would cause the grammar to have two harmony-drivers that operate over the same set of vowels, making them redundant with one another.

(135) Input: /a-√buk-ar$_R$/

/a-√buk-ɔr$_R$/	*[−H,+A]C$_o$[−H,−A]	Id[+A]	Align-R[+A]	Rec-Id[−A]	Align-L[+Atr]	Id[−A]	
a. [a-√bʊk-ɔr$_R$]		*!					*Dominance Reversal*
b. [a-√buk-ɔr$_R$]			*!		*		*Faithful Cand.*
☞ c. [ə-√buk-or$_R$]				*		**	*[+Atr] Harmony*

However, in this instance, Rec-Ident[−Atr] is ranked so low that the grammar will simply never instantiate dominance reversal, driving full harmony uniformly even when the trigger is non-high, as seen in (136c).

(136) Input: /a-√pet-ar$_R$/

/a-√pet-ɔr$_R$/	*[−H,+A]C$_o$[−H,−A]	Id[+A]	Align-R[+A]	Rec-Id[−A]	Align-L[+Atr]	Id[−A]	
a. [a-√pet-ɔr$_R$]	*!		*		*		*Faithful Cand.*
b. [a-√pɛt-ɔr$_R$]		*!					*Dominance Reversal*
☞ c. [ə-√pet-or$_R$]				*		*	*[+Atr] Harmony*

And so, while the sequential markedness constraint on the Medium variant of Turkana allows high [+ATR] vowels to remain [+ATR], there is no possible permutation of these constraints that would simultaneously allow both dominance reversal and full harmony. In other words, while the surface outcome between the Medium variant of Turkana and Ateso is similar, the underlying principles are different.

In Turkana, the ranking REC-IDENT[−ATR] >> IDENT[+ATR] >> ALIGN-R[+ATR] creates a situation in which protecting the inalterably recessive suffixes' [−ATR] value through REC-IDENT[−ATR] and protecting input [+ATR] through IDENT[+ATR] are prioritized over [ATR] harmony. This results in partial harmony, which simultaneously avoids violations of the two faithfulness constraints while minimizing violations of ALIGN-R[+ATR]. On the other hand, dominance reversal occurs for the mid vowels because it is licensed by a separate subranking where the pressure to harmonize governed by *[−HI,+ATR]C_o[−HI,−ATR] does not subsume high vowels. This creates a context where, while the strength of the blocker is uniform, the pressure to harmonize isn't.

Ateso's situation is the opposite. The pressure to harmonize is equally uniform, yielding *either* full dominance reversal or full harmony, but the strength of the blocker isn't. Ateso must maintain constant pressure to harmonize by employing the same constraint for both high vowels and mid vowels to allow the grammar to override inalterably recessive suffixes' [−ATR] value where necessary, while allowing dominance reversal in the mid vowels.

The solution involves two modifications. First, REC-IDENT[−ATR] must be demoted below ALIGN-R[+ATR]. Second, an additional faithfulness constraint IDENT[+ATR]/[+HI] must be introduced above REC-IDENT[−ATR].

This creates two subrankings. First, when the [+ATR] vowel is also high, the ranking IDENT[+ATR]/[+HI], ALIGN-R[+ATR] >> REC-IDENT[−ATR] drives full harmony. The high vowel's [+ATR] value is preserved and causes the inalterably recessive suffix to

harmonize (137c). Second, when the [+ATR] vowel is non-high, IDENT[+ATR]/[+HI] is vacuously satisfied. The evaluation falls to the ranking *[−HI,+ATR]C_o[−HI,−ATR], ALIGN-R[+ATR] >> REC-IDENT[−ATR] >> IDENT[+ATR], which yields dominance reversal (138c).

(137) Input: /a-√buk-ar$_R$/

/a-√buk-ɔr$_R$/	*[−H,+A]C$_o$[−H,−A]	ID[+A]/[+H]	ALIGN-R[+A]	REC-ID[−A]	ID[+A]	ALIGN-L[+ATR]	ID[−A]	
a. [a-√bʊk-ɔr$_R$]		*!			*			*Dominance Reversal*
b. [a-√buk-ɔr$_R$]			*!			*		*Faithful Cand.*
☞ c. [ɔ-√buk-or$_R$]				*			**	*[+Atr] Harmony*

(138) Input: /a-√pet-ar$_R$/

/a-√pet-ɔr$_R$/	*[−H,+A]C$_o$[−H,−A]	ID[+A]/[+H]	ALIGN-R[+A]	REC-ID[−A]	ID[+A]	ALIGN-L[+ATR]	ID[−A]	
a. [a-√pet-ɔr$_R$]	*!		*			*		*Faithful Cand.*
b. [ɔ-√pet-or$_R$]				*!			**	*[+Atr] Harmony*
☞ c. [a-√pɛt-ɔr$_R$]					*			*Dominance Reversal*

Additionally, since ALIGN-R[+ATR] and REC-IDENT[−ATR] are now ranked above IDENT[+ATR], the constraint *[−HI, +ATR]C_o[−HI,−ATR] becomes redundant, since the outcome is the same whether or not it dominates IDENT[+ATR]. The Ateso ranking therefore does not require it in a high position, and the tableau in (139) omits it.

(139) Input: /a-√pet-ar$_R$/

/a-√pet-ɔr$_R$/	ID[+A]/[+H]	ALIGN-R[+A]	REC-ID[−A]	ID[+A]	ALIGN-L[+ATR]	ID[−A]	
a. [a-√pet-ɔr$_R$]		*!			*		*Faithful Cand.*
b. [ɔ-√pet-or$_R$]			*!			*	*[+Atr] Harmony*
☞ c. [a-√pɛt-ɔr$_R$]				*			*Dominance Reversal*

And so, Ateso shows that dominance reversal can coexist with full harmony in the same structure by ranking ALIGN-R[+ATR] above REC-IDENT[−ATR] and introducing an additional faithfulness constraint protecting a subset of [+ATR] vowels, ranked above REC-IDENT[−ATR].

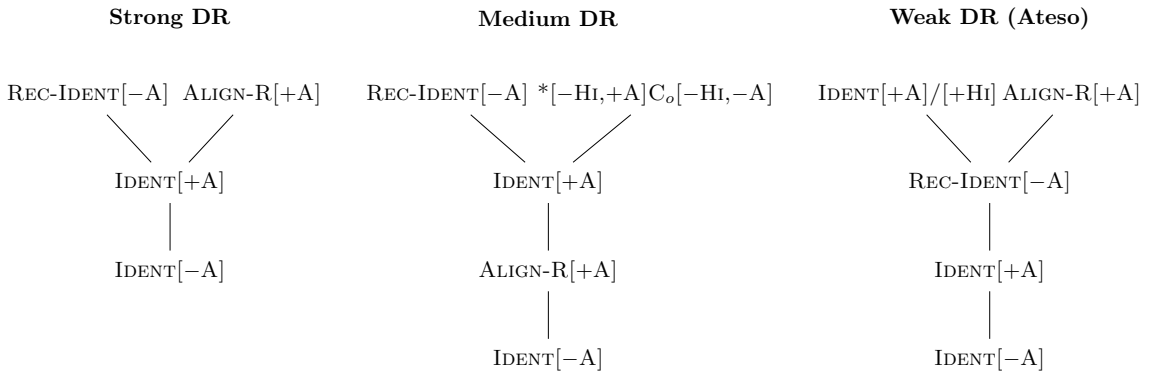
In other words, the structure that allows dominance reversal can coexist with different resolutions that initially appear incompatible depending on minor variations of both the ranking involved and the surrounding constraints, resulting in a hierarchy of subrankings competing for different resolutions to satisfy the same harmony imperative.

4.5 Discussion

This chapter has shown how the Teso-Turkana systems vary in their harmony patterns. These differences stem fundamentally from the constraints added to the basic grammar that models [+ATR] harmony and how those constraints are ranked relative to one another. Each modification thus has traceable consequences for how the grammar resolves structures containing both a [+ATR] vowel and an inalterably recessive suffix. In other words, what might seem like arbitrary cross-linguistic variation follows from three grammatical differences: the identity of the harmony-driving constraint, the position of the blocker relative to other constraints, and any partition on the set of potential triggers.

The three systems examined here correspond to a hierarchy that is defined by the relationship between dominance reversal and the resolution it can coexist with, as shown in (4.2).

Figure 4.2: Hasse diagrams for the three dominance reversal rankings. Lines indicate strict domination.



In the Strong variant, dominance reversal is the uniform resolution, where all [+ATR] vowels surface as [−ATR] in the presence of an inalterably recessive suffix. In the Medium variant, dominance reversal applies to non-high vowels, a subset of triggers, while high [+ATR] vowels retain their [+ATR] specification and yield partial harmony. In the Weak variant (Ateso), dominance reversal again targets non-high vowels, but the high vowels now drive full [+ATR] harmony, overtaking the dominant [−ATR] suffix vowels.

The variation across the three systems occurs due to differences in three parameters: (1) which constraint fills the harmony-driver slot, (2) how the blocker is ranked relative to the harmony-driver, and (3) whether additional constraints partition the set of triggers into distinct subsets.

First, the application space of dominance reversal ultimately depends on how restricted and specific the harmony-driving constraint is. For example, the Strong and Weak variants use a general alignment constraint targeting *all* [+ATR] vowels, whereas the Medium variant uses a sequential markedness constraint that bans a specific sequence of non-high vowels disagreeing in their ATR value.

Second, in most cases, when the blocking constraint is ranked above or alongside the harmony-driving constraint, it creates a situation where that blocking constraint is inviolable and the only available resolutions are those which preserve the blocker's input value. However, if the blocking constraint is ranked *below* the harmony-driving constraint, the grammar can allow changing the blocker's input value as a potential resolution.

And finally, while a general ranking IDENT[+ATR] >> IDENT[−ATR] defines the full set of trigger vowels, more restricted constraints like IDENT[+ATR]/[+HI] can further partition that set, such that certain vowels can be exempt from undergoing dominance reversal, the same way that constraints like REC-IDENT[−ATR] partition the set of [−ATR] vowels such that some are exempt from [+ATR] harmony.

Thus, once the ranking BLOCKER, HARMONY-DRIVER >> IDENT[+ATR] >> IDENT[−ATR] is in place, the internal configuration of the grammar determines not only whether dominance reversal occurs, but which vowel it targets, what resolutions it coexists with, and whether the blocker itself can be overridden. The three Teso-Turkana language varieties examined here represent three configurations of the same architecture, each producing a distinct surface outcome emerging from modifications to the same set of constraints.

To conclude, if Chapter 2 identified the formal conditions that make dominance reversal possible and established the schematic ranking, and if Chapter 3 validated that schema across typologically diverse systems, this chapter instead focuses on microvariation among related languages to show that the grammar that can produce dominance reversal can also instantiate different resolutions depending on the type of harmony-driving constraint and how the constraints are ranked relative to one another.

CHAPTER 5
BEYOND ATR

5.1 Introduction

Dominance reversal has, until now, been defined by its surface description, where [+ATR] spreading is overridden in a limited context by [−ATR] dominance. Chapters 2–4 established that this pattern reduces to a ranking configuration, {BLOCKER, HARMONY-DRIVER} >> IDENT-TRIGGER >> IDENT-TARGET, recoverable across typologically diverse ATR systems and independently proposed analyses. Because the configuration is defined over constraint roles rather than over specific constraints or features, it provides a formal identity condition: any analysis whose ranking maps to this configuration necessarily produces dominance reversal, regardless of the phonological domain. This chapter tests that prediction and explores its consequences.

I begin with two non-ATR systems: Stanton's (2020) analysis of nasal harmony in Gurindji, where she assumes that the trigger loses its [NASAL] specification when spreading is blocked (§5.2.1), and Hyman's (1998) analysis of Yaka's height harmony, where a trigger of vowel lowering raises to a high vowel if the initial vowel of the stem is high (§5.2.2). I show that both rankings map to the dominance reversal configuration, confirming that the pattern is not restricted to ATR harmony. I then turn to Wilson (2011) and Mullin & Pater (2015), both of whom independently derived the same ranking from general properties of constraint interaction in OT, treating the resulting outcome as a pathological prediction and proposing mechanisms to prevent it (§5.3). The configuration shows that what they sought to rule out is dominance reversal. Finally, I show that the configuration maps onto Pater's (1999) analysis of the OshiKwanyama *N̥Ç conspiracy, connecting dominance reversal to the broader

class of conspiracy effects identified by Kisseberth (1970) (§5.4). §5.5 concludes.

5.2 Beyond ATR

In this section, I examine two independently proposed analyses of non-ATR systems, Gurindji nasal harmony (Stanton 2020) and Yaka vowel lowering (Hyman 1998). I show that the ranking for these two systems maps to the configuration that yields dominance reversal, and that both instantiate a conflict between a harmony-driver and a blocking constraint preventing the more general repair, yielding a dominance reversal effect where the trigger's specification is sacrificed. I first examine Gurindji, then Yaka.

5.2.1 Gurindji (Stanton 2020)

Gurindji (Pama-Nyungan; McConvell 1988; Stanton 2020) exhibits a pattern that Stanton (2020) analyses as regressive [NASAL] harmony triggered by coda nasals. Spreading is unbounded through vowels, glides and liquids (140a), but blocked by any [−CONT] segment (*i.e.,* oral stops and nasals), yielding partial harmony when an oral stop (140b) or an onset nasal (140c) intervenes between the trigger and the left edge of the domain. When the blocker is a coda nasal, however, the [NASAL] feature on the rightmost NC is deleted (140d), a process she motivates on perceptual grounds since the contrast between nasal and oral consonants is weakened in coda position (Stanton 2020:165-167; see also Walker (2005) on weak triggers).[27]

(140) Gurindji Dissimilation (Stanton 2020:165)[28]

[27] Stanton acknowledges limited phonetic evidence for the perceptual motivation, though she also argues that alternative accounts fail to capture the full empirical pattern.

[28] Stanton's form (140d) is /wampa⟅umpa/→ [wampa⟅upa]. Following her analysis, however, the output for the input /wampa⟅umpa/ is expected to be [w̃ãmpa⟅upa] and not *[wampa⟅upa], since the coda nasal in the leftmost cluster is not deleted, while *VN]$_\sigma$ (as will be shown below) prevents an oral vowel immediately preceding a coda nasal. Under Stanton's ranking, this would, then, subsequently drive nasalization onto the preceding glide. I therefore assume the nasalization is present but unmarked in (140d).

	UR	SR
a.	/waɭuɾumpa/	[w̃ãl̃ũr̃ũmpa]
b.	/waturumpa/	[watũr̃ũmpa]
c.	/wanuɾumpa/	[wanũr̃ũmpa]
d.	/wampaɭumpa/	[w̃ãmpaɭupa]

To drive [NASAL] spreading, Stanton posits two markedness constraints, *VN]$_\sigma$, which penalizes an oral vowel immediately preceding a coda nasal (Stanton 2020:169), and SPREAD, which penalizes a non-nasal continuant immediately followed by a nasal continuant (Stanton 2020:170), while MAX preserves both links to the nasal autosegment and the segments themselves (Stanton 2020:170-171). Ranking MAX over DEP-LINK[NASAL] drives nasal spreading as the optimal repair (141).

(141) Input: /waɭuɾumpa/ (adapted from Stanton 2020:172)

/waɭuɾumpa/	*VN]$_\sigma$	SPREAD	MAX	DEP	
a. [waɭuɾumpa]	*!				*Faithful Cand.*
b. [waɭuɾũmpa]		*!		*	*Local Nasalization*
c. [waɭuɾupa]			*!		*Denasalization*
☞ d. [w̃ãl̃ũr̃ũmpa]				******	*Full Nasalization*

The tableau in (141) shows nasal harmony, where the harmony-driving constraints *VN]$_\sigma$ and SPREAD eliminate the faithful candidate (141a) and the local nasalization candidate (141b), while MAX protects the [NASAL] feature (141c), yielding full nasalization.

However, when the input contains two NC clusters NC$_1$...NC$_2$ with [−CONT] between them, the nasal in NC$_2$ loses its [NASAL] feature. To account for this pattern, Stanton introduces an additional constraint, *NCṼ, which penalizes a nasalized vowel immediately following a coda nasal (Stanton 2020:171).

Ranking this constraint alongside the harmony-driving constraint, and above MAX, yields denasalization, as shown in (142).

(142) Input: /wampa͡umpa/ (adapted from Stanton 2020:172)

/wampa͡umpa/	*VN]$_\sigma$	*NC$\tilde{\text{V}}$	SPREAD	MAX	DEP	
a. [wampa͡umpa]	*!*					*Faithful Cand.*
b. [wãmpa͡umpa]	*!		*		*	*Part. Nasalization*
c. [w̃ãmpã͡ũmpa]		*!			*****	*Full Nasalization*
d. [wãmpa͡ũmpa]			*!*		**	*Local Nasalization*
☞ e. [w̃ãmpa͡upa]				*	**	*Denasalization*

Thus, full nasalization (142c) now fatally violates high-ranked *NC$\tilde{\text{V}}$, while partial nasalization still violates SPREAD (142d). The winner, then, is (142e), where the rightmost nasal is deleted, despite incurring a violation of MAX.

Stanton's ranking, *NC$\tilde{\text{V}}$, *VN]$_\sigma$, SPREAD >> MAX >> DEP-LINK[NASAL] maps onto the ranking configuration identified throughout this dissertation: {BLOCKER, HARMONY-DRIVER} >> IDENT-TRIGGER >> IDENT-TARGET. *NC$\tilde{\text{V}}$ acts as the blocking constraint preventing the more general repair strategy, while *VN]$_\sigma$ and SPREAD act as harmony-driving constraints imposing a change from the input in order to avoid a marked structure. MAX and DEP, then, act as IDENT-TRIGGER and IDENT-TARGET respectively, creating the asymmetry where spreading nasality is generally seen as preferable as a repair strategy over removing links to a NASAL autosegment. Thus, when no blocking context is present (*i.e.,* no $\text{NC}_1 \ldots \text{NC}_2$ structure), the ranking produces full or partial nasal spreading. When the blocking context *is* present, however, the grammar opts to delete nasality on NC_2 to satisfy both the higher-ranked blocking and harmony-driving constraints, yielding dominance reversal.[29]

[29]Stanton herself connects this pattern to *use it or lose it* (Mullin and Pater 2015), though she has not recognized the connection between use it or lose it and dominance reversal.

5.2.2 Yaka Height Harmony (Hyman 1998)

Yaka (Western Bantu; Eynde 1968; Guthrie 1967/1971; Goldsmith 1985; Hyman 1998) exhibits a pattern of height harmony that Hyman (1998) analyzes as regressive [−Hi] spreading. In Hyman's account, a final mid vowel triggers lowering on preceding vowels (*contra* Goldsmith's (1985) proposal that the pattern involves progressive [Hi] spreading), where high vowels lower to mid when they occur between an initial and a final mid vowel.[30] The data in (143) show the pattern. In (143a,b), where no final mid vowel is present, high vowels in the suffixes surface faithfully. In (143c,d), the combination of the initial mid vowel and the presence of the perfective suffix /-ile/ triggers regressive lowering on the preceding high vowels.

(143) Vowel Lowering in Yaka (Hyman 1998:44)

	UR	SR	Gloss
a.	/√zib-uk-a/	[√zib-uk-a]	'be open'
b.	/√yek-uk-a/	[√yek-uk-a]	'be separated'
c.	/√yek-uk-ile/	[√yek-ok-ele]	'be separated'
d.	/√bet-ilik-ile/	[√bet-elek-ele]	'lower'

To drive regressive harmony, Hyman (1998:49) posits a Plateau (*HM, *LM) constraint, where H M L refer to High, Mid, Low respectively, which penalizes a high vowel immediately preceding a mid vowel, while IdentFV (Hyman 1998:49) preserves the features of the final vowel. Together, these favor a resolution in which preceding vowels lower to match the final mid vowel rather than the final vowel raising to match the preceding one (144).

[30] Yaka also has a low vowel /a/ which interacts with this pattern in complex ways: /a/ in root-initial position triggers a similar effect on the final vowel as high vowels (Hyman 1998:43), but trough /a/, a specific position in the verb stem between CV(V)C and the final vowel (Hyman 1998:62) undergoes M-spreading from the final vowel (Hyman 1998:61), and also surfaces as [e] when coalescing with an infixed /i/ (Hyman 1998:60). Since my purpose here is to show that Hyman's ranking maps onto the dominance reversal configuration, I restrict the discussion to the high/mid interaction.

(144) Input: /√keb-ile/ (Hyman 1998:50)

/√keb-ile/	PLATEAU	IDENTFV	
a. [√keb-ile]	*!		*Faithful Cand.*
b. [√keb-idi]		*!	*Part. [+Hi] Harmony*
c. [√kib-idi]		*!	*[+Hi] Harmony*
☞ d. [√keb-ele]			*[−Hi] Harmony*

In (144), PLATEAU eliminates the faithful candidate (144a), while IDENTFV protects the identity of the final mid vowel (144b,c), thus resulting in the lowering of the suffix vowel /-i/ (144d).

Note, however, that Hyman does not formalize a faithfulness constraint preventing gratuitous changes on high vowels. His analysis necessarily presupposes one, since non-final high vowels surface in contexts that are not penalized by PLATEAU. As shown in (144), the winning candidate (144d) incurs no violations, which means that the lowering of /i/ to [e] appears costless. Without a faithfulness constraint penalizing this change, the grammar has no mechanism to preserve non-final high vowels from converging to some unmarked outputs (see Prince and Smolensky 1993/2004:27 on the Economy Property). We can thus assume some kind of ranking where IDENT[FV] outranks a hypothetical IDENT[+HI] constraint (145).

(145) Input: /√keb-ile/

/√keb-ile/	PLATEAU	IDENTFV	IDENT[+HI]	
a. [√keb-ile]	*!			*Faithful Cand.*
b. [√keb-idi]		*!		*Part. [+Hi] Harmony*
c. [√kib-idi]		*!		*[+Hi] Harmony*
☞ d. [√keb-ele]			*	*[−Hi] Harmony*

Additionally, the data for Yaka show that when the initial vowel is non-mid, such

as in (146b), the final mid vowel raises to a high vowel.

(146) Raising in Yaka (Hyman 1998:44)

	UR	SR	Gloss
a.	/√yel-uk-ile/	[√yel-ok-ele]	'taste'
b.	/√zib-uk-ile/	[√zib-uk-idi][31]	'be open'

Hyman accounts for this behavior with an IDENTV$_1$ constraint, which preserves the identity of the first vowel of the stem, a position of established prominence in Bantu languages (Hyman 1998:49; see also Hyman 1990; Beckman 1997). Ranked above IDENTFV, this constraint prevents lowering of the first vowel as a resolution of violations of PLATEAU. When lowering is blocked by IDENTV$_1$, the combined ranking of IDENTV$_1$ and PLATEAU outranking IDENTFV forces the raising of the final mid vowel (147e) as the optimal resolution, while the ranking IDENTV$_1$ >> PLATEAU is determined by the eliminated candidate [kek-ele] (Hyman 1998:50), since it violates IDENTV$_1$ but incurs no violations on PLATEAU.

(147) Input: /√zib-uk-ile/

/√zib-uk-ile/	IDENT V$_1$	PLATEAU	IDENTFV	IDENT[+HI]	
a. [√zeb-ok-ele]	*!			***	*Full Lowering*
b. [√zib-ok-ele]		*!		**	*Part. [−Hi] Harmony*
c. [√zib-uk-ele]		*!		*	*Part. [−Hi] Harmony*
d. [√zib-uk-ile]		*!			*Faithful Cand.*
☞ e. [√zib-uk-ili]			*		*[+Hi] Harmony*

In (147), full lowering (147a) is eliminated by IDENTV$_1$, while the faithful candidate (147d) and partial lowering (147b,c) both leave violations of PLATEAU unresolved. The winner (147e) raises the final vowel, incurring violations of IDENTFV, but avoiding violations of both the higher-ranked IDENTV$_1$ and PLATEAU.

[31]In the perfective suffix, /l/ becomes [d] when preceding an [i] (Hyman 1998:43)

As with Gurindji, this ranking maps onto the configuration for dominance reversal: {Blocker, Harmony-Driver} >> Ident-Trigger >> Ident-Target, resolving the pressure between the blocking constraint IdentV$_1$ and the harmony-driving constraint Plateau by sacrificing the height of the final vowel /e/ despite incurring violations of IdentFV. IdentV$_1$ acts as the blocking constraint, preventing the more general repair strategy of spreading [−Hi] to avoid violations of Plateau. When the first stem vowel is mid, IdentV$_1$ is vacuously satisfied, and lowering proceeds due to the ranking IdentFV >> Ident[+Hi]. However, in cases where the first stem vowel is non-mid, such as a high vowel, the ranking of IdentV$_1$ and Plateau forces the grammar to sacrifice the final vowel's [−Hi] specification, producing the raising pattern seen in data like /√zib-uk-ile/ → [√zib-uk-idi] in order to satisfy the higher-ranked constraints.

5.2.3 Discussion

In this section, I have shown that the ranking configuration that yields dominance reversal is not restricted to Atr harmony. Stanton's (2020) ranking of Gurindji and Hyman's (1998) ranking of Yaka both instantiate dominance reversal effects, with their rankings mapping to {Blocker, Harmony-Driver} >> Ident-Trigger >> Ident-Target, despite involving different features and different constraints (148).

(148) Comparison to the dominance reversal structure

Language	Blocker		Harmony-Driver		Ident-Trigger		Ident-Target
Gurindji	*NC$\tilde{V}$	,	*VN]$_\sigma$, Spread	>>	Max	>>	Dep
Yaka	IdentV$_1$	>>	Plateau	>>	IdentFV	>>	Ident[+Hi]

Mapping those analyses to the ranking configuration provides an identity condition that a comparison of surface data cannot. If dominance reversal is defined by Blocker, Harmony-Driver >> Ident-Trigger >> Ident-Target, then any ranking that

maps to that configuration necessarily produces dominance reversal. Looking at the rankings of Gurindji and Yaka, both instantiate the necessary ranking that identifies them as dominance reversal systems.

If these patterns all instantiate a single ranking configuration, then they are not independent phenomena. They are, instead, instances of one grammatical resolution that surfaces whenever the structural ingredients (a harmony driver, a blocker, and an asymmetric faithfulness ranking) coexist in the required configuration. A result about this configuration in one system therefore carries over to every other system that instantiates it. Without it, denasalization in Gurindji is a fact about Gurindji, to be explained by Gurindji's constraints and Gurindji's data; height harmony in Yaka is a fact about Yaka; and dominance reversal in ATR harmony is a fact about ATR harmony systems. Each pattern receives its own analysis in its own literature. Focusing on the structural configuration underlying those patterns, however, shifts the question from why any individual language exhibits the pattern, and whether its attestation in specific patterns is meaningful, to how grammars produce it at all. The answer is that the resolution follows from the ranking. With harmony-driving constraints being indifferent to how violations are avoided, it is the ranking of the surrounding constraints that ultimately determines the repair strategy: spreading of the trigger feature when full harmony is available, and sacrificing the trigger's specification when full harmony of the trigger feature isn't available. The configuration also functions as a diagnostic. By reducing dominance reversal to a specific ranking that focuses on the structure and the roles that each constraint occupies, it becomes possible to take independently proposed analyses and unrelated surface facts, and determine whether the ranking instantiates the same pattern, as was done in this section with Gurindji and Yaka.

The following section turns to two cases in which the dominance reversal ranking was independently derived, though treated as a pathology.

5.3 On avoiding reversal

§5.2 showed that dominance reversal is recoverable in independently proposed analyses of non-ATR harmony. This section examines two cases in which the configuration emerges from a different direction. Both Wilson (2011) and Mullin & Pater (2015) derive, from general properties of constraint interaction in OT, a ranking in which a blocked harmony-driving constraint produces deletion of the trigger's marked feature, treating this outcome as pathological and proposing distinct mechanisms to prevent it. Mapping their rankings to the structure that yields dominance reversal, I show that the ranking they derive is the same one that produces dominance reversal.

5.3.1 Nasal Place Deletion (Wilson 2011)

Wilson (2011) shows that, in nasal place assimilation (NPA)with complex segments like [g͡b], ranking a markedness constraint such as AGREE[PLACE] and a blocking constraint such as NOLINK[PLACE], which assigns one violation mark for each association between a PLACE feature and a segment in the output whose input correspondents are not themselves associated (McCarthy 2008:278), above MAX[PLACE] produces deletion of the unshared place feature (149c) rather than total assimilation (149b). This follows from the indifference of AGREE to how violations are avoided (see also Ch. 1). While it correctly identifies two adjacent consonants that do not share a PLACE feature and specifically penalizes that configuration, both spreading and deleting the unshared feature are considered satisfactory resolutions of AGREE; and when both AGREE and NOLINK[Place] (the harmony-driver and the blocking constraint) together outrank MAX[Place], deletion of the complex place feature is considered optimal (149).

(149) Input: /ŋg͡ba/ (Wilson 2011:3)

/ŋg͡ba/	Agree[Place]	NoLink[Place]	Max[Place]	*CompSeg	
a. ŋg͡ba	*!			*	*Faithful Cand.*
b. n̼mg͡ba		*!		**	*Place Assimilation*
☞ c. ŋga			*		*Deletion*

Wilson treats this deletion as both unattested and undesirable, and proposes t-Agree[place] as a replacement. t-Agree[Place] is part of the targeted constraint family (Wilson 2000, 2001), a class of markedness constraints that depart from standard OT markedness constraints in fundamental ways: while standard markedness constraints identify a marked structure and can be satisfied in multiple ways, targeted constraints both identify a marked structure and specify a designated repair, rewarding outputs that undergo the specified change (+1), while penalizing outputs that resolve the violation in any other way (−1). In Wilson's own terms, targeted constraints combine properties of OT constraints and rules by being able to assign violations and participate in strict-domination relations, but also by specifying a change relating inputs to outputs (Wilson 2011:5). Under t-Agree[Place] total NPA wins because spreading is the designated repair and receives a +1 score (150c), while deletion of the Place feature receives −1 (150b), the same score as the faithful candidate (150a).

(150) Input: /ŋg͡ba/ (Wilson 2011:7)

/ŋg͡ba/	t-Agree[Place]	NoLink[Place]	Max[Place]	*CompSeg	
a. [ŋg͡ba]	-1!			*	*Faithful Cand.*
b. [ŋga]	-1!		*		*Deletion*
☞ c. [n̼mg͡ba]	+1	*		**	*Place Assimilation*

Wilson's ranking is t-Agree[Place] >> NoLink[Place] >> Max[Place] >> *CompSeg. Mapping this ranking to the dominance reversal configuration, t-Agree functions as the harmony-driving constraint, while NoLink[Place] functions as the blocker, preventing the more general total assimilation resolution as shown in (151).

(151) Ranking Comparison

Language	Harmony-Driver		Blocker		Ident-Trigger		Ident-Target
NPA	Agree	,	NoLink[Place]	>>	Max[Place]	>>	*CompSeg
Wilson	t-Agree[Place]	>>	NoLink[Place]	>>	Max[Place]	>>	*CompSeg

The critical difference, however, is that even though the ranking is the same, the harmony-driving constraints differ in what they treat as a satisfactory resolution. Chapter 1, specifically, focused on showing how harmony-driving constraints are indifferent to how violations are avoided. So long as full spreading could be achieved, a constraint like Agree did not ultimately distinguish between [+F] or [−F] spreading. t-Agree, however, works in such a way that the constraint penalizes *deletion*, treating it on the same footing as disagreement in the structure, while specifically rewarding spreading of the desired feature, necessarily encoding a feature asymmetry between [+F] and [−F] that is not present in other harmony-driving constraints. Because they specify the desired repair rather than merely identifying the marked structure, targeted constraints thus function more like rules than standard markedness constraints, dictating which resolution the grammar must select rather than leaving the choice to the faithfulness hierarchy. In this context, deletion (or spreading of the target value) is demoted as a potential resolution, becomes less preferred than spreading of the trigger value, making dominance reversal undesirable even under the right configuration.

5.3.2 Use it or lose it (Mullin & Pater 2015)

Mullin & Pater (2015) independently encounter the same deletion outcome, which they term *use it or lose it*, where a marked feature spreads if there is an available target (152a,b), but if spreading is blocked, the feature is deleted (152c).

(152) Use it or lose it (Mullin and Pater 2015:2)

a. /ã-i/ [ãĩ]

b. /ã-waya/ [ãw̃ãỹã]

c. /ã-paya/ [a-paya]

To model this outcome, Mullin & Pater posit SHARE, a Harmonic Serialism (*e.g.,* McCarthy 2000; 2009, 2010) constraint which assigns one violation for every pair of adjacent segments not linked to the same [F] autosegment in the output (Mullin and Pater 2015:6), *[F], which penalizes any surface instance of [F] (Mullin and Pater 2015:8), and IDENT([F]), which penalizes changes to the input specification of [F].[32] Ranking SHARE >> *[F] >> IDENT([F]) thus drives assimilation to the marked feature (153 Step 2.d).

(153) Input: /Tuu/[33] (Mullin and Pater 2015:7)

Step 1 /Tuu/ → [TUu]

/Tuu/	SHARE	*[F]	IDENT([F])	
a. [Tuu]	**!	*		*Faithful Cand.*
b. [tuu]	**!		*	*Use It or Lose It*
☞ c. [TUu]	*	**	*	*Spreading*

Step 2 /TUu/ → [TUU]

/TUu/	SHARE	*[F]	IDENT([F])	
a. [TUu]	*!	**		*Faithful Cand.*
b. [Tuu]	*!*	*	*	*Use It or Lose It*
c. [tUu]	*!*	*	*	*Redocking*
☞ d. [TUU]		***	*	*Spreading*

To derive the blocking effect that induces use it or lose it, Mullin & Pater introduce

[32]Note that Mullin & Pater use a single IDENT([F]) constraint protecting both triggers and targets, collapsing the asymmetry that is necessary for harmony to be properly induced (see Chapter 1).

[33]Note that in this system, T/t stands for "trigger", U/u stands for "undergoer" and B/b is for "blocker" with the capitalization indicating whether the particular segment bears a feature [F] or not.

*B, penalizing [F] on /b/ (Mullin and Pater 2015:8). Ranked above IDENT([F]), *B prevents [F] from spreading to /b/ to avoid violations of SHARE (154a). With [F] spreading unavailable, the ranking of *[F] above IDENT([F]) makes deletion of [F] on the trigger optimal (154c), eliminating the faithful candidate (154b).[34]

(154) Input: /Tbu/ (Mullin and Pater 2015:7)

Step 1 /Tbu/ → [tbu]

/Tbu/	*B	SHARE	*[F]	IDENT([F])	
a. [TBu]	*!	*	**	*	*Spreading*
b. [Tbu]		**	*!		*Faithful Cand.*
☞ c. [tbu]		**[35]		*	*Use It or Lose It*

Like Wilson, they treat this outcome as unattested and pathological, proposing a solution (PARSE) under Containment Theory (Prince & Smolensky 1993/2004), which prevents deletion altogether as a potential resolution and requires all input features to receive phonetic interpretation (Mullin & Pater 2015:3), thus preventing *use it or lose it* as a possible outcome (155).

(155) Input: /Tbu/

Step 1 /Tbu/ → [Tbu]

[34]There is an interesting interaction between *[F], a markedness constraint that prevents any F feature in the output, and IDENT([F]), which protects input [F]. With the ranking used by Mullin & Pater, ranking *[F] above IDENT([F]) has the consequence that avoiding any F segment in the output is preferable to protecting input [F]. Under SHARE this outcome can be avoided since SHARE indirectly preserves [F] due to how its violations are calculated. In any structure, however, where spreading [F] cannot minimize or avoid violations on SHARE, it is expected that the ranking *[F] >> IDENT([F]) would prevent [F] from surfacing anyhow. The simplest solution would be to remove *[F] and split IDENT([F]) into MAX-LINK[F] and DEP-LINK[F].

[35]SHARE[F] assigns a violation for every *pair* of adjacent segments not linked to the same token of [F] (Mullin and Pater 2015:6; see also McCarthy 2009). Under this definition, a candidate in which [F] has been deleted doesn't reduce SHARE violations.

/Tbu/	*B	PARSE	*[F]	IDENT([F])	
a. [TBu]	*!	*	**	*	*Spreading*
b. [tbu]		***!	*	*	*Use It or Lose It*
☞ c. [Tbu]		**	*		*Faithful Cand.*

In (155), candidate (155a), where [F] spreads to the blocker, fatally violates *B. Candidate (155b), on the other hand, incurs three violations of PARSE due to the delinking of the feature [F]. The result is the faithful candidate, where [F] surfaces on the trigger without spreading to the blocker /b/, satisfying *B and partially satisfying PARSE.

Compare this outcome with the SHARE ranking in (154), where the same ranking over the same input /Tbu/ instead selects the *use it or lose it* candidate (154c) [tbu] as the optimal outcome. Under SHARE, the faithful candidate and the *use it or lose it* candidate incur the same number of violations, leaving *[F] to break the tie in favor of deletion. Under PARSE (155), however, violations are attributed to every segment not linked to an [F] feature that cannot be deleted (as a property of Containment Theory). Delinking of the feature, then, incurs more violations than the faithful candidate since the feature is still present, thus preventing *use it or lose it*.

5.3.3 Discussion

Both Wilson (2011) and Mullin & Pater (2015) successfully prevent the deletion outcome from surfacing under their respective systems. Wilson's t-AGREE does so by making deletion as costly as disagreement, ensuring that spreading is always preferred. Mullin & Pater's PARSE does so by making deletion impossible as a property of their formalism. Both of their rankings, however, map onto the ranking that induces dominance reversal (156).

(156) Ranking Comparison

Language	Harmony-Driver		Blocker		Ident-Trigger		Ident-Target
Wilson	t-Agree[Place]	>>	NoLink[Place]	>>	Max[Place]	>>	*CompSeg
Mullin & Pater	Parse	<<	*B	>>	Ident([F])	>>	Ident([F])

As has been established throughout this dissertation, any ranking that maps to the dominance reversal configuration should, necessarily, produce dominance reversal effects, because standard markedness constraints are agnostic to the resolution. A constraint like Agree[Place], for example, penalizes a structure in which two adjacent consonants disagree in Place and is equally satisfied by place assimilation and deletion.

Thus, because Wilson (2011) and Mullin & Pater (2015) change the property that makes markedness constraints indifferent to the resolution, they can avoid modeling dominance reversal even though their rankings map to the ranking architecture that would normally yield it. This, in turns, forces a ranking between the harmony-driver and the blocking constraint. Wilson's t-Agree[Place] encodes the featural asymmetry directly into the markedness constraint, with targeted constraints functioning closer to a rule mandating a specific repair strategy rather than merely identifying a marked structure. Mullin & Pater, by adopting Containment Theory and ranking Parse above the relevant faithfulness constraints, remove deletion as a possible resolution entirely. Under this formalism, input features can never be deleted, thus also mandating spreading of a specific value as a repair strategy since spreading of the other value (or deletion of the feature) would never be able to satisfy the markedness constraint.[36] Thus, while both solutions successfully prevent the outcomes they target, the configuration identified in this dissertation shows that what Wilson and Mullin & Pater treated as pathological predictions of OT are actually instances of dominance reversal, a pattern that is attested cross-linguistically.

[36]This does raise questions for binary feature systems, since if delinking of [F] is formally equivalent to spreading [-F], then this analysis universally rules out spreading of the recessive value, with potentially broad consequences for patterns like vowel raising ([+Lo] to [−Lo]) or glide formation ([+Syl] to [−Syl])

5.4 On Conspiracies

Kisseberth (1970) observed that formally unrelated rules in Yawelmani (now generally known as Yowlumne) (Newman 1944; Kuroda 1967; Kisseberth 1969) conspired toward the same surface goal of eliminating illicit consonant clusters. These rules, which included vowel epenthesis, consonant deletion, and vowel deletion (Kisseberth 1970, *cf.* Kisseberth 1969), differ in their structural descriptions and in the changes they produce, but share a common function of avoiding triconsonantal clusters (Kisseberth 1970:7). Within rule-based phonology, this functional unity presented a formal problem, since the theory lacked a mechanism to link independent rules to a shared surface target. Optimality Theory, however, offered a trivial solution by replacing multiple rules with a single, high-ranked markedness constraint identifying the illicit structure, and leaving the choice of the resolution to be determined by the interaction between that markedness constraint and the rest of the hierarchy. As established in Chapter 1, standard markedness constraints are indifferent to the nature of the repair: different faithfulness rankings yield different surface outcomes, all serving to satisfy the same high-ranked imperative. The traditional concept of a conspiracy thus reduces, in OT, to a high-ranked markedness constraint whose violations can be resolved in multiple ways depending on the language-specific ranking (Kager 1999; McCarthy 2001; see also Heterogeneity of Process (McCarthy 2001).

Pater (1999) extends this logic with his *N$\underset{\circ}{\text{C}}$ constraint, showing that a single markedness constraint can drive multiple repairs, not only cross-linguistically, but also within a single language. I show that Pater's analysis of OshiKwanyama, which features two repairs for *N$\underset{\circ}{\text{C}}$ (post-nasal voicing and nasal substitution), has the same ranking as dominance reversal, connecting dominance reversal to the broader class of conspiracy effects.

5.4.1 *NC̥ (Pater 1999)

Pater (1999) argues that a single markedness constraint, *NC̥, which bans nasal-voiceless obstruent sequences, can be satisfied by a range of formally distinct repairs cross-linguistically. Which repair surfaces in a given language thus depends on which faithfulness constraint is ranked lowest beneath *NC̥, and the indifference of the markedness constraint is what allows for these multiple resolutions cross-linguistically. Pater's analysis of OshiKwanyama (Western Bantu; Steinbergs 1985) extends his argument intra-linguistically. The language tolerates no NC̥ clusters on the surface, exhibiting multiple ways to avoid that structure depending on the morphological domain. I show that Pater's ranking for OshiKwanyama maps to the dominance reversal configuration.[37]

5.4.1.1 OshiKwanyama

As mentioned above, OshiKwanyama is a western Bantu language (Steinbergs 1985) which Pater describes as "demonstrat[ing] a conspiracy between nasal substitution and post-nasal voicing" (Pater 1999:27). While the evidence for post-nasal voicing root-internally comes from the complementary distribution of [k] and [g], with [g] occurring exclusively after nasals within a root and [k] occurring word-initially and intervocalically (Pater 1999:27), loanwords from English (157) confirm that this process is productive, with post-nasal voiceless obstruents systematically surfacing as voiced.

(157) Post-nasal voicing from English Loanwords (Pater 1999:27)

[37]Kager (1999:83-86) also provides an independent and detailed walkthrough of Pater's analysis of OshiKwanyama, explicitly identifying the role of Root-Lin as the constraint that blocks the primary repair (nasal substitution) root-internally, forcing the grammar onto a secondary repair (post-nasal voicing), though neither Kager nor Pater connects it to dominance reversal or to the broader ranking configuration discussed in this dissertation.

	OshiKwanyama	English transl.
a.	[√sitamba]	'stamp'
b.	[√pelenda]	'print'
c.	[√oinga]	'ink'

On the other hand, when a prefix nasal attaches to a root-initial voiceless consonant (/C̥/), nasal substitution occurs to prevent a prohibited NC̥ sequence, a process which Pater analyzes as fusion of the nasal and the voiceless obstruent (158).

(158) Root-initial nasal substitution in OshiKwanyama (Pater 1999:28)

	UR	SR	Gloss
a.	/eːN+√pati/	[eːmati]	'ribs'
b.	/oN+√pote/	[omote]	'good-for-nothing'
c.	/oN+√tana/	[onana]	'calf'

Pater's analysis of the pattern uses four constraints: *NC̥, which bans sequences of a nasal followed by a voiceless obstruent (Pater 1999:4); Linearity, a correspondence constraint demanding that the precedence structure of the input be preserved in the output (Pater 1999:6); Root-Lin, a root-specific version of Linearity which protects the precedence structure within roots (Pater 1999:7); and Ident-O→I[Exp], where [Exp] is an obstruent-specific feature (Pater 1999:19, *cf.* Steriade 1995), ensuring that any voicing present on an output obstruent must be present underlyingly (Pater 1999:19). With the ranking Root-Lin, *NC̥ >> Ident-O→I[Exp] >> Linearity, Pater shows that both root-internal post-nasal voicing (160) and substitution (159) can be modeled within the same grammar.

(159) Input: /N_1#T_2/ (Pater 1999:28)

/N_1#T_2/	Root-Lin	*NC̥	IdentO→I[Exp]	Lin	
a. N_1#T_2		*!			*Faithful Cand.*
b. N_1#D_2			*!		*Post-Nasal Voicing*
☞ c. #N_{12}				*	*Nasal Substitution*

Thus, in (159), when fusion maps N_1+T_2 → #N_{12} (159c), Root-Lin is vacuously satisfied since the precedence relations among root segments remain unchanged, leaving the evaluation to the lower-ranked constraints, where *NC̥ eliminates the faithful candidate (159a), and IdentO→I[Exp] eliminates the voicing candidate (159b), leaving nasal substitution as the optimal repair. However, in root-internal NC̥ sequences, as in (160), fusion would disrupt the root's precedence relations, fatally violating Root-Lin (160a) and leaving post-nasal voicing as the optimal repair strategy (160c).

(160) Input: /N_1T_2/ (Pater 1999:28)

/N_1T_2/	Root-Lin	*NC̥	IdentO→I[Exp]	Lin	
a. N_{12}	*!			*	*Nasal Substitution*
b. N_1T_2		*!			*Faithful Cand.*
☞ c. N_1D_2			*		*Post-Nasal Voicing*

OshiKwanyama exhibits two different repair strategies which depend on the morphological position of the segments involved. Substitution functions as the general repair, where an NC̥ fuses to a nasal with the place of the following consonant outside of a root, while root-internally, this repair strategy is blocked by the high-ranked Root-Lin, resulting in a resolution where post-nasal voicing occurs to avoid violations of *NC̥ at the expense of IdentO→I[Exp].

Comparing, for example, the ranking that yields Komo's dominance reversal (see Chapter 3) with the OshiKwanyama conspiracy shows that the rankings are

structurally identical and map to the configuration established in Chapter 2 (161).[38]

(161) Ranking Comparison

Language	Blocker		Markedness Constraint		Ident-Trigger		Ident-Target
Komo	Ident[+Hi,−Atr]	,	*[−Atr]C$_o$[+Hi,+Atr]	>>	Ident[+Atr]	>>	Ident[−Atr]
OshiKwanyama Conspiracy	Root-Lin	,	*NC̥	>>	IdentO→I[Exp]	>>	Lin.

In Komo, *[−Atr]C$_o$[+Hi,+Atr] drives [+Atr] harmony when the blocking constraint Ident[+Hi,−Atr], which protects high [−Atr] vowels /ɪ ʊ/ from surfacing as [+Atr], is vacuously satisfied. When that constraint is not vacuously satisfied, however, [−Atr] harmony occurs as a compromise between Ident[+Hi,−Atr] and *[−Atr]C$_o$[+Hi,+Atr]. The same logic applies in OshiKwanyama, where *NC̥ drives substitution when Root-Lin is vacuously satisfied and post-nasal voicing when it is not. Dominance reversal, then, can be considered a type of conspiracy effect, where two formally distinct resolutions ([+Atr] and [−Atr] harmony) satisfy the same harmony imperative within the same grammar, with the choice between them conditioned by whether the blocking constraint is active in the structure under evaluation.

5.5 Discussion

The configuration identified in Chapter 2 has functioned throughout this dissertation as a way to compare OT analyses that would otherwise be impossible to compare. By abstracting away from specific constraints, features, and formalisms, I reduce the comparison to the roles each constraint occupies in the ranking and their relationship to one another. In Chapters 3 and 4, this allowed analyses of Komo, Kinande, Liko, Turkana, and Ateso to be unified despite differences in language family, directionality, blocking domain, and formal assumptions, and further allowed prior analyses (Baković

[38]The labels of Ident-Trigger and Ident-Target do not accurately characterize IdentO→I[Exp] and Linearity, since nasal substitution and post-nasal voicing do not involve feature spreading from a trigger to a target. It is more accurate to view the ranking Ident-Trigger >> Ident-Target as a relation between two faithfulness constraints that determines the primary repair strategy.

2000; Noske 2000; Downing and Krämer 2022) to be shown to instantiate the same ranking. Chapter 5 shows that this configuration ultimately extends beyond ATR harmony and beyond harmony proper and is recoverable in Stanton's (2020) analysis of Gurindji nasal harmony, Hyman's (1998) analysis of Yaka height harmony, nasal place deletion and *use it or lose it*. The comparison between the ranking from Pater's (1999) analysis of OshiKwanyama *NC̥ conspiracy and dominance reversal expanded the results of this dissertation to show that the ranking {BLOCKER, MARKEDNESS} >> IDENT-TRIGGER >> IDENT-TARGET corresponds to intra-linguistic conspiracies (see also **bakovic2007**). Consequently, we can consider dominance reversal to be an instance of an intra-linguistic conspiracy.

The OshiKwanyama result, in particular, connects the configuration to the broader class of conspiracy effects. In the OT sense, conspiracies can be considered as cases where a single markedness constraint drives multiple distinct repair strategies, with the specific repair determined by the faithfulness hierarchy (Kager 1999; McCarthy 2001). Cross-linguistically, this requires nothing beyond a high-ranked markedness constraint and a language-specific ranking of some faithfulness constraints. Since different languages can rank different faithfulness constraints lowest, producing different repairs is trivial. Intra-linguistically, however, two repair strategies must coexist within the same fixed ranking, since a language cannot rerank constraints. The position of the markedness constraint alone does not determine which repair surfaces for which structure and ranking it higher only strengthens the imperative to avoid the marked configuration without necessarily changing the outcome.

Thus, for two repairs to coexist within the same ranking, the grammar needs a constraint that is active in some input structures, while vacuously satisfied in others, partitioning the input space into two evaluation paths. Consider, for example, Pater's ranking for OshiKwanyama once again, for example, repeated from (159) and (160). In one path (162), the general subranking applies and determines that substitution is

the least costly repair strategy, since ROOT-LIN is vacuously satisfied.

(162) Input: /N_1#T_2/ (Pater 1999:28)

	/N_1#T_2/	ROOT-LIN	*NC̥	IDENTO→I[EXP]	LIN	
	a. N_1#T_2		*!			*Faithful Cand.*
	b. N_1#D_2			*!		*Post-Nasal Voicing*
☞	c. #N_{12}				*	*Nasal Substitution*

In the other path (163), because ROOT-LIN is not vacuously satisfied, the evaluation falls to a different subranking, producing post-nasal voicing as the optimal strategy instead:

(163) Input: /N_1T_2/ (Pater 1999:28)

	/N_1T_2/	ROOT-LIN	*NC̥	IDENTO→I[EXP]	LIN	
	a. N_{12}	*!			*	*Nasal Substitution*
	b. N_1T_2		*!			*Faithful Cand.*
☞	c. N_1D_2			*		*Post-Nasal Voicing*

While the ranking itself is fixed, different subrankings can become responsible for the evaluation of the optimal candidate depending on the input structure, essentially giving the illusion that one language has two different rankings.

The field has long recognized that OT can handle conspiracies trivially through high-ranked markedness constraints because they are indifferent to the resolution chosen by the grammar. Kager (1999:83-86), in particular, shows explicitly through Pater's analysis of OshiKwanyama how a domain-specific blocking constraint can partition the evaluation between two repairs within a single grammar. The configuration provided in this dissertation extends the empirical reach, showing that it is recoverable not only in OshiKwanyama, but across a multitude of languages and different phonological

phenomena that have not been previously connected. Intra-linguistic conspiracies correspond to the ranking {BLOCKER, MARKEDNESS} >> IDENT-TRIGGER >> IDENT-TARGET, of which the dominance reversal configuration established in Chapter 2 is an instance.[39]

[39]It is interesting to note that Baković (2007:243) defined the following ranking for conspiracies: $M_X, C_{Y/\ldots} >> F_Z >> F_Y$ that maps perfectly to the ranking configuration for dominance reversal, where M_X corresponds to a markedness constraint driving a change from input to output, $F_Z >> F_Y$ represents the relative ranking that penalizes the alternative repair over the primary repair strategy, corresponding to IDENT-TRIGGER >> IDENT-TARGET, while $C_{Y/\ldots}$ corresponds to the blocking constraint that prevents the primary repair.

CHAPTER 6
CONCLUDING THOUGHTS

While the preceding chapters showed that the ranking for dominance reversal was both necessary and sufficient to derive dominance reversal effects across a typologically diverse range of harmony systems, they did not address why grammars should require this particular configuration in the first place.

In other words, why should a grammar that permits both [+ATR] and [−ATR] harmony converge on exactly this ranking rather than any other? I argue that the answer follows from a more general consequence of contradirectional spreading: the grammar must resolve a contradiction whenever opposing values of the same feature spread in overlapping domains.

To begin, there is nothing inherently problematic about a grammar having contradirectional spreading patterns. Consider, for example, a hypothetical language in which [+RND] spreads rightward (164a) while [+NAS] spreads leftward (164b).

(164)

	UR	SR
a.	/o-ii/	[o-uu]
b.	/ii-ẽ/	[ĩĩ-ẽ]

In such a system, while the two patterns are operating in different directions, they ultimately define independent trigger-target relations. If [+NAS] spreads leftward, causing other vowels to surface as [+NAS], the set of triggers for this pattern consists of all the vowels which are [+NAS], while the set of targets consists of all the vowels which are [−NASAL]. These sets are entirely independent of the trigger-target set pertaining to [+RND] spreading. Their application spaces are fully disjoint, meaning

that they can apply together over the same structure trivially (165a), and the same segment can even act as a trigger for *both* processes without issue (165b).

(165)

	UR	SR
a.	/o-ii-ẽ/	[õ-ũũ-õ]
b.	/ee-õ-ii/	[ẽẽ-õ-uu]

In other words, a vowel that undergoes rounding harmony does not, by virtue of becoming rounded, lose the property that would allow it to trigger or undergo nasal harmony. The same segment can stand in different relations to the two patterns at the same time, allowing it to be a trigger of one, but a target of the other; or a target for both; or a trigger for both, without any contradiction.

Consider, however, a structure such as /ɛ-i/ in a language where [+ATR] harmony spreads leftward and [−ATR] harmony spreads rightward. In this case, /i/ is a trigger for leftward [+ATR] harmony, and /ɛ/ is a trigger for rightward [−ATR] harmony. Simultaneously, *both* of these vowels are also targets for the other since, unlike the rounding/nasal case above, these relations are defined over the same featural dimension.

The result is that the application space of these patterns fully overlap. If [+ATR] harmony applies, /ɛ/ surfaces as [e], no longer bearing the [−ATR] value that would allow it to trigger rightward [−ATR] harmony (166a). Conversely, if [−ATR] harmony applies, /i/ surfaces as [ɪ], no longer bearing the [+ATR] value that would allow it to trigger leftward [+ATR] harmony (166b). In both cases, the application of one removes the conditions required by the other.

(166)

	UR	SR
a.	/ɛ-i/	[e-i]
b.	/ɛ-i/	[ɛ-ɪ]

This follows from the partitioning of the feature space itself. For any binary feature [F], the set of segments that may trigger [+F] harmony is also the set of segments that may serve as targets for [−F] harmony, and the set of segments that may trigger [−F] harmony is likewise the set of segments that may serve as targets for [+F] harmony. In such a system, the trigger set of one pattern and the target set of the other are necessarily defined in relation to each other.[40]

The result is that the grammar faces a contradiction that does not arise with different features. If we take our previous rounding/nasal case, a segment can surface as both rounded and nasal because [+Rnd] and [+Nas] are independent dimensions of representation. A segment cannot, however, surface as both [+Atr] and [−Atr] in the same featural dimension. If both harmonies make demands on the same segment for the same feature, the grammar must determine which demand is satisfied, and unless something partitions the trigger set of one pattern from the target set of the other, only one value can survive as the output value.

The problem, then, isn't just that two processes spread in overlapping directions, but that opposing values of the same feature define overlapping trigger-target relations over the same domain. Once this overlap is present, the grammar cannot realize both patterns by treating them as independent, and must either allow one process to bleed the other, yielding a single harmony system uniformly across the language, or introduce some way of partitioning the space so that the two patterns apply under different conditions. This can be done in a few different ways. For example, by using underspecification and therefore introducing a ternary contrast in the representation between [+F], [−F] and [∅F], we can split the overlapping trigger-target relation by having, say, underspecified segments behave as targets of [±F] spreading. Other ways,

[40]It is important to mention that the relevant property here is not the use of binary [±F] features per se, but the fact that the feature partitions the segment inventory into exactly two complementary sets, regardless of whether those two sets are labelled [+F]/[−*F*] or [F]/∅. A privative system in which segments are either specified or unspecified for [F] produces the same partition: the set of segments bearing [F] is the trigger set for one harmony pattern and the target set for the other, and vice versa.

such as lexical specification or morphological position, can also be used to introduce a partition by creating a subset of [±F] segments which can act as blockers based on some kind of idiosyncratic property (as seen in the Teso-Turkana family) or by appealing to their morphological position (as seen in Kinande).

In OT, the default resolution of this overlap is determined by the faithfulness hierarchy. As discussed in Chapter 1, harmony-driving constraints do not, on their own, determine which repair is prioritized. A constraint like AGREE[ATR] penalizes disagreement between adjacent vowels for the feature ATR, but is satisfied equally well by a candidate in which all relevant vowels surface as [+ATR] and by one in which they all surface as [−ATR]. Thus, while the markedness constraint creates the pressure for uniformity, it does not determine, on its own, the value of that uniformity.

The choice between the two outcomes is determined, as previously established, by the ranking between faithfulness constraints protecting the two values of the feature. If IDENT[+F] >> IDENT[−F], then changing an input [+F] segment is considered more costly by the grammar than changing an input [−F] segment, and the grammar resolves the pressure imposed by the markedness constraint in favor of [+F] spreading. If the ranking is reversed, then the grammar resolves the same pressure in favor of [−F] spreading. In either case, the result is a uniform outcome across the entire language where one value is preserved at the expense of the other. Once harmony is defined over opposite values of the same featural dimension, the grammar cannot treat the two spreading processes as independent. They are competing over the same representational space. Thus, a single ranking cannot simultaneously privilege [+F] in some structures and [−F] in others if the only dimension available to the grammar is [F]. To keep both [±F] harmony as potential resolutions, the grammar must partition the application space over a dimension that is independent of the feature entirely, because the feature itself offers no basis for differentiation: any ranking between IDENT[+F] and IDENT[−F] applies uniformly across all structures and across the full

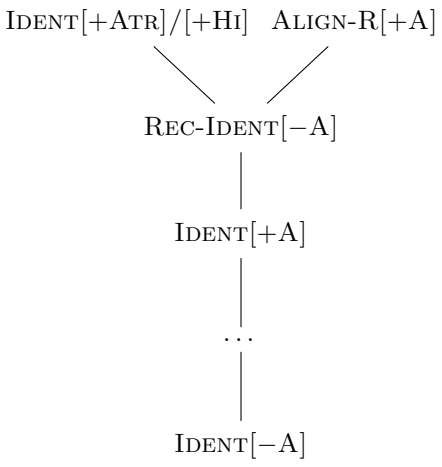
set of segments involved, and so the only way to license a different resolution in a subset of structures is to condition that subset on something outside [F].

The configuration that produces dominance reversal is, then, the shape this resolution takes in OT: the grammar partitions the application space through a constraint whose conditions are independent of [F], and the ranking of that constraint relative to the faithfulness hierarchy determines what resolution is licensed within the partitioned subspace.

The term "blocking constraint" is, in this sense, somewhat misleading. Throughout the preceding chapters, constraints like Rec-Ident[−Atr] and Ident[−Atr]/[+Hi] were defined as "blocking" constraints, implicitly treating the structure they govern as blockers. What these constraints actually do in the configuration, however, is carve out a subspace in which the general repair governed by Ident-Trigger >> Ident-Target isn't available, opening that subspace to an alternative resolution, regardless of the constraint's identity itself.

The ranking for Ateso established in Chapter 4 (6.1) is particularly revealing in this regard, offering three different resolutions in the same structural context to resolve violations of Align-R[+Atr]. At the broadest level, Align-R[+Atr] >> Ident[+Atr] >> Ident[−Atr] establishes a general resolution: across all vowels between a [+Atr] vowel and the right edge of the prosodic word, [Atr] disagreement resolves toward [+Atr]. Thus, any conflict between a [+Atr] and a [−Atr] vowel in this domain is repaired by changing the [−Atr] vowel, because the grammar treats preserving [+Atr] as more costly to violate than preserving [−Atr]. Within a subset structures, how-

Figure 6.1: Full Ateso ranking



ever, the ranking of ALIGN-R[+ATR] >> REC-IDENT[−ATR] >> IDENT[+ATR] >> IDENT[−ATR] narrows the resolution. While the pressure to avoid [ATR] disagreement remains the same, with ALIGN-R[+ATR] ranked highest, the ranking of REC-IDENT[−ATR] >> IDENT[+ATR] ensures that, when there is an [ATR] disagreement involving a [+ATR] vowel and a morphologically idiosyncratic [−ATR] suffix, the grammar will now treat changing that suffix's [−ATR] value as more costly than changing the [+ATR] to [−ATR] as a way to avoid violations of ALIGN-R[+ATR]. The purpose of REC-IDENT[−ATR] is to carve out a subspace along the morphological dimension, in which the general resolution toward [+ATR] is no longer available, and in which the grammar resolves the same pressure towards [−ATR] instead, yielding dominance reversal. Crucially, the partition that REC-IDENT[−ATR] introduces references the morphological class membership of a lexical item, a property that is entirely independent of the featural dimension (*i.e.,* ATR) generating the conflict. Additionally, the ranking of IDENT[+ATR]/[+HI] >> REC-IDENT[−ATR] narrows the resolution space a second time, specifying that in a structure that disagrees in ATR (ALIGN-R[+ATR]) and which contains a morphologically idiosyncratic [−ATR] suffix (REC-IDENT[−ATR]), *if* the [+ATR] vowel is [+HI], then changing the [+ATR] value of that vowel is now more costly than overriding the suffix's [−ATR] specification, creating an additional phonological dimension over the feature HI, which causes the grammar to now treat high and non-high [+ATR] vowels differently.

The result, then, is three nested domains of resolutio: the general domain favoring [+ATR] contains the more restricted morphological domain favoring [−ATR], which in turn contains the still narrower phonological domain favoring [+ATR] when the trigger is high.

Thus, once individual constraints are abstracted to their roles and the diagnostic is situated at the level of the relationships between those roles, it becomes possible to distinguish between two fundamentally different situations: (1) analyses that are

distinct implementations of the same grammatical generalization and (2) phonological patterns that appear unrelated at the descriptive level but share the same grammatical architecture. This configuration, then, functions as both a definition and a diagnostic, making it possible to determine whether an analysis instantiates a specific pattern regardless of its descriptive or analytic details. This makes the question of whether different languages share a grammatical identity traceable. The method itself is reproducible since, given a constraint set and an input space, the full typological space can be generated; the rankings that yield a specific outcome can be isolated, and those rankings can be compared for a shared relational structure. Whether the same kind of architectural identity can be recovered for stress assignment, tonal patterns, or segmental processes more broadly is an empirical question, and one that this dissertation does not attempt to answer, but makes concrete enough to pursue.

Furthermore, the ranking architecture discussed in this dissertation is internal to OT. The relational structure over constraint roles (*i.e.,* Blocker, Harmony-Driver, Ident-Trigger, Ident-Target) is defined over elements that exist only within a constraint-based grammar, with no obvious equivalent in a rule-based framework, where the grammar consists of ordered operations rather than ranked constraints. Whether an equivalent structural object exists in such frameworks, something that would allow the same kind of abstraction over individual rules and their ordering relations, remains to be determined, though preliminary work in different frameworks such as Logical Phonology (*e.g.,* Bale and Reiss 2018; Volenec and Reiss 2020; Reiss 2021) suggests that it likely does (Leduc 2026).

A related question concerns the relationship between the ranking architecture and the computational classification of phonological patterns. The subregular hierarchy (*e.g.,* Heinz 2011a, 2011b; Heinz and Lai 2013; Jardine 2016) classifies phonological patterns by the computational properties of the functions that map inputs to outputs, situating each pattern within a hierarchy of formal language classes that differ in

expressive power. The ranking architecture identified in this dissertation classifies grammars by their constraint relations, independently of which constraints instantiate those relations. Both abstract away from the surface description of a pattern toward a structural property that is invariant across implementations, and in this sense, they are similar. Whether there is a formal relationship between the two, or whether they are capturing independent properties of the same system, is an open question, and one that I leave for future work.

6.1 Conclusion

This dissertation has argued that dominance reversal in ATR harmony corresponds to a single ranking configuration in Optimality Theory: {BLOCKER, HARMONY-DRIVER} >> IDENT-TRIGGER >> IDENT-TARGET. The configuration was identified through a factorial typology over a minimal ATR system consisting of nine constraints, nine inputs, and fifty input-output pairs, computed over all $362,880$ ranking permutations. Of the twenty-six distinct languages generated by the typology, exactly seven exhibit dominance reversal, all sharing the same ranking: a blocking constraint and a harmony-driving constraint jointly dominate IDENT-TRIGGER, which in turn dominates IDENT-TARGET.

This configuration was then tested against five typologically diverse ATR harmony systems. First, Komo, Kinande and Liko share little beyond the surface generalization that [+ATR] triggers can surface as [−ATR] in restricted contexts, belonging to different language families and exhibiting different phonological properties in their respective vowel harmony pattern. Despite this variation, the ranking from the factorial typology is recoverable from all three, with the differences between them reducing to differences in the blocking condition and in the vowels involved in the harmony-driving constraint. Second, Turkana and Ateso shifted the question from whether the configuration is cross-linguistically robust to what it predicts about the space of possible resolutions

once harmony is blocked. The ranking that yields dominance reversal does not inherently force the grammar to that outcome. Given a pressure to harmonize and a pressure that some segment remain invariant, the grammar must choose which part of the structure to sacrifice, and the resolution that surfaces ultimately depends on the type of harmony-driving constraint, the way the grammar partitions the set of potential triggers, and the ranking of the blocker. The three systems examined across these two closely related languages correspond to a typology defined by the relationship between dominance reversal and the resolution it can coexist with, such as partial harmony or even full harmony overriding the blocker's [−ATR] value. These differences reduce to independently motivated changes in constraint ranking within the same architecture.

The configuration was further shown to extend beyond ATR harmony. Stanton's analysis of nasal harmony in Gurindji and Hyman's analysis of height harmony in Yaka both map to the dominance reversal configuration, confirming that the pattern is a property of the ranking rather than of the feature. Wilson (2011) and Mullin & Pater (2015) independently derived the same ranking from general properties of constraint interaction in OT, treating the resulting outcome as a pathological prediction and proposing formal mechanisms to prevent it. However, I showed that what they sought to rule out is dominance reversal, an attested pattern. Finally, Pater's analysis of the OshiKwanyama *N̥Ç conspiracy was shown to instantiate the same ranking, connecting dominance reversal to the broader class of conspiracy effects originally identified by Kisseberth. In that system, two formally distinct repairs, nasal substitution and post-nasal voicing, coexist within a single grammar, with the choice between them conditioned by whether the blocking constraint ROOT-LIN is active in the structure under evaluation, thereby instantiating the same logic that produces dominance reversal in ATR systems.

The results, then, establish that dominance reversal is a grammatical resolution,

defined by a relational structure over its constraints, that arises whenever a grammar must resolve competing and seemingly contradictory pressures over the same application space.

At the level of phonological theory, the ranking architecture provides a way to cut across formalisms and representational assumptions within OT, offering a tractable way to compare analyses and to unify grammars into objects defined over their relational structure rather than over the identity of individual constraints or over their individual phonological properties. At the computational level, there is a potential relationship between the ranking architecture provided in this dissertation and the algebraic classification of computational objects, where identity is likewise defined over the structure of operations, raising the possibility that abstract ranking structures could be classified computationally without requiring each grammar to be individually compiled into a transducer. Lastly, at the empirical level, this dissertation has unified phonological patterns across different features, language families, and phonological domains, cutting across the language-specific machinery that would otherwise make such comparison difficult to conduct.

Appendices

APPENDIX A

COMPLETE INPUT-CANDIDATE SET

Set	Input	Candidates
1	/ɛɛiɛɛ/	[ɛɛiɛɛ], [eeiee], [ɛɛɪɛɛ], [eeiɛɛ], [ɛɛiee]
2	/aɛiɛa/	[aɛiɛa], [aeiea], [əeiəə], [əeiɛa], [aɛiea], [əeiea], [aeiəə], [aɛieə], [aɛɪɛa], [aɛiea], [aeiɛa]
3	/ɛaiaɛ/	[ɛaiaɛ], [ɛəiəɛ], [eəiəe], [eəiaɛ], [ɛaiəe], [ɛaɪaɛ]
4	/aɛiɛɛ/	[aɛiɛɛ], [aeiee], [əeiee], [aɛiee], [aɛɪɛɛ], [aeiɛɛ], [əeiɛɛ]
5	/ɛɛiɛa/	[ɛɛiɛa], [eeiea], [eeiəə], [eeiɛa], [ɛɛɪɛa], [ɛɛiea], [ɛɛiəə]
6	/iɛɛɛɛ/	[iɛɛɛɛ], [ieeee], [ɪɛɛɛɛ]
7	/ɛɛɛɛi/	[ɛɛɛɛi], [eeeei], [ɛɛɛɛɪ]
8	/aɛɛɛi/	[aɛɛɛi], [aeeei], [əeeei], [aɛɛɛɪ]
9	/iɛɛɛa/	[iɛɛɛa], [ieeea], [ieeeə], [ɪɛɛɛa]

APPENDIX B

FULL TYPOLOGICAL OUTPUT

Languages / Inputs :	ɛɛiɛɛ	aɛiaɛ	ɛaiaɛ	aɛiɛɛ	ɛɛiɛa	iɛɛɛɛ	ɛɛɛɛi	aɛɛɛi	iɛɛɛa
L#1	ɛɛiɛɛ	aɛiaɛ	ɛaiaɛ	aɛiɛɛ	ɛɛiɛa	iɛɛɛɛ	ɛɛɛɛi	aɛɛɛi	iɛɛɛa
L#2	eeiee	aɛiɛa	ɛaiaɛ	aɛiee	eeiɛa	ieeee	eeeei	aɛɛɛi	iɛɛɛa
L#3	eeiee	aeiea	ɛaiaɛ	aeiee	eeiea	ieeee	eeeei	aeeei	ieeea
L#4	eeiee	əeieə	eəiəe	əeiee	eeieə	ieeee	eeeei	əeeei	ieeeə
L#5	eeiee	əeiɛa	eəiaɛ	əeiee	eeiɛa	ieeee	eeeei	əeeei	iɛɛɛa
L#6	eeiee	aɛiea	ɛaiaɛ	aɛiee	eeiea	ieeee	eeeei	aɛɛɛi	ieeea
L#7	eeiee	əeiea	eəiaɛ	eəiee	eeiea	ieeee	eeeei	əeeei	ieeea
L#8	eeiee	aeieə	ɛaiəe	aeiee	eeieə	ieeee	eeeei	aeeei	ieeeə
L#9	eeiee	aɛieə	ɛaiəe	aɛiee	eeieə	ieeee	eeeei	aɛɛɛi	ieeeə
L#10	eeiee	aɛɪɛa	ɛaɪaɛ	aeiee	ɛɛɪɛa	ieeee	eeeei	aeeei	ɪɛɛɛa
L#11	eeiee	aɛɪɛa	ɛaɪaɛ	aɛiee	ɛɛɪɛa	ieeee	eeeei	aɛɛɛi	ɪɛɛɛa
L#12	eeiee	aɛɪɛ	ɛaɪaɛ	aɛɪɛɛ	eeiea	ieeee	eeeei	aɛɛɛɪ	ieeea
L#13	eeiee	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	eeiɛa	ieeee	eeeei	aɛɛɛɪ	iɛɛɛa

Languages / Inputs :	ɛɛiɛɛ	aɛiaɛ	ɛaiaɛ	aɛiɛɛ	ɛɛiɛa	iɛɛɛɛ	ɛɛɛɛi	aɛɛɛi	iɛɛɛa
L#14	eeiee	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	ɛɛɪɛa	ieeee	eeeei	aɛɛɛɪ	ɪɛɛɛa
L#15	eeiee	aeiɛa	ɛaiaɛ	aeiee	eeiɛa	ieeee	eeeei	aeeei	iɛɛɛa
L#16	ɛɛɪɛɛ	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	ɛɛɪɛa	iɛɛɛ	ɛɛɛɪ	aɛɛɛɪ	iɛɛɛa
L#17	ɛɛɪɛɛ	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	ɛɛɪɛa	ɪɛɛɛɛ	ɛɛɛɛi	aɛɛɛi	ɪɛɛɛa
L#18	ɛɛɪɛɛ	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	ɛɛɪɛa	ɪɛɛɛɛ	ɛɛɛɛɪ	aɛɛɛɪ	ɪɛɛɛa
L#19	eeiɛɛ	aɛiɛa	ɛaiaɛ	aɛiɛɛ	eeiɛa	iɛɛɛɛ	eeeei	aɛɛɛi	iɛɛɛa
L#20	eeiɛɛ	əeiɛa	ɛəiaɛ	əeiɛɛ	eeiɛa	iɛɛɛɛ	eeeei	əeeei	iɛɛɛa
L#21	eeiɛɛ	aɛɪɛa	ɛaɪaɛ	aɛɪɛɛ	eeiɛa	iɛɛɛɛ	eeeei	aɛɛɛɪ	iɛɛɛa
L#22	eeiɛɛ	aeiɛa	ɛaiaɛ	aeiɛɛ	eeiɛa	iɛɛɛɛ	eeeei	aeeei	iɛɛɛa
L#23	ɛɛiee	aɛiɛa	ɛaiaɛ	aɛiee	ɛɛiɛa	ieeee	ɛɛɛɛi	aɛɛɛi	iɛɛɛa
L#24	ɛɛiee	aɛiea	ɛaiaɛ	aɛiee	ɛɛiea	ieeee	ɛɛɛɛi	aɛɛɛi	ieeea
L#25	ɛɛiee	aɛieə	ɛaiəe	aɛiee	ɛɛieə	ieeee	ɛɛɛɛi	aɛɛɛi	ieeeə
L#26	ɛɛiee	aɛɪɛa	ɛaɪaɛ	aɛiee	ɛɛɪɛa	ieeee	ɛɛɛɛi	aɛɛɛi	ɪɛɛɛa

APPENDIX C

COMPLETE WINNER-LOSER PAIRS

C.1 W∼L pair L#10

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	Faith[+A]	*[−A]C_o[+A]	A-L[+A]	Faith[−A]	A-R[−A]
a.	aɛiɛɛ	aeiee	əeiɛɛ	W	W	W	W		L	L	L	L
b.	aɛiɛa	aɛɪɛa	əeiɛa	W	W	W	W	L			W	
c.	aɛiɛa	aɛɪɛa	əeiea	W	W	W	W	L			W	
d.	ɛaiaɛ	ɛaɪaɛ	eəiaɛ	W	W	W	W	L			W	
e.	aɛiɛɛ	aeiee	əeiee	W					L	L	W	L
f.	aɛɛɛi	aeeei	əeeei	W					L	L	W	L
g.	aɛiɛa	aɛɪɛa	aeieə	W				L	W	W	W	W
h.	aɛiɛa	aɛɪɛa	aɛieə	W				L	W	W	W	W
i.	ɛaiaɛ	ɛaɪaɛ	ɛaiəe	W				L	W	W	W	W
j.	ɛɛiɛa	ɛɛɪɛa	ɛɛieə	W				L	W	W	W	W
k.	aɛiɛa	aɛɪɛa	əeieə	W				L		W		
l.	ɛaiaɛ	ɛaɪaɛ	eəiəe	W				L			W	
m.	ɛɛiɛa	ɛɛɪɛa	eeieə	W				L			W	
n.	iɛɛɛa	ɪɛɛɛa	ieeeə	W				L		W		
o.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ		W	W	W		W	W	L	W
p.	aɛiɛɛ	aeiee	aɛiɛɛ		W	W	W			W	L	L
q.	ɛɛiɛɛ	eeiee	eeiɛɛ		W	W	W				L	
r.	aɛiɛɛ	aeiee	aeiɛɛ		W	W	W				L	
s.	iɛɛɛɛ	ieeee	iɛɛɛɛ		W	W	W				L	
t.	aɛiɛa	aɛɪɛa	aeiea		W	W	W	L	W	W	W	W
u.	aɛiɛa	aɛɪɛa	aɛiea		W	W	W	L	W	W	W	W
v.	aɛiɛa	aɛɪɛa	aeiɛa		W	W	W	L	W	W	W	W
w.	ɛɛiɛa	ɛɛɪɛa	ɛɛiea		W	W	W	L	W	W	W	W
x.	aɛiɛa	aɛɪɛa	aɛiɛa		W	W	W	L	W	W		W
y.	ɛaiaɛ	ɛaɪaɛ	ɛaiaɛ		W	W	W	L	W	W		W
z.	ɛɛiɛa	ɛɛɪɛa	ɛɛiɛa		W	W	W	L	W	W		W
aa.	ɛɛiɛa	ɛɛɪɛa	eeiea		W	W	W	L			W	
ba.	ɛɛiɛa	ɛɛɪɛa	eeiɛa		W	W	W	L			W	
ca.	iɛɛɛa	ɪɛɛɛa	ieeea		W	W	W	L			W	
da.	iɛɛɛa	ɪɛɛɛa	iɛɛɛa		W	W	W	L				

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	Faith[+A]	*[−A]C_o[+A]	A-L[+A]	Faith[−A]	A-R[−A]
a.	ɛɛiɛɛ	eeiee	ɛɛɪɛɛ					W			L	
b.	iɛɛɛɛ	ieeee	ɪɛɛɛɛ					W			L	
c.	ɛɛɛɛi	eeeei	ɛɛɛɛɪ					W			L	
d.	aɛiɛɛ	aeiee	aɛɪɛɛ					W	L	L	L	L
e.	aɛɛɛi	aeeei	aɛɛɛɪ					W	L	L	L	L

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	Faith[+A]	*[−A]C_o[+A]	A-L[+A]	Faith[−A]	A-R[−A]
a.	ɛɛiɛɛ	eeiee	ɛɛiee						W	W	L	W
b.	ɛɛɛɛi	eeeei	ɛɛɛɛi						W	W	L	W
c.	aɛiɛɛ	aeiee	aɛiee							W	L	L
d.	aɛɛɛi	aeeei	aɛɛɛi							W	L	L

C.2 W∼L pair L#11

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	Faith[+A]	*[−A]C_o[+A]	A-R[−A]	Faith[−A]	A-L[+A]
a.	aɛiɛɛ	aɛiee	əeiɛɛ	W	W	W	W		L	L		L
b.	aɛiɛa	aɛiɛa	əeiɛa	W	W	W	W	L			W	
c.	aɛiɛa	aɛiɛa	əeiea	W	W	W	W	L			W	
d.	ɛaiaɛ	ɛaiaɛ	eəiaɛ	W	W	W	W	L			W	
e.	aɛiɛɛ	aɛiee	əeiee	W					L	L	W	L
f.	aɛɛɛi	aɛɛɛi	əeeei	W					L	L	W	L
g.	aɛiɛa	aɛiɛa	aeieə	W				L	W	W	W	W
h.	aɛiɛa	aɛiɛa	aɛieə	W				L	W	W	W	W
i.	ɛaiaɛ	ɛaiaɛ	ɛaiəe	W				L	W	W	W	W
j.	ɛɛiɛa	ɛɛiɛa	ɛɛieə	W				L	W	W	W	W
k.	aɛiɛa	aɛiɛa	əeieə	W				L			W	
l.	ɛaiaɛ	ɛaiaɛ	eəiəe	W				L			W	
n.	ɛɛiɛa	ɛɛiɛa	eeieə	W				L			W	
o.	iɛɛɛa	ɪɛɛɛa	ieeeə	W				L			W	
p.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ		W	W	W		W	W	L	W
q.	aɛiɛɛ	aɛiee	aeiɛɛ		W	W	W			W	L	L
r.	ɛɛiɛɛ	eeiee	eeiɛɛ		W	W	W				L	
s.	aɛiɛɛ	aɛiee	aɛiɛɛ		W	W	W				L	
t.	iɛɛɛɛ	ieeee	iɛɛɛɛ		W	W	W				L	
u.	aɛiɛa	aɛiɛa	aeiea		W	W	W	L	W	W	W	W
v.	aɛiɛa	aɛiɛa	aɛiea		W	W	W	L	W	W	W	W
w.	aɛiɛa	aɛiɛa	aeiɛa		W	W	W	L	W	W	W	W
x.	ɛɛiɛa	ɛɛiɛa	ɛɛiea		W	W	W	L	W	W	W	W
y.	aɛiɛa	aɛiɛa	aɛiɛa		W	W	W	L	W	W		W
z.	ɛaiaɛ	ɛaiaɛ	ɛaiaɛ		W	W	W	L	W	W		W
aa.	ɛɛiɛa	ɛɛiɛa	ɛɛiɛa		W	W	W	L	W	W		W
ba.	ɛɛiɛa	ɛɛiɛa	eeiɛa		W	W	W	L			W	
ca.	ɛɛiɛa	ɛɛiɛa	eeiɛa		W	W	W	L			W	
da.	iɛɛɛa	ɪɛɛɛa	ieeɛa		W	W	W	L			W	
ea.	iɛɛɛa	ɪɛɛɛa	iɛɛɛa		W	W	W	L				

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	Faith[+A]	*[−A]C_o[+A]	A-R[−A]	Faith[−A]	A-L[+A]
a.	ɛɛiɛɛ	eeiee	ɛɛɪɛɛ					W			L	
b.	iɛɛɛɛ	ieeee	ɪɛɛɛɛ					W			L	
c.	ɛɛɛɛi	eeeei	ɛɛɛɛɪ					W			L	
d.	aɛɛɛi	aɛɛɛi	aɛɛɛɪ					W	L	L		L
e.	aɛiɛɛ	aɛiee	aɛɪɛɛ					W	L	L	L	L

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C$_o$[−A]	A-L[−A]	A-R[+A]	Faith[+A]	*[−A]C$_o$[+A]	A-R[−A]	Faith[−A]	A-L[+A]
a.	ɛɛiɛɛ	eeiee	ɛɛiee						W	W	L	W
b.	ɛɛɛɛi	eeeei	ɛɛɛɛi						W	W	L	W
c.	aɛiɛɛ	aɛiee	aeiee							W	W	L
d.	aɛɛɛi	aɛɛɛi	aeeei							W	W	L

C.3 W∼L pair L#12

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]	Faith[+A]	*[+A]C$_o$[−A]	A-R[+A]	Faith[−A]	A-L[−A]
a.	ɛɛiɛa	eeiea	ɛɛieə	W	W	W	W		L	L	L	L
b.	aɛiɛa	aɛɪɛa	aeieə	W	W	W	W	L			W	
c.	aɛiɛa	aɛɪɛa	aɛieə	W	W	W	W	L			W	
d.	ɛaiaɛ	ɛaɪaɛ	ɛaiəe	W	W	W	W	L			W	
e.	ɛɛiɛa	eeiea	eeieə	W					L	L	W	L
f.	iɛɛɛa	ieeea	ieeeə	W					L	L	W	L
g.	aɛiɛa	aɛɪɛa	əeiɛa	W				L	W	W	W	W
h.	aɛiɛa	aɛɪɛa	əeiea	W				L	W	W	W	W
i.	ɛaiaɛ	ɛaɪaɛ	eəiaeɛ	W				L	W	W	W	W
j.	aɛiɛɛ	aɛɪɛɛ	əeiɛɛ	W				L	W	W	W	W
k.	aɛiɛa	aɛɪɛa	əeieə	W				L			W	
l.	ɛaiaɛ	ɛaɪaɛ	eəiəe	W				L			W	
m.	aɛiɛɛ	aɛɪɛɛ	əeiee	W				L			W	
n.	aɛɛɛi	aɛɛɛɪ	əeeei	W				L			W	
o.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ		W	W	W		W	W	L	W
p.	ɛɛiɛa	eeiea	ɛɛiɛa		W	W	W			W	L	L
q.	ɛɛiɛɛ	eeiee	ɛɛiee		W	W	W				L	
r.	ɛɛiɛa	eeiea	ɛɛiea		W	W	W				L	
s.	ɛɛɛɛi	eeeei	ɛɛɛɛi		W	W	W				L	
t.	aɛiɛa	aɛɪɛa	aeiea		W	W	W	L	W	W	W	W
u.	aɛiɛa	aɛɪɛa	aɛiea		W	W	W	L	W	W	W	W
v.	aɛiɛa	aɛɪɛa	aeiɛa		W	W	W	L	W	W	W	W
w.	aɛiɛɛ	aɛɪɛɛ	aeiɛɛ		W	W	W	L	W	W	W	W
x.	aɛiɛa	aɛɪɛa	aɛiɛa		W	W	W	L	W	W		W
y.	ɛaiaɛ	ɛaɪaɛ	ɛaiaɛ		W	W	W	L	W	W		W
z.	aɛiɛɛ	aɛɪɛɛ	aɛiɛɛ		W	W	W	L	W	W		W
aa.	aɛiɛɛ	aɛɪɛɛ	aeiee		W	W	W	L			W	
ba.	aɛiɛɛ	aɛɪɛɛ	aɛiee		W	W	W	L			W	
ca.	aɛɛɛi	aɛɛɛɪ	aeeei		W	W	W	L			W	
da.	aɛɛɛi	aɛɛɛɪ	aɛɛɛi		W	W	W	L				

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]	Faith[+A]	*[+A]C$_o$[−A]	A-R[+A]	Faith[−A]	A-L[−A]
a.	ɛɛiɛɛ	eeiee	ɛɛɪɛɛ					W			L	
b.	iɛɛɛɛ	ieeee	ɪɛɛɛɛ					W			L	
c.	ɛɛɛɛi	eeeei	ɛɛɛɛɪ					W			L	
d.	ɛɛiɛa	eeiea	ɛɛɪɛa					W	L	L	L	L
e.	iɛɛɛa	ieeea	ɪɛɛɛa					W	L	L	L	L

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]	Faith[+A]	*[+A]C$_o$[−A]	A-R[+A]	Faith[−A]	A-L[−A]
a.	ɛɛiɛɛ	eeiee	eeiɛɛ						W	W	L	W
b.	i ɛɛɛɛ	ieeee	iɛɛɛɛ						W	W	L	W
c.	ɛɛiɛa	eeiea	eeiɛa							W	L	L
d.	iɛɛɛa	ieeea	iɛɛɛa							W	L	L

C.4 W∼L pair L#13

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]	Faith[+A]	*[+A]C$_o$[−A]	A-L[−A]	Faith[−A]	Align-Right[+Atr]
a.	ɛɛiɛa	eeiɛa	ɛɛieə	W	W	W	W		L	L		L
b.	aɛiɛa	aɛɪɛa	aeieə	W	W	W	W	L			W	
c.	aɛiɛa	aɛɪɛa	aɛieə	W	W	W	W	L			W	
d.	ɛaiaɛ	ɛaɪaɛ	ɛaiəe	W	W	W	W	L			W	
e.	ɛɛiɛa	eeiɛa	eeieə	W					L	L	W	L
f.	iɛɛɛa	iɛɛɛa	ieeeə	W					L	L	W	L
g.	aɛiɛa	aɛɪɛa	əeiɛa	W				L	W	W	W	W
h.	aɛiɛa	aɛɪɛa	əeiea	W				L	W	W	W	W
i.	ɛaiaɛ	ɛaɪaɛ	eəiaɛ	W				L	W	W	W	W
j.	aɛiɛɛ	aɛɪɛɛ	əeiɛɛ	W				L	W	W	W	W
k.	aɛiɛa	aɛɪɛa	əeieə	W				L			W	
l.	ɛaiaɛ	ɛaɪaɛ	eəiəe	W				L			W	
m.	aɛiɛɛ	aɛɪɛɛ	əeiee	W				L			W	
n.	aɛɛɛi	aɛɛɛɪ	əeeei	W				L			W	
o.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ		W	W	W		W	W	L	W
p.	ɛɛiɛa	eeiɛa	ɛɛiea		W	W	W			W	L	L
q.	ɛɛiɛɛ	eeiee	ɛɛiee		W	W	W				L	
r.	ɛɛiɛa	eeiɛa	ɛɛiɛa		W	W	W				L	
s.	ɛɛɛɛi	eeeei	ɛɛɛɛi		W	W	W				L	
t.	aɛiɛa	aɛɪɛa	aeiea		W	W	W	L	W	W	W	W
u.	aɛiɛa	aɛɪɛa	aɛiea		W	W	W	L	W	W	W	W
v.	aɛiɛa	aɛɪɛa	aeiɛa		W	W	W	L	W	W	W	W
w.	aɛiɛɛ	aɛɪɛɛ	aeiɛɛ		W	W	W	L	W	W	W	W
x.	aɛiɛa	aɛɪɛa	aɛiɛa		W	W	W	L	W	W		W
y.	ɛaiaɛ	ɛaɪaɛ	ɛaiaɛ		W	W	W	L	W	W		W
z.	aɛiɛɛ	aɛɪɛɛ	aɛiɛɛ		W	W	W	L	W	W		W
aa.	aɛiɛɛ	aɛɪɛɛ	aeiee		W	W	W	L			W	
ba.	aɛiɛɛ	aɛɪɛɛ	aɛiee		W	W	W	L			W	
ca.	aɛɛɛi	aɛɛɛɪ	aeeei		W	W	W	L			W	
da.	aɛɛɛi	aɛɛɛɪ	aɛɛɛi		W	W	W	L				

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]	Faith[+A]	*[+A]C$_o$[−A]	A-L[−A]	Faith[−A]	A-L[−A]
a.	ɛɛiɛɛ	eeiee	ɛɛɪɛɛ					W			L	
b.	iɛɛɛɛ	ieeee	ɪɛɛɛɛ					W			L	
c.	ɛɛɛɛi	eeeei	ɛɛɛɛɪ					W			L	
d.	iɛɛɛa	iɛɛɛa	ɪɛɛɛa					W	L	L		L
e.	ɛɛiɛa	eeiɛa	ɛɛɪɛa					W	L	L	L	L

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C_o[+A]	A-L[+A]	A-R[−A]	Faith[+A]	*[+A]C_o[−A]	A-L[−A]	Faith[−A]	A-L[−A]
a.	ɛɛiɛɛ	eeiee	eeiɛɛ						W	W	L	W
b.	iɛɛɛɛ	ieeee	iɛɛɛɛ						W	W	L	W
c.	ɛɛiɛa	eeiɛa	eeiea							W	W	L
d.	iɛɛɛa	iɛɛɛa	ieeea							W	W	L

C.5 W∼L pair L#14

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	*[−A]C_o[+A]	A-L[+A]	A-R[−A]	Faith[+A]	Faith[−A]
a.	aɛiɛa	aɛɪɛa	əeiɛa	W	W	W	W				L	W
b.	aɛiɛa	aɛɪɛa	əeiea	W	W	W	W				L	W
c.	ɛaiaɛ	ɛaɪaɛ	eəiaɛ	W	W	W	W				L	W
d.	aɛiɛɛ	aɛɪɛɛ	əeiɛɛ	W	W	W	W				L	W
e.	aɛiɛa	aɛɪɛa	aeieə	W				W	W	W	L	W
f.	aɛiɛa	aɛɪɛa	aɛieə	W				W	W	W	L	W
g.	ɛaiaɛ	ɛaɪaɛ	ɛaiəe	W				W	W	W	L	W
h.	ɛɛiɛa	ɛɛɪɛa	ɛɛieə	W				W	W	W	L	W
i.	aɛiɛa	aɛɪɛa	əeieə	W							L	W
j.	ɛaiaɛ	ɛaɪaɛ	eəiəe	W							L	W
k.	aɛiɛɛ	aɛɪɛɛ	əeiee	W							L	W
l.	ɛɛiɛa	ɛɛɪɛa	eeieə	W							L	W
m.	aɛɛɛi	aɛɛɛɪ	əeeei	W							L	W
n.	iɛɛɛa	ɪɛɛɛa	ieeeə	W							L	W
o.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ		W	W	W	W	W	W		L
p.	aɛiɛa	aɛɪɛa	aeiea		W	W	W	W	W	W	L	W
q.	aɛiɛa	aɛɪɛa	aɛiea		W	W	W	W	W	W	L	W
r.	aɛiɛa	aɛɪɛa	aeiɛa		W	W	W	W	W	W	L	W
s.	aɛiɛɛ	aɛɪɛɛ	aeiɛɛ		W	W	W	W	W	W	L	W
t.	ɛɛiɛa	ɛɛɪɛa	ɛɛiea		W	W	W	W	W	W	L	W
u.	aɛiɛa	aɛɪɛa	aɛiɛa		W	W	W	W	W	W	L	
v.	ɛaiaɛ	ɛaɪaɛ	ɛaiaɛ		W	W	W	W	W	W	L	
w.	aɛiɛɛ	aɛɪɛɛ	aɛiɛɛ		W	W	W	W	W	W	L	
x.	ɛɛiɛa	ɛɛɪɛa	ɛɛiɛa		W	W	W	W	W	W	L	
y.	ɛɛiɛɛ	eeiee	eeiɛɛ		W	W	W					L
z.	iɛɛɛ	ieeee	iɛɛɛ		W	W	W					L
aa.	ɛɛiɛa	ɛɛɪɛa	eeiɛa		W	W	W				L	W
ba.	ɛɛiɛa	ɛɛɪɛa	eeiɛa		W	W	W				L	W
ca.	iɛɛɛa	ɪɛɛɛa	ieeea		W	W	W				L	W
da.	iɛɛɛa	ɪɛɛɛa	iɛɛɛa		W	W	W				L	
ea.	ɛɛiɛɛ	eeiee	ɛɛiee					W	W	W		L
fa.	ɛɛɛɛi	eeeei	ɛɛɛɛi					W	W	W		L
ga.	aɛiɛɛ	aɛɪɛɛ	aeiee					W	W	W	L	W
ha.	aɛiɛɛ	aɛɪɛɛ	aɛiee					W	W	W	L	W
ia.	aɛɛɛi	aɛɛɛɪ	aeeei					W	W	W	L	W
ja.	aɛɛɛi	aɛɛɛɪ	aɛɛɛi					W	W	W	L	

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C$_o$[−A]	A-L[−A]	A-R[+A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]	Faith[+A]	Faith[−A]
a.	ɛɛiɛɛ	eeiee	ɛɛiɛɛ								W	L
b.	iɛɛɛɛ	ieeee	iɛɛɛɛ								W	L
c.	ɛɛɛɛi	eeeei	ɛɛɛɛi								W	L

C.6 L∼W pair L#21

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]	Faith[+A]	Faith[−A]	*[+A]C$_o$[−A]	A-L[−A]	A-R[+A]
a.	ɛɛiɛa	eeiɛa	ɛɛieə	W	W	W	W			L	L	L
b.	aɛiɛa	aɛiɛa	aɛieə	W	W	W	W	L	W			
c.	aɛiɛa	aɛiɛa	aɛieə	W	W	W	W	L	W			
d.	ɛaiaɛ	ɛaiaɛ	ɛaiəe	W	W	W	W	L	W			
e.	ɛɛiɛa	eeiɛa	eeieə	W					W	L	L	L
f.	iɛɛɛa	iɛɛɛa	ieeeə	W					W	L	L	L
g.	aɛiɛa	aɛiɛa	əeiɛa	W				L	W	W	W	W
h.	aɛiɛa	aɛiɛa	əeiea	W				L	W	W	W	W
i.	ɛaiaɛ	ɛaiaɛ	eəiaɛ	W				L	W	W	W	W
j.	aɛiɛɛ	aɛiɛɛ	əeiɛɛ	W				L	W	W	W	W
k.	aɛiɛa	aɛiɛa	əeieə	W				L	W			
l.	ɛaiaɛ	ɛaiaɛ	eəiəe	W				L	W			
m.	aɛiɛɛ	aɛiɛɛ	əeiee	W				L	W			
n.	aɛɛɛi	aɛɛɛi	əeeei	W				L	W			
o.	ɛɛiɛɛ	eeiɛɛ	ɛɛiee		W	W	W			L	L	L
p.	ɛɛiɛa	eeiɛa	ɛɛiea		W	W	W		L		W	L
q.	ɛɛiɛɛ	eeiɛɛ	ɛɛiɛɛ		W	W	W		L			
r.	ɛɛiɛa	eeiɛa	ɛɛiɛa		W	W	W		L			
s.	ɛɛɛɛi	eeeei	ɛɛɛɛi		W	W	W		L			
t.	aɛiɛa	aɛiɛa	aɛiea		W	W	W	L	W	W	W	W
u.	aɛiɛa	aɛiɛa	aɛiɛa		W	W	W	L	W	W	W	W
v.	aɛiɛa	aɛiɛa	aɛiɛa		W	W	W	L	W	W	W	W
w.	aɛiɛɛ	aɛiɛɛ	aɛiɛɛ		W	W	W	L	W	W	W	W
x.	aɛiɛɛ	aɛiɛɛ	aɛiee		W	W	W	L	W			
y.	aɛiɛɛ	aɛiɛɛ	aɛiee		W	W	W	L	W			
z.	aɛɛɛi	aɛɛɛi	aɛeei		W	W	W	L	W			
aa.	aɛiɛa	aɛiɛa	aɛiɛa		W	W	W	L		W	W	W
ba.	ɛaiaɛ	ɛaiaɛ	ɛaiaɛ		W	W	W	L		W	W	W
ca.	aɛiɛɛ	aɛiɛɛ	aɛiɛɛ		W	W	W	L		W	W	W
da.	aɛɛɛi	aɛɛɛi	aɛɛɛi		W	W	W	L				

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]	Faith[+A]	Faith[−A]	*[+A]C$_o$[−A]	A-L[−A]	A-R[+A]
a.	iɛɛɛɛ	iɛɛɛɛ	iɛɛɛɛ					W		L	L	L
b.	iɛɛɛa	iɛɛɛa	iɛɛɛa					W		L	L	L
c.	ɛɛɛɛi	eeeei	ɛɛɛɛi					W	L			
d.	ɛɛiɛɛ	eeiɛɛ	ɛɛiɛɛ					W	L	L	L	L
e.	ɛɛiɛa	eeiɛa	ɛɛiɛa					W	L	L	L	L

Pair	**Input**	Winner	Loser	*[+L,+A]	*[−A]C_o[+A]	A-L[+A]	A-R[−A]	Faith[+A]	Faith[−A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]
a.	ɛɛiɛa	eeiɛa	eeiea						W		W	L
b.	iɛɛɛa	iɛɛɛa	ieeea						W		W	L
c.	ɛɛiɛɛ	eeiɛɛ	eeiee						W	L	L	L
d.	iɛɛɛɛ	iɛɛɛɛ	ieeee						W	L	L	L

C.7 W∼L pair L#26

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	Faith[+A]	Faith[−A]	*[−A]C_o[+A]	A-L[+A]	A-R[−A]
a.	aɛiɛɛ	aɛiee	əeiɛɛ	W	W	W	W			L	L	L
b.	aɛiɛa	aɛɪɛa	əeiɛa	W	W	W	W	L	W			
c.	aɛiɛa	aɛɪɛa	əeiea	W	W	W	W	L	W			
d.	ɛaiaɛ	ɛaɪaɛ	eəiaɛ	W	W	W	W	L	W			
e.	aɛiɛɛ	aɛiee	əeiee	W					W	L	L	L
f.	aɛɛɛi	aɛɛɛi	əeeei	W					W	L	L	L
g.	aɛiɛa	aɛɪɛa	aeiəa	W				L	W	W	W	W
h.	aɛiɛa	aɛɪɛa	aɛiəa	W				L	W	W	W	W
i.	ɛaiaɛ	ɛaɪaɛ	ɛaiəe	W				L	W	W	W	W
j.	ɛɛiɛa	ɛɛɪɛa	ɛɛiəa	W				L	W	W	W	W
k.	aɛiɛa	aɛɪɛa	əeiəa	W				L	W			
l.	ɛaiaɛ	ɛaɪaɛ	eəiəe	W				L	W			
m.	ɛɛiɛa	ɛɛɪɛa	eeiəa	W				L	W			
n.	iɛɛɛa	iɛɛɛa	ieeəa	W				L	W			
o.	ɛɛiɛɛ	ɛɛiee	eeiɛɛ		W	W	W			L	L	L
p.	ɛɛiɛɛ	ɛɛiee	ɛɛiɛɛ		W	W	W		L			
q.	aɛiɛɛ	aɛiee	aɛiɛɛ		W	W	W		L			
r.	iɛɛɛɛ	ieeee	iɛɛɛɛ		W	W	W		L			
s.	aɛiɛɛ	aɛiee	aeiɛɛ		W	W	W		L		L	W
t.	aɛiɛa	aɛɪɛa	aeiea		W	W	W	L	W	W	W	W
u.	aɛiɛa	aɛɪɛa	aɛiea		W	W	W	L	W	W	W	W
v.	aɛiɛa	aɛɪɛa	aeiɛa		W	W	W	L	W	W	W	W
w.	ɛɛiɛa	ɛɛɪɛa	ɛɛiea		W	W	W	L	W	W	W	W
x.	ɛɛiɛa	ɛɛɪɛa	eeiea		W	W	W	L	W			
y.	ɛɛiɛa	ɛɛɪɛa	eeiɛa		W	W	W	L	W			
z.	iɛɛɛa	iɛɛɛa	ieeea		W	W	W	L	W			
aa.	aɛiɛa	aɛɪɛa	aɛiɛa		W	W	W	L		W	W	W
ba.	ɛaiaɛ	ɛaɪaɛ	ɛaiaɛ		W	W	W	L		W	W	W
ca.	ɛɛiɛa	ɛɛɪɛa	ɛɛiɛa		W	W	W	L		W	W	W
da.	iɛɛɛa	iɛɛɛa	iɛɛɛa		W	W	W	L				

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C_o[−A]	A-L[−A]	A-R[+A]	Faith[+A]	Faith[−A]	*[−A]C_o[+A]	A-L[+A]	A-R[−A]
a.	ɛɛɛi	ɛɛɛi	ɛɛɛɪ					W		L	L	L
b.	aɛɛi	aɛɛi	aɛɛɪ					W		L	L	L
c.	iɛɛɛ	ieeee	iɛɛɛ					W	L			
d.	ɛɛiɛ	ɛɛiee	ɛɛiɛ					W	L	L	L	L
e.	aɛiɛ	aɛiee	aɛiɛ					W	L	L	L	L

Pair	**Input**	Winner	Loser	*[+L,+A]	*[+A]C$_o$[−A]	A-L[−A]	A-R[+A]	Faith[+A]	Faith[−A]	*[−A]C$_o$[+A]	A-L[+A]	A-R[−A]
a.	aɛiɛɛ	aɛiee	aeiee						W		L	W
b.	aɛɛɛi	aɛɛɛi	aeeei						W		L	W
c.	ɛɛiɛɛ	ɛɛiee	eeiee						W	L	L	L
d.	ɛɛɛɛi	ɛɛɛɛi	eeeei						W	L	L	L

APPENDIX D
VBA SCRIPTS

Note on AI assistance. The VBA scripts reproduced in this appendix were developed with the assistance of Claude/Anthropic. AI assistance was used to generate and debug portions of the code, which I subsequently reviewed and tested.

```
'Author: You
Option Explicit
Sub mySTUFF()
'insert your subs and funcs in this module
'export & re-insert upon update
End Sub
Sub markPair()
Dim L0 As String
Dim L1 As String
Dim L2 As String
Dim cel As Range
Dim myMOAT As Range
Set myMOAT = Range(Cells(3, 4), Cells(45, 14))
myMOAT.Interior.ColorIndex = 0
L0 = InputBox("Give 2 lgs., comma sep")
L0 = Trim(L0)
L1 = Chomp(L0, ",")
L2 = endChomp(L0, ",")
'MsgBox "L1 is " & L1 & "L2 is " & L2
For Each cel In myMOAT
   If Trim(cel.Value) = L1 Then
        cel.Interior.ColorIndex = 23
   End If
```

```
    If Trim(cel.Value) = L2 Then
        cel.Interior.ColorIndex = 22
    End If
Next
End Sub
Function GetATR(char As String) As Integer ' Assign code pointer to an ATR feature
    ' Returns 1 for [+ATR], 0 for [-ATR], -1 for non-vowel
    Dim c As Variant
    c = AscW(char)
    Select Case c
        Case 105, 117, 101, 111, 601 ' i, u, e, o, ?
            GetATR = 1
        Case 618, 650, 603, 596, 97 ' ?, ?, ?, ?, a
            GetATR = 0
        Case Else
            GetATR = -1
    End Select
End Function
Function AlignLPlusATR(target As String) As Variant ' Align Function FIX
    Dim i As Variant
    Dim count As Variant
    Dim inSpan As Boolean
    Dim vowelPos As Variant

    count = 0
    inSpan = False
    vowelPos = 0

    For i = 1 To Len(target)
        ' Only count vowel positions
        If GetATR(Mid(target, i, 1)) <> -1 Then
            If GetATR(Mid(target, i, 1)) = 1 Then
                If Not inSpan Then
```

```
57                      count = count + vowelPos
58                      inSpan = True
59                  End If
60              Else
61                  inSpan = False
62              End If
63              vowelPos = vowelPos + 1
64          End If
65      Next i
66
67      AlignLPlusATR = count
68  End Function
69  Function BanSchwa(target As String) As Variant ' Faith[a] equivalent (ban
        [+atr,+lo] in the output)
70      Dim i As Variant
71      Dim count As Variant
72
73      count = 0
74
75      For i = 1 To Len(target)
76          If AscW(Mid(target, i, 1)) = 601 Then
77              count = count + 1
78          End If
79      Next i
80
81      BanSchwa = count
82  End Function
83
84  Function AFLAtr(sOutput As String) As Long 'OG defined Align constraint (it's
        wrong)
85  'Cumulative distance version - counts position of every [+ATR] vowel
86  'This will penalize spreading (more [+ATR] vowels = more violations)
87  Dim i As Long
```

```
88  Dim alignL As Long
89  Dim vowelPos As Long
90
91  alignL = 0
92  vowelPos = 0
93
94  For i = 1 To Len(sOutput)
95      If GetATR(Mid(sOutput, i, 1)) <> -1 Then 'it's a vowel
96          If GetATR(Mid(sOutput, i, 1)) = 1 Then '[+ATR]
97              alignL = alignL + vowelPos
98          End If
99          vowelPos = vowelPos + 1
100     End If
101 Next i
102
103 AFLAtr = alignL
104 End Function
105
106
107 Function Faithi(sIn As String, sOut As String) As Variant ' Preserves [+hi,+atr]
        in the output
108     Dim i As Variant
109     Dim count As Variant
110
111     If Len(sIn) = 0 Or Len(sOut) = 0 Then
112     Faithi = ""
113     Exit Function
114 End If
115
116     If Len(sIn) <> Len(sOut) Then
117         Faithi = ""
118         Exit Function
119     End If
```

```
120
121     count = 0
122
123     For i = 1 To Len(sIn)
124         If AscW(Mid(sIn, i, 1)) = 105 And AscW(Mid(sOut, i, 1)) <> 105 Then
125             count = count + 1
126         End If
127     Next i
128
129     Faithi = count
130 End Function
131
132 Function FaithPlusATR(sIn As String, sOut As String) As Variant ' Preserve [+atr]
        in the output
133     Dim i As Variant
134     Dim count As Variant
135
136     If Len(sIn) = 0 Or Len(sOut) = 0 Then
137     FaithPlusATR = ""
138     Exit Function
139 End If
140
141     If Len(sIn) <> Len(sOut) Then
142         FaithPlusATR = ""
143         Exit Function
144     End If
145
146     For i = 1 To Len(sIn)
147         If GetATR(Mid(sIn, i, 1)) = 1 And GetATR(Mid(sOut, i, 1)) = 0 Then
148             count = count + 1
149         End If
150     Next i
151
```

```
152     FaithPlusATR = count
153 End Function
154
155
156 Function FaithMinusATR(sIn As String, sOut As String) As Variant ' Preserves
        [-atr] in the output
157     Dim i As Variant
158     Dim count As Variant
159
160     If Len(sIn) = 0 Or Len(sOut) = 0 Then
161     FaithMinusATR = ""
162     Exit Function
163     End If
164
165     If Len(sIn) <> Len(sOut) Then
166         FaithMinusATR = ""
167         Exit Function
168     End If
169
170     For i = 1 To Len(sIn)
171         If GetATR(Mid(sIn, i, 1)) = 0 And GetATR(Mid(sOut, i, 1)) = 1 Then
172             count = count + 1
173         End If
174     Next i
175
176     FaithMinusATR = count
177 End Function
178
179 Function AlignRPlusATR(target As String) As Variant
180 ' ALIGN-R([+ATR]): for each [+ATR] span, count vowels to its RIGHT (from span end)
181     Dim i As Long, vowelPos As Long, totalVowels As Long
182     Dim spanEnd As Long
183     Dim inSpan As Boolean
```

```
    Dim spans() As Long
    Dim spanCount As Long

    ' First pass: count total vowels
    totalVowels = 0
    For i = 1 To Len(target)
        If GetATR(Mid(target, i, 1)) <> -1 Then totalVowels = totalVowels + 1
    Next i

    ' Second pass: collect span-end vowel positions
    spanCount = 0
    inSpan = False
    vowelPos = 0

    For i = 1 To Len(target)
        If GetATR(Mid(target, i, 1)) <> -1 Then
            If GetATR(Mid(target, i, 1)) = 1 Then
                inSpan = True
                spanEnd = vowelPos
            Else
                If inSpan Then
                    spanCount = spanCount + 1
                    ReDim Preserve spans(1 To spanCount)
                    spans(spanCount) = spanEnd
                    inSpan = False
                End If
            End If
            vowelPos = vowelPos + 1
        End If
    Next i

    ' Close final span if string ends in [+ATR]
    If inSpan Then
```

```
217         spanCount = spanCount + 1
218         ReDim Preserve spans(1 To spanCount)
219         spans(spanCount) = spanEnd
220     End If
221
222     Dim total As Long
223     total = 0
224     For i = 1 To spanCount
225         total = total + (totalVowels - 1 - spans(i))
226     Next i
227
228     AlignRPlusATR = total
229 End Function
230
231 Function AlignLMinusATR(target As String) As Variant
232 ' ALIGN-L([-ATR]): for each [-ATR] span, count vowels to its LEFT (from span
        start)
233     Dim i As Long, vowelPos As Long
234     Dim spanStart As Long
235     Dim inSpan As Boolean
236     Dim total As Long
237
238     total = 0
239     inSpan = False
240     vowelPos = 0
241
242     For i = 1 To Len(target)
243         If GetATR(Mid(target, i, 1)) <> -1 Then
244             If GetATR(Mid(target, i, 1)) = 0 Then
245                 If Not inSpan Then
246                     total = total + vowelPos
247                     inSpan = True
248                 End If
```

```
            Else
                inSpan = False
            End If
            vowelPos = vowelPos + 1
        End If
    Next i

    AlignLMinusATR = total
End Function

Function AlignRMinusATR(target As String) As Variant
' ALIGN-R([-ATR]): for each [-ATR] span, count vowels to its RIGHT (from span end)
    Dim i As Long, vowelPos As Long, totalVowels As Long
    Dim spanEnd As Long
    Dim inSpan As Boolean
    Dim spans() As Long
    Dim spanCount As Long

    totalVowels = 0
    For i = 1 To Len(target)
        If GetATR(Mid(target, i, 1)) <> -1 Then totalVowels = totalVowels + 1
    Next i

    spanCount = 0
    inSpan = False
    vowelPos = 0

    For i = 1 To Len(target)
        If GetATR(Mid(target, i, 1)) <> -1 Then
            If GetATR(Mid(target, i, 1)) = 0 Then
                inSpan = True
                spanEnd = vowelPos
            Else
```

```
282                 If inSpan Then
283                     spanCount = spanCount + 1
284                     ReDim Preserve spans(1 To spanCount)
285                     spans(spanCount) = spanEnd
286                     inSpan = False
287                 End If
288             End If
289             vowelPos = vowelPos + 1
290         End If
291     Next i
292
293     If inSpan Then
294         spanCount = spanCount + 1
295         ReDim Preserve spans(1 To spanCount)
296         spans(spanCount) = spanEnd
297     End If
298
299     Dim total As Long
300     total = 0
301     For i = 1 To spanCount
302         total = total + (totalVowels - 1 - spans(i))
303     Next i
304
305     AlignRMinusATR = total
306 End Function
307
308 Function SeqMarkProg(target As String) As Variant
309 ' Counts [+ATR][-ATR] bigrams on the vowel tier (progressive disharmony)
310     Dim i As Long
311     Dim prevATR As Integer
312     Dim currATR As Integer
313     Dim count As Long
314
```

```
315     count = 0
316     prevATR = -1 ' no previous vowel yet
317
318     For i = 1 To Len(target)
319         currATR = GetATR(Mid(target, i, 1))
320         If currATR <> -1 Then ' it's a vowel
321             If prevATR = 1 And currATR = 0 Then
322                 count = count + 1
323             End If
324             prevATR = currATR
325         End If
326     Next i
327
328     SeqMarkProg = count
329 End Function
330
331 Function SeqMarkReg(target As String) As Variant
332 ' Counts [-ATR][+ATR] bigrams on the vowel tier (regressive disharmony)
333     Dim i As Long
334     Dim prevATR As Integer
335     Dim currATR As Integer
336     Dim count As Long
337
338     count = 0
339     prevATR = -1
340
341     For i = 1 To Len(target)
342         currATR = GetATR(Mid(target, i, 1))
343         If currATR <> -1 Then
344             If prevATR = 0 And currATR = 1 Then
345                 count = count + 1
346             End If
347             prevATR = currATR
```

```
348         End If
349     Next i
350
351     SeqMarkReg = count
352 End Function
```

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